Hydrodynamics

Contents

1 Continuum dynamics 4
- 1.1 The velocity field . . . 4
  - 1.1.1 Streamlines . . . 4
  - 1.1.2 Pathlines . . . 4
  - 1.1.3 Streaklines . . . 5
  - 1.1.4 Steady flow case . . . 5
- 1.2 Mass conservation . . . 7
- 1.3 Incompressible flow . . . 8
  - 1.3.1 Global and local forms . . . 8
  - 1.3.2 Leonardo's Law. . . . 8
- 1.4 Lagrangian and Eulerian derivatives . . . 11
  - 1.4.1 Lagrangian description . . . 11
  - 1.4.2 Eulerian description . . . 11
- 1.5 Cauchy's equation . . . 12
- 1.6 Big Bang and Newtonian Cosmology . . . 12
  - 1.6.1 Explosion field and Hubble law . . . 12
  - 1.6.2 Newtonian cosmology . . . 13
  - 1.6.3 Cosmological equations . . . 13
  - 1.6.4 Cosmic scale factor . . . 13
  - 1.6.5 Critical density . . . 13
  - 1.6.6 Cosmological constant . . . 13
- 1.7 Problems . . . 14

2 Incompressible frictionless (Euler) fluids 18
- 2.1 Euler equation for incompressible ideal flow . . . 18
- 2.2 Adiabatic and isentropic flow . . . 19
  - 2.2.1 Boundary conditions . . . 20
  - 2.2.2 Irrotational flow . . . 21
- 2.3 Bernoulli's equation . . . 21
  - 2.3.1 Flow tube . . . 21
  - 2.3.2 The Venturi effect . . . 22
  - 2.3.3 Torricelli's law . . . 22
  - 2.3.4 Quasi-stationary emptying . . . 23
  - 2.3.5 Case: The Pitot tube . . . 24
  - 2.3.6 Rotating fluid in steady state . . . 24
  - 2.3.7 Exercises . . . 26
- 2.4 Vorticity . . . 29
  - 2.4.1 The vorticity field . . . 29
  - 2.4.2 Vortex lines . . . 30
  - 2.4.3 Equation of motion for vorticity . . . 31
- 2.5 Circulation . . . 32
  - 2.5.1 Conservation of circulation . . . 32

2.5.2 Stokes' theorem and steady flow . . . 33
2.6 Potential flow . . . 34
2.6.1 Incompressible potential flow . . . 34
2.7 Application: Cylinder in uniform crosswind . . . 34
2.7.1 Velocity potential . . . 35
2.7.2 Velocity and pressure fields . . . 36
2.7.3 Lift on a half-cylinder . . . 36
2.7.4 The Magnus effect . . . 37
2.8 Problems . . . 38

3 Compresible flow **40**
3.1 Sound Waves in an fluids . . . 40
3.1.1 Adiabatic (isentropic) perturbations . . . 41
3.1.2 Sound speed in the atmosphere . . . 41
3.1.3 Plane wave solution . . . 42
3.1.4 The Jeans instability . . . 44
3.2 Steady compressible flow . . . 46
3.2.1 Effective incompressibility . . . 46
3.2.2 Bernoulli's theorem for barotropic fluids . . . 48
3.2.3 Stagnation temperature rise . . . 48
3.3 Problems . . . 49

4 Viscosity **51**
4.1 Shear viscosity . . . 51
4.1.1 Molecular origin of viscosity in gases . . . 52
4.1.2 Temperature dependence of viscosity . . . 52
4.1.3 Kinematic viscosity . . . 53
4.1.4 Velocity-driven planar flow . . . 53
4.1.5 Case: Steady planar flow . . . 53
4.2 Newtonian and Non-Newtonian Fluids . . . 53
4.2.1 Newtonian fluids . . . 54
4.2.2 Non-Newtonian fluids . . . 54
4.2.3 Power-law model . . . 56
4.3 Dynamics of incompressible Newtonian fluids . . . 57
4.3.1 Isotropic viscous stress . . . 57
4.3.2 The Navier–Stokes equations . . . 57
4.3.3 Boundary conditions . . . 58
4.4 Classification of flows . . . 59
4.4.1 The Reynolds number . . . 59
4.4.2 Examples . . . 59
4.4.3 Hydrodynamic similarity . . . 60
4.4.4 Example: Robofly . . . 61
4.4.5 Example: High-pressure wind tunnels . . . 61
4.5 Dynamics of compressible Newtonian fluids . . . 61
4.5.1 The Navier–Stokes equations . . . 61
4.5.2 Boundary conditions . . . 62
4.6 Problems . . . 62

5 Steady, incompressible, viscous flow **65**
5.1 Pressure-driven channel flow 65
5.2 Gravity-driven planar flow 68
5.3 Inclined flow with an open surface 69
5.4 Laminar pipe flow 70
5.5 Laminar cylindric flow 73
5.6 Application: Accretion Disks 77
5.7 Problems 79

6 Creeping flow: the Stokes flow **88**
6.1 Creeping flow around a solid ball 88
6.2 Life at Low Reynolds Number 92
6.3 Problems 93

Disclaimer

Discussions taken from Alonso Sepúlveda [1], Landau & Lifschits [2], Lautrup [3] books

1 Continuum dynamics

Two fundamental principles govern the motion of continuous matter: conservation of mass and Newton's Second Law (momentum balance) applied to a continuous system. With appropriate expressions for internal and external forces, these lead to the equations of motion for the mass density and velocity fields.

1.1 The velocity field

The velocity field $\vec{v}(\vec{x}, t)$ represents the center-of-mass velocity of the collection of molecules in a material particle with center-of-mass position $\vec{x}$ at time t.

For a material particle of mass dM and volume dV, the total momentum is

$$d\vec{P} = \vec{v}(x, t)\, dM = \vec{v}\, \rho\, dV, \tag{1}$$

where $\rho(x, t)$ is the mass density field and $\vec{v}(x, t)$ is the center-of-mass velocity of the material particle. Because ρ and $\vec{v}$ are averages over many molecules, their fluctuations become negligible provided the particle size is larger than the microscopic scale L'_{micro}.

1.1.1 Streamlines

Streamlines are curves everywhere tangent to the velocity field at a fixed time t_0. They satisfy the ordinary differential equation

$$\frac{d\vec{x}}{dt} = \vec{v}(\vec{x}, t_0). \tag{2}$$

A streamline shows the instantaneous direction of flow and depends on the chosen reference frame. At a fixed time, there is one and only one streamline passing through each point in space, and streamlines cannot intersect (see Fig. 1).

1.1.2 Pathlines

A particle trajectory (or particle orbit) is the actual path followed by an infinitesimal tracer particle advected by the flow. It satisfies

$$\frac{d\vec{x}}{dt} = \vec{v}(\vec{x}, t). \tag{3}$$

Given initial data $\vec{x}(t_0) = \vec{x}_0$, the trajectory is defined for all times. Unlike streamlines, different trajectories may intersect since they correspond to different times.

A pathline is the locus of points traversed by a given particle as it travels in a field of flow; the path- line provides us with a "history" of the particle's locations. A photograph of a pathline would require a time exposure of an illuminated particle. A photograph showing pathlines of particles below a water surface with waves is given in Fig. 2.

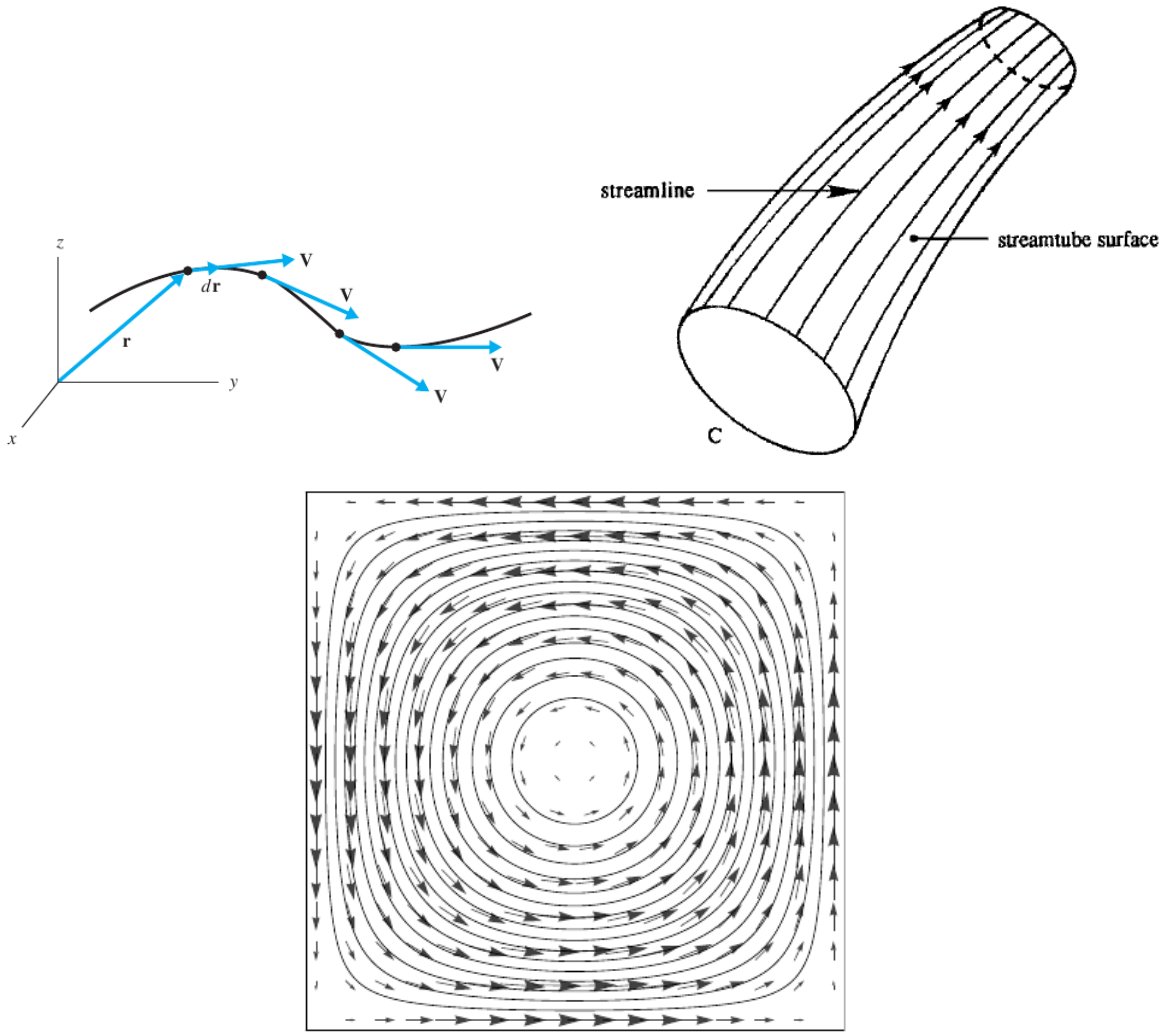


Figure 1: Top: Streamline and streamtube in a flow field. Bottom: Plots of streamlines and little arrows for the stationary incompressible flow $\vec{v} = (\sin x \cos y, -\cos x \sin y)$. Figures taken from Ref. [4] and Ref. [3].

1.1.3 Streaklines

A streakline is formed by all particles that have passed through the same fixed point $\vec{x}_0$, but at different emission times. Practically, this is obtained by continuously injecting dye or smoke at $\vec{x}_0$.

A streakline is defined as an instantaneous line whose points are occupied by all particles originating from some specified point in the flow field. Streaklines tell us where the particles are "right now." A photograph of a streakline would be a snapshot of the set of illuminated particles that passed a certain point. Figure 3.2 shows streaklines produced by the continuous release of a small diameter stream of smoke as it moves around a cylinder.

1.1.4 Steady flow case

For steady flow, where $\vec{v}(\vec{x}, t) = \vec{v}(\vec{x})$,

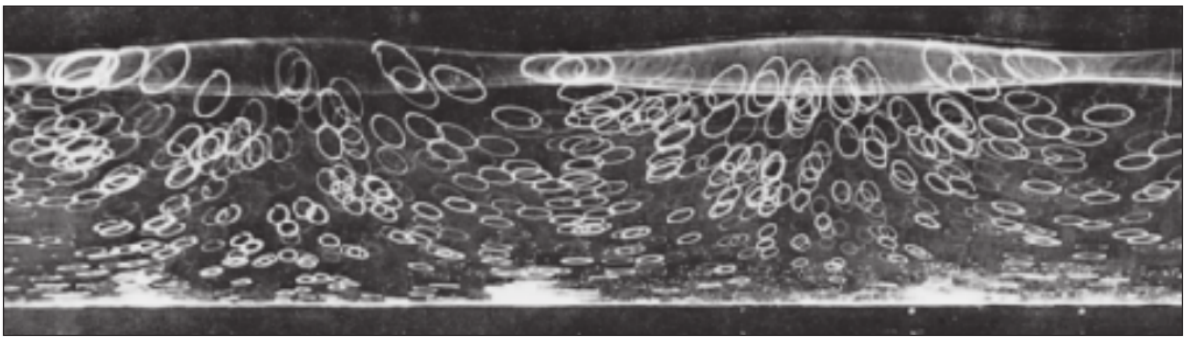


Figure 2: Pathlines underneath a wave in a tank of water. Figure taken from Ref. [4].



Figure 3: Streaklines in the unsteady flow around a cylinder. Figure taken from Ref. [4].

- streamlines,
- particle trajectories, and
- streaklines

all coincide.

For unsteady flow, these three types of flow visualization differ significantly, and streaklines may be misleading if interpreted as instantaneous flow directions.

1.2 Mass conservation

Let's consider some volume V_0 of space. The mass of fluid in this volume is $\int \rho \, dV$, where ρ is the fluid density, and the integration is taken over the volume V_0. The mass of fluid flowing in unit time through an element $d\vec{S}$ of the surface bounding this volume is $\rho \, \vec{v} \cdot d\vec{S}$; the magnitude of the vector $d\vec{S}$ is equal to the area of the surface element, and its direction is along the normal. By convention, we take $d\vec{S}$ along the outward normal. Then $\rho \, \vec{v} \cdot d\vec{S}$ is positive if the fluid is flowing out of the volume, and negative if the flow is into the volume. The total mass of fluid flowing out of the volume V_0 in unit time is therefore

$$\oint_S \rho \, \vec{v} \cdot d\vec{S}, \tag{4}$$

where the integration is taken over the whole of the closed surface surrounding the volume in question.

Next, the decrease per unit time in the mass of fluid in the volume V_0 can be written

$$-\frac{\partial}{\partial t} \int \rho \, dV.$$

Equating the two expressions, we have

$$\frac{\partial}{\partial t} \int \rho \, dV \; = \; - \oint \rho \, \vec{v} \cdot d\vec{S}. \tag{5}$$

This is the global equation of mass conservation for an arbitrary fixed control volume.

The surface integral can be transformed by Green's formula to a volume integral:

$$\oint \rho \, \vec{v} \cdot d\vec{S} = \int \nabla \cdot (\rho \vec{v}) \, dV. \tag{6}$$

Thus

$$\int \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) \right] dV = 0. \tag{7}$$

Since this equation must hold for any volume, the integrand must vanish, i.e.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0. \tag{8}$$

This is the local equation of mass conservation, also called the *equation of continuity.* equation of continuity. Although derived from global mass conservation applied to fixed volumes, it is itself

a local relation completely without reference to volumes. Expanding the expression $\nabla \cdot (\rho\vec{v})$, we can also write

$$\frac{\partial \rho}{\partial t} + \rho\, \nabla \cdot\ \vec{v} + \vec{v} \cdot \nabla\, \rho = 0. \tag{9}$$

The vector

$$\vec{j} = \rho\, \vec{v} \tag{10}$$

is called the *current density of mass* (or mass flux density). Its direction is that of the motion of the fluid, while its magnitude equals the mass of fluid flowing in unit time through unit area perpendicular to the velocity.

1.3 Incompressible flow

In a great many cases of the flow of liquids (and also of gases), their density may be supposed invariable, i.e. constant throughout the volume of the fluid and throughout its motion. In other words, there is no noticeable compression or expansion of the fluid in such cases. We then speak of incompressible flow.

Most liquids are perceived as incompressible under ordinary circumstances. A fluid is, however, effectively incompressible when flow speeds are much smaller than the velocity of sound.

All materials must nevertheless in principle be compressible. For in truly incompressible matter the sound velocity would be infinite, and that violates the relativistic injunction against any signal moving faster than the speed of light. Incompressibility is always an approximation and should, strictly speaking, be viewed as a condition on the flow rather than an absolute material property.

1.3.1 Global and local forms

Using $\rho(\vec{r}, t) = \rho_0 = constant$ in Eq. (5) we arrive at the global incompressibility condition

$$\oint_S \vec{v} \cdot d\vec{S} = 0, \tag{11}$$

Incompressible matter cannot accumulate anywhere, and equal volumes of incompressible fluid must enter and leave through any fixed surface per unit of time. Since the above condition refers to a single instant of time, it does in fact not matter whether the surface is fixed or moves in any way you may desire. On the other hand, using Eq. 8 we obtain the local incompressibility condition

$$\nabla \cdot \vec{v} = 0. \tag{12}$$

The divergence of the velocity field vanishes in incompressible flow. Notice that this local incompressibility condition does not refer to any volume of matter, only to the velocity field itself. A divergence-free field is sometimes called solenoidal.

1.3.2 Leonardo's Law.

Leonardo da Vinci observed that in an incompressible flow, the product of the cross-sectional area of a canal and the average velocity of the water remains constant. If A_1 and A_2 are two cross-sections of a canal and v_1, v_2 the corresponding average velocities, incompressibility implies

$$A_1 v_1 = A_2 v_2. \tag{13}$$

This follows from the fact that no fluid can pass through the canal walls. Applying the incompressibility condition to the closed surface composed of the two cross-sections and the canal walls gives

$$\oint \vec{v} \cdot d\vec{S} = \int_{A_2} \vec{v} \cdot d\vec{S} - \int_{A_1} \vec{v} \cdot d\vec{S} = 0.$$

The average velocity through any cross-section A is defined as

$$v = \frac{1}{A} \int_A \vec{v} \cdot d\vec{S}.$$

Hence, the quantity Av is constant along the canal for incompressible fluids, regardless of whether the flow is laminar or turbulent.

Leonardo's law, however, does *not* apply to compressible flows. In such cases the product of area and average velocity cannot remain constant, and a stronger law—accounting for density variations—is required.

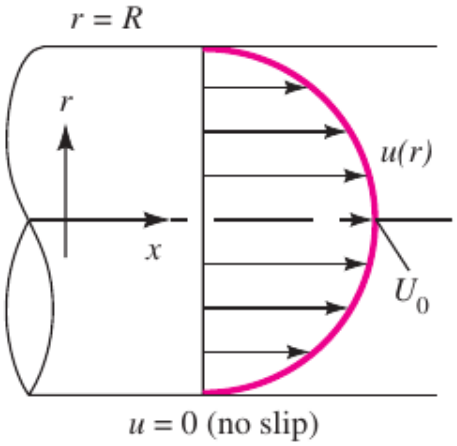


Figure 4: Steady viscous flow through a circular tube. Figure taken from Ref. [5].

Example

For steady viscous flow through a circular tube (Fig. E3.4), the axial velocity profile is given approximately by

$$u = U_0 \left(1 - \frac{r}{R}\right)^m \tag{14}$$

so that u varies from zero at the wall ($r = R$), or no slip, up to a maximum $u = U_0$ at the centerline $r = 0$. For highly viscous (laminar) flow $m \approx \frac{1}{2}$, while for less viscous (turbulent) flow $m \approx \frac{1}{7}$. Compute the average velocity if the density is constant.

Here $\vec{v} = \hat{i}\, u$ and $\vec{n} = \hat{i}$, and thus $\vec{v} \cdot \vec{n} = u$. Since the flow is symmetric, the differential area can be taken as a circular strip

$$dA = 2\pi r\, dr. \tag{15}$$

It follows that

$$\bar{v} = \frac{1}{A}\int u\, dA \tag{16}$$

$$= \frac{1}{\pi R^2}\int_0^R U_0\left(1-\frac{r}{R}\right)^m 2\pi r\, dr, \tag{17}$$

or

$$\bar{v} = U_0\frac{2}{(1+m)(2+m)}. \tag{18}$$

For the laminar flow approximation, $m \approx \frac{1}{2}$ and

$$\bar{v} \approx 0.53\, U_0. \tag{19}$$

(The exact laminar theory gives $\bar{v} = 0.50\, U_0$.) For turbulent flow, $m \approx \frac{1}{7}$ and

$$\bar{v} \approx 0.82\, U_0. \tag{20}$$

(There is no exact turbulent theory, and so we accept this approximation.) The turbulent velocity profile is more uniform across the section, and thus the average velocity is only slightly less than the maximum.

Example

Determine the rate at which the water level rises in an open container if the water coming in through a 0.10-m^2 pipe has a velocity of 0.5 m/s and the flow rate going out is 0.2 m^3/s (Fig. 5). The container has a circular cross section with a diameter of 0.5 m.

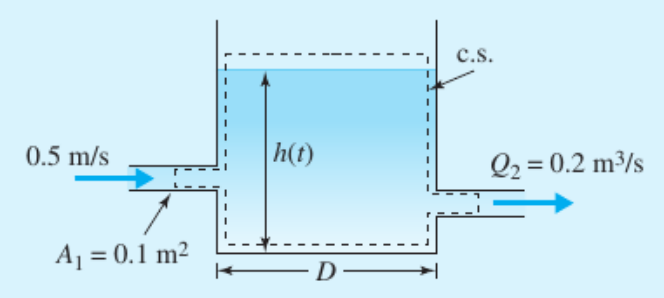


Figure 5: Steady viscous flow through a circular tube. Figure taken from Ref. [5].

First we select a control volume that extends above the water surface as shown in Fig. 5. Apply the continuity equation:

$$\frac{d}{dt}\int_{\text{c.v.}} \rho\, V + \rho(-v_1)A_1 + \rho v_2 A_2 = 0 \tag{21}$$

in which the first term describes the rate of change of mass in the control volume. Hence, neglecting the air mass above the water, we have

$$\frac{d(\rho h \pi D^2/4)}{dt} - \rho v_1 A_1 + \rho Q_2 = 0. \tag{22}$$

Divide by the constant ρ,

$$\frac{\pi D^2}{4}\frac{dh}{dt} - v_1 A_1 + Q_2 = 0. \tag{23}$$

The rate at which the water level is rising is then

$$\frac{dh}{dt} = \frac{v_1 A_1 - Q_2}{\pi D^2/4}. \tag{24}$$

Thus

$$\frac{dh}{dt} = \frac{0.5 \times 0.1 - 0.2}{\pi \times 0.5^2/4} = -0.764 \text{ m/s}. \tag{25}$$

The negative sign indicates that the water level is actually decreasing.

1.4 Lagrangian and Eulerian derivatives

Fluid motion may be described in two complementary ways.

1.4.1 Lagrangian description

In the *Lagrangian* viewpoint we follow a specific fluid element of mass Δm. Its motion is governed by Newton's law, and the time derivative d/dt represents the change experienced by that material element. This is the *material* or *Lagrangian* derivative. The velocity and acceleration of a fluid particle are written as

$$\vec{v} \equiv \frac{D\vec{r}}{Dt} = \frac{d\vec{r}}{dt}, \qquad \vec{a} \equiv \frac{D\vec{v}}{Dt} = \frac{d\vec{v}}{dt}. \tag{26}$$

1.4.2 Eulerian description

In the *Eulerian* viewpoint the fluid is treated as a field $f(\vec{r}, t)$ defined at fixed spatial positions. Changes in time may arise from variations at a fixed point, measured by $\partial f/\partial t$, or from the transport of fluid properties by the motion, represented by spatial derivatives.

According to differential calculus, a function $f(\vec{r}, t)$ has, in Cartesian coordinates, a differential of the form:

$$df = \frac{\partial f}{\partial t}\,dt + \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy + \frac{\partial f}{\partial z}\,dz = \frac{\partial f}{\partial t}\,dt + d\vec{r}\cdot\nabla f\,. \tag{27}$$

The temporal derivative of any fluid quantity is

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \vec{v}\cdot\nabla f. \tag{28}$$

Thus the relation between Eulerian and Lagrangian derivatives is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \vec{v}\cdot\nabla. \tag{29}$$

Applied to the position vector,

$$\frac{D\vec{r}}{Dt} = \vec{v}, \tag{30}$$

showing that the material derivative of position yields the velocity. Applied to the velocity field,

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}, \tag{31}$$

where the first term is the *local acceleration* and the second term is the *convective acceleration*, arising from velocity gradients in space.

For an incompressible fluid ($\nabla \cdot \vec{v} = 0$), the continuity equation gives

$$\frac{D\rho}{Dt} = 0 \qquad \Rightarrow \qquad \frac{\partial \rho}{\partial t} = -\vec{v} \cdot \nabla \rho.$$

This shows that although density may vary in space, each fluid element preserves its density as it moves.

1.5 Cauchy's equation

Applying Newton's Second Law to any comoving material particle with mass $dM = \rho\, dV$, its equation of motion becomes

$$dM\, \vec{a} = d\vec{\mathcal{F}}, \tag{32}$$

where $d\vec{\mathcal{F}}$ is the total force acting on the particle. Dividing by dV we arrive at the seductively simple equation, $\rho\, \vec{a} = \vec{f}^{\,*}$, where $\vec{f}^{\,*} = d\vec{\mathcal{F}}/dV$ is the *effective force density*. It was defined earlier:

$$f_i^* = f_i + \sum_j \nabla_j \sigma_{ij}, \tag{33}$$

where f_i is the true body force density and σ_{ij} the stress tensor. Finally, inserting the acceleration field we have arrived at Cauchy's equation:

$$\rho\, \frac{D\vec{v}}{Dt} = \rho \left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{v} \right) = \vec{f}^{\,*} \tag{34}$$

1.6 Big Bang and Newtonian Cosmology

1.6.1 Explosion field and Hubble law

For a cloud of non-interacting particles, inertia implies vanishing comoving acceleration,

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = 0. \tag{35}$$

For an explosion at the origin, the velocity field is

$$\vec{v}(\vec{x}, t) = \frac{\vec{x}}{t}, \tag{36}$$

which has the same form as Hubble's law,

$$\vec{v} = H_0 \vec{x}. \tag{37}$$

1.6.2 Newtonian cosmology

Homogeneity and isotropy imply

$$\rho(\vec{x},t)=\rho(t), \qquad \vec{v}(\vec{x},t)=H(t)\vec{x}. \tag{38}$$

The gravitational field inside a homogeneous universe is

$$\vec{g}(\vec{x},t)=-\frac{4\pi G}{3}\rho(t)\vec{x}. \tag{39}$$

1.6.3 Cosmological equations

The continuity and Euler equations reduce to

$$\dot{\rho}=-3H\rho, \qquad \dot{H}+H^2=-\frac{4\pi G}{3}\rho. \tag{40}$$

1.6.4 Cosmic scale factor

Introducing the scale factor $a(t)$ via

$$\frac{\dot{a}}{a}=H, \tag{41}$$

mass conservation gives $\rho a^3=\text{const}$, and the evolution equation becomes

$$\ddot{a}=-\frac{GM}{a^2}. \tag{42}$$

1.6.5 Critical density

The conserved energy can be written as

$$E=\frac{4\pi}{3}Ga^2(\rho_c-\rho), \tag{43}$$

defining the critical density

$$\rho_c=\frac{3H^2}{8\pi G}. \tag{44}$$

Universes with $\rho<\rho_c$ are open, $\rho>\rho_c$ closed, and $\rho=\rho_c$ flat.

1.6.6 Cosmological constant

Including a vacuum density ρ_v modifies the equation of motion to

$$\ddot{a}=-\frac{4\pi G}{3}(\rho-2\rho_v)a, \tag{45}$$

leading to accelerated expansion when $\rho<2\rho_v$.

1.7 Problems

1. Determine the equation of a general streamline of the flow

$$u_\phi = a, \qquad u_R = b, \qquad u_z = 0 \tag{46}$$

in cylindrical polar coordinates, and sketch the flow.

Repeat for the flow

$$u_\phi = aR^2, \qquad u_R = bR^2, \qquad u_z = 0. \tag{47}$$

If the flows are steady, and the density at a given radius is independent of ϕ, find the radial dependence of the density in both cases.

2. Show that for a steady flow with

$$\vec{\nabla} \cdot \vec{u} = 0, \tag{48}$$

the density ρ is constant along the streamlines. Need ρ be constant throughout the medium?

3. Define streamlines and particle paths and write down the differential equations for each of these.

A flow is described by

$$\vec{u} = (u_x,\, u_y e^{-t/\tau},\, 0) \tag{49}$$

where u_x, u_y, and τ are constants. Determine the streamlines and particle paths for such a flow and show that they are not the same at all times.

4. If

$$\vec{R} = (x, y, 0) \tag{50}$$

and

$$\hat{\vec{R}} = \frac{\vec{R}}{|\vec{R}|}, \tag{51}$$

and a flow velocity is given for $R \geq a$ by

$$\vec{u} = U\hat{\vec{x}}\left(1 + \frac{a^2}{R^2}\right) - 2Ua^2 x R^{-3}\hat{\vec{R}} \tag{52}$$

(where $\hat{\vec{x}} = \frac{x}{|x|}$), show that the streamlines obey

$$U\left(R - \frac{a^2}{R}\right)\sin\phi = \text{constant} \tag{53}$$

(where $\phi = \tan^{-1}(y/x)$). Sketch the streamlines and explain what the flow represents physically.

5. A fluid of density $\rho(x,t)$ flows in one dimension with a velocity

$$u(\rho) > 0 \tag{54}$$

which depends only on the density. Show that

$$\frac{\partial \rho}{\partial t} + c(\rho)\frac{\partial \rho}{\partial x} = 0, \tag{55}$$

where

$$c(\rho) = \frac{d(\rho u)}{d\rho}. \tag{56}$$

Show that this implies that $\rho(x,t)$ is constant along a line in the (x,t) plane with slope

$$\frac{dx}{dt} = c(\rho). \tag{57}$$

At $t = 0$,

$$\rho(x,0) = \begin{cases} \rho_0, & x < -a, \\ \rho_0 + \rho_1\left(1 - \dfrac{x^2}{a^2}\right), & -a < x < a, \\ \rho_0, & x > a. \end{cases} \tag{58}$$

Sketch the approximate behaviour of density $\rho(x,t)$ at subsequent times for the cases

(a) $c =$ constant,

(b) $\dfrac{dc}{d\rho} > 0$.

6. Determine the streamline equation in spherical coordinates.

7. Show that for a steady flow, the streamline containing a point P coincides with the pathline for a particle that passes through the point P at some time t.

8. Given the two-dimensional velocity field

$$v_1 = \frac{kx_1x_2}{1 + kx_2t}, \qquad v_2 = 0. \tag{59}$$

(a) Find the streamline at time t, which passes through the spatial point (α_1, α_2).

(b) Find the pathline equation $\vec{x} = \vec{x}(t)$ for a particle that is at (X_1, X_2) at time t_0.

9. Given the two-dimensional flow

$$v_1 = kx_2, \qquad v_2 = 0. \tag{60}$$

(a) Obtain the streamline passing through the point (α_1, α_2).

(b) Obtain the pathline for the particle that is at (X_1, X_2) at $t = 0$, including the time history of the particle along the pathline.

10. Do previous problem for the following velocity field:

$$v_1 = \omega x_2, \qquad v_2 = -\omega x_1. \tag{61}$$

11. Given the following velocity field in polar coordinates (r, θ):

$$v_r = \frac{Q}{2\pi r}, \qquad v_\theta = 0. \tag{62}$$

(a) Obtain the streamline passing through the point (r_0, θ_0).

(b) Obtain the pathline for the particle that is at (R, Θ) at $t = 0$, including the time history of the particle along the pathline.

12. Do previous problem for the following velocity field in polar coordinates (r, θ):

$$v_r = 0, \qquad v_\theta = \frac{C}{r}. \tag{63}$$

13. A fire is started and the smoke from the chimney goes straight up; no wind is present. After a few minutes a wind arises but the smoke continues to rise slowly. Sketch the streakline of the smoke, the pathline of the first particles leaving the chimney, and a few streamlines, assuming that the wind blows parallel to the ground in a constant direction.

14. A researcher has a large number of small flotation devices each equipped with a battery and light bulb. Explain how she would determine the pathlines and streaklines near the surface of a stream with some unknown currents that vary with time.

15. The hot-air balloon of Fig. P3.13 travels with the wind. The wind velocity vector is

$$\vec{V} = 6\hat{\vec{i}} + 10\hat{\vec{j}} \text{ ft/s} \tag{64}$$

for the first hour and then it is

$$\vec{V} = 10\hat{\vec{i}} + 5\hat{\vec{j}} \text{ ft/s} \tag{65}$$

for 2 h. On xy-coordinates, sketch the pathline of the balloon and streamlines at $t = 2$ h. If a number of hot-air balloons started from the same location, sketch the streakline formed by the balloons at $t = 3$ h. The balloons start at the origin.

16. A velocity field is given by

$$\vec{V} = (2t + 2)\hat{\vec{i}} + 2t\hat{\vec{j}} \text{ m/s}. \tag{66}$$

Sketch the pathlines of two particles up to $t = 5$ s, one that originates from the origin at $t = 0$, and another that originates from the origin at $t = 2$ s. Also, sketch the streamlines at $t = 5$ s.

17. The traffic situation on Mackinac Island, Michigan, where no automobiles are allowed (bikes are permitted), is to be studied. Comment on how such a study could be performed using a Lagrangian approach and an Eulerian approach.

18. Determine the speed of a fluid particle at the origin and at the point $(1, -2, 0)$ for each of the following velocity fields when $t = 2$ s. All distances are in feet and t is in seconds.

    (a)

$$\vec{V} = (x+2)\hat{\vec{i}} + xt\hat{\vec{j}} - \frac{z}{2}\hat{\vec{k}} \quad \text{ft/s.} \tag{67}$$

    (b)

$$\vec{V} = xy\hat{\vec{i}} - 2y^2\hat{\vec{j}} + tyz\hat{\vec{k}} \quad \text{ft/s.} \tag{68}$$

    (c)

$$\vec{V} = x^2t\hat{\vec{i}} - (xz + 2t)\hat{\vec{j}} + xyt\hat{\vec{k}} \quad \text{ft/s.} \tag{69}$$

19. Calculate the angle that the velocity vector makes with the x-axis and a unit vector normal to the streamline at $(1, -2)$ for the following velocity fields when $t = 2$ s. All distances are in meters and t is in seconds.

    (a)

$$\vec{V} = (x+2)\hat{\vec{i}} + xt\hat{\vec{j}} \quad \text{m/s.} \tag{70}$$

    (b)

$$\vec{V} = xy\hat{\vec{i}} - 2y^2\hat{\vec{j}} \quad \text{m/s.} \tag{71}$$

    (c)

$$\vec{V} = (x^2+4)\hat{\vec{i}} - y^2t\hat{\vec{j}} \quad \text{m/s.} \tag{72}$$

2 Incompressible frictionless (Euler) fluids

2.1 Euler equation for incompressible ideal flow

In an ideal homogeneous fluid, the only contact force is pressure, but in distinction to hydrostatics where the pressure is in balance with the body forces (like gravity), the effective density of force

$$\vec{f}^{*} = \vec{f_e} - \vec{\nabla} p \tag{73}$$

may now be non-vanishing. Inserted into the general dynamic equation we obtain, after dividing by the constant density $\rho(x,t) = \rho_0$,

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\vec{\nabla})\vec{v} = \frac{\vec{f_e}}{\rho_0} - \frac{\vec{\nabla} p}{\rho_0}, \tag{74}$$

$$\vec{\nabla}\cdot\vec{v} = 0. \tag{75}$$

These two *Euler equations* govern the dynamics of incompressible ideal fluid, also called *Euler fluid*.

From the identity

$$\vec{\nabla}(\vec{A}\cdot\vec{B}) = (\vec{A}\cdot\vec{\nabla})\vec{B} + (\vec{B}\cdot\vec{\nabla})\vec{A} + \vec{A}\times(\vec{\nabla}\times\vec{B}) + \vec{B}\times(\vec{\nabla}\times\vec{A}), \tag{76}$$

with $\vec{A} = \vec{B} = \vec{v}$ it follows that

$$\frac{1}{2}\,\vec{\nabla} v^2 = \vec{v}\cdot\vec{\nabla}\vec{v} + \vec{v}\times(\vec{\nabla}\times\vec{v}), \tag{77}$$

so that, replacing $\vec{v}\cdot\vec{\nabla}\vec{v}$ in Euler's equation, we obtain

$$\rho_0\,\frac{\partial \vec{v}}{\partial t} + \frac{\rho_0}{2}\,\vec{\nabla} v^2 - \rho_0\,\vec{v}\times(\vec{\nabla}\times\vec{v}) + \vec{\nabla} p = \vec{f}_{\text{ext}}. \tag{78}$$

Unlike the hydrostatic case, it is no longer a necessary condition that external forces be derivable from a potential, i.e. that the forces be conservative. External forces may now include, in addition to electrical forces (including time-dependent ones), gravitational and inertial forces (centrifugal, Coriolis and linear acceleration), as well as magnetic forces.

In what follows we restrict ourselves to conservative forces, and more specifically to gravitational forces, centrifugal forces, and constant acceleration. Later, Coriolis forces will be introduced, although no conservative scalar potential exists for them. For the external forces we may then write

$$\vec{f} = -\rho\,\vec{\nabla}\Phi, \quad \text{with} \quad \Phi = \Phi_{\text{grav}} - \frac{\omega^2 r^2}{2} + \vec{a}\cdot\vec{r}.$$

Thus, the equation of motion for non-viscous fluids in the presence of these forces is

$$\rho_0\,\frac{\partial \vec{v}}{\partial t} + \frac{\rho_0}{2}\,\vec{\nabla} v^2 - \rho_0\,\vec{v}\times(\vec{\nabla}\times\vec{v}) + \vec{\nabla} p + \rho_0\,\vec{\nabla}\Phi = 0. \tag{79}$$

2.2 Adiabatic and isentropic flow

The absence of heat exchange between different parts of the fluid (and also, of course, between the fluid and bodies adjoining it) means that the motion is adiabatic throughout the fluid. Thus the motion of an ideal fluid must necessarily be supposed adiabatic.

In adiabatic motion the entropy of any particle of fluid remains constant as that particle moves about in space. Denoting by s the entropy per unit mass, we can express the condition for adiabatic motion as

$$\frac{Ds}{Dt} = 0 = \frac{\partial s}{\partial t} + \vec{v} \cdot \vec{\nabla} s. \tag{80}$$

This is the general equation describing adiabatic motion of an ideal fluid. Using the continuity equation, we may write it as an "equation of continuity" for entropy:

$$\frac{\partial(\rho s)}{\partial t} + \vec{\nabla} \cdot (\rho s \vec{v}) = 0. \tag{81}$$

The product $\rho s \vec{v}$ is the *entropy flux density.*

The adiabatic equation usually takes a much simpler form. If, as usually happens, the entropy is constant throughout the volume of the fluid at some initial instant, it retains everywhere the same constant value at all times and for any subsequent motion of the fluid. In this case we can write the adiabatic equation simply as

$$S = \rho s = \text{constant}. \tag{82}$$

Such a motion is said to be *isentropic.*

We may use the fact that the motion is isentropic to put the equation of motion in a somewhat different form. To do so, we employ the familiar thermodynamic relation

$$d\mathcal{H} = T\, ds + V\, dp, \tag{83}$$

where $\mathcal{H}$ is the heat function per unit mass (enthalpy), $V = 1/\rho$ is the specific volume, and T is the temperature. Since $s = \text{constant}$, we have simply

$$d\mathcal{H} = V\, dp = \frac{dp}{\rho}. \tag{84}$$

Thus

$$\frac{\vec{\nabla} p}{\rho} = \vec{\nabla} \mathcal{H},$$

and Euler's equation can therefore be written in the form

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{v} = \frac{\vec{f_e}}{\rho_0} - \vec{\nabla} \mathcal{H}. \tag{85}$$

Or

$$\rho_0 \frac{\partial \vec{v}}{\partial t} + \frac{\rho_0}{2} \vec{\nabla} v^2 - \rho_0\, \vec{v} \times (\vec{\nabla} \times \vec{v}) + \rho_0 \vec{\nabla} \mathcal{H} + \rho_0\, \vec{\nabla} \Phi = 0, \tag{86}$$

$$\frac{\partial \vec{v}}{\partial t} + \vec{\nabla} \left(\tfrac{1}{2} v^2 + \mathcal{H} + \Phi \right) - \vec{v} \times (\vec{\nabla} \times \vec{v}) = 0. \tag{87}$$

From the isentropic condition $\vec{\nabla}\mathcal{H} = \vec{\nabla}p/\rho$ it follows, by taking the curl, that

$$\vec{\nabla}p \times \vec{\nabla}\rho = 0,$$

so that $p = p(\rho)$. That is, *isentropic flow is associated with a barotropic fluid*. Also, from

$$\rho\,\vec{\nabla}\mathcal{H} = \vec{\nabla}p,$$

taking the curl we obtain

$$\vec{\nabla}\rho \times \vec{\nabla}\mathcal{H} = 0,$$

from which $\mathcal{H} = \mathcal{H}(\rho)$.

The description of the state of a moving fluid is given by $\vec{v}$, p, and ρ, which altogether amount to five quantities. Five equations are therefore required, namely:

- Equation of motion (three equations when expressed in components),
- Continuity equation (one equation),
- Although it appears that we miss an explicit equation of motion for the pressure, it is in fact determined indirectly by the divergence condition. To see this we calculate the divergence of the first Euler equation, and obtain,

$$\nabla^2 p = \rho_0\,\vec{\nabla}\cdot\vec{g} - \rho_0\,\vec{\nabla}\cdot\left[(\vec{v}\cdot\vec{\nabla})\vec{v}\right]. \tag{88}$$

  This is nothing but the Poisson equation, so well known from electrostatics and Newtonian gravity. Together with suitable boundary conditions, it can always be solved and determines the pressure field at a given time from the velocity field (and gravity) at the same time.

2.2.1 Boundary conditions

Partial differential equations need boundary conditions to connect the fields on both sides of a material interface. As in hydrostatics, Newton's Third Law demands that the pressure—in the absence of surface tension—must be continuous across the interface. Furthermore, since fluid cannot accumulate in a material interface, the velocity field must be continuous along the normal to the interface. There is no restriction on the density, which usually jumps at a material interface. The complete set of *jump conditions* valid at a material interface may accordingly be written (with a bracket $[\cdot]$ to indicate the difference between the two sides) as

$$[\vec{v}\cdot\vec{n}] = 0, \qquad [p] = 0. \tag{13.3}$$

where $\hat{\vec{n}}$ is the local normal to the interface. In truly ideal flow, the tangential velocity component is unrestricted. Ideal fluids are thus able to slip along container surfaces with finite tangential velocity, and there may even be internal surfaces of discontinuity with different tangential flow velocities on the two sides.

2.2.2 Irrotational flow

Now, taking the curl of the equation of motion, the pressure gradient term disappears and we obtain an equation involving only the velocity. It describes the time evolution of $\vec{\nabla} \times \vec{v}$, which we shall call the *vorticity*:

$$\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{v}) - \vec{\nabla} \times \left[\vec{v} \times (\vec{\nabla} \times \vec{v})\right] = 0. \tag{89}$$

A solution—the simplest one—of this equation is

$$\vec{\nabla} \times \vec{v} = 0,$$

which is known as *irrotational flow*.

2.3 Bernoulli's equation

From the equation of motion, specialized to the steady state ($\partial \vec{v}/\partial t = 0$), taking the scalar product with $d\vec{r} = \vec{v}\, dt$ and noting that

$$d\vec{r} \cdot \left[\vec{v} \times (\vec{\nabla} \times \vec{v})\right] = \vec{v} \cdot \left[\vec{v} \times (\vec{\nabla} \times \vec{v})\right] dt = 0,$$

and since $\vec{v} \cdot (\vec{v} \times \vec{\xi}) = 0$, we conclude that

$$d\vec{r} \cdot \vec{\nabla} \left(\frac{1}{2}v^2 + \mathcal{H} + \Phi\right) = \frac{d}{dl}\left(\frac{1}{2}v^2 + \mathcal{H} + \Phi\right) = 0, \tag{90}$$

where d/dl denotes differentiation along the direction of the velocity, that is, along a streamline. It then follows that along a streamline the following condition holds, known as Bernoulli's equation:

$$H \equiv \frac{1}{2}v^2 + \mathcal{H} + \Phi = \text{constant}. \tag{91}$$

This field—which we shall call the Bernoulli field H—may be viewed as an extension of the effective potential to fluid in motion. The constant has the same value at all points on a given streamline, but may differ from one streamline to another. In the incompressible case, we have $\mathcal{H} = p/\rho$.

The tangent to a streamline at any point is in the direction of the velocity at that point. The equation of a streamline is therefore

$$\vec{v} \times d\vec{r} = 0.$$

In steady state these streamlines do not change with time and coincide with the trajectories of the fluid particles.

2.3.1 Flow tube

The path of an individual particle in a moving fluid is called a flow line. The flow lines passing through the edge of an imaginary element of area, such as area A in Fig. 6, form a tube called a flow tube. From the definition of a flow line, in steady flow no fluid can cross the side walls of a given flow tube.

If a fluid of constant density flows in a horizontal tube, the term $\Phi = gz$ remains constant, so that Bernoulli's equation reduces to

$$\frac{1}{2}\rho v^2 + p = \text{constant}. \tag{92}$$

Consequently, in a horizontal tube, the greater the velocity, the lower the pressure, and vice versa.

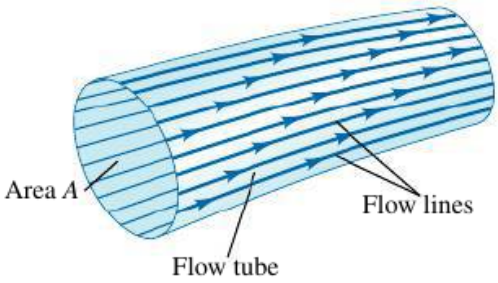


Figure 6: A flow tube bounded by flow lines. In steady flow, fluid cannot cross the walls of a flow tube. Figure taken from Ref. [6].

2.3.2 The Venturi effect

A simple duct with gently varying cross-section carries a constant volume flow rate Q of incompressible fluid (see Fig. 7). For simplicity we assume the duct is horizontal, such that gravity can be disregarded. For a streamline running through the duct, Bernoulli's theorem tells us that

$$H = \frac{1}{2}v^2 + \frac{p}{\rho_0} \tag{93}$$

takes the same value everywhere along the streamline. Approximating the velocity by its average value $U = Q/A$ over the cross-section A, the pressure becomes

$$p = \rho_0 \left(H - \frac{Q^2}{2A^2} \right). \tag{94}$$

This demonstrates the Venturi effect: the pressure rises when the duct cross-section increases, and conversely. The Venturi effect is used in many technological devices, from carburetors in car engines to perfume atomizers, as well as pumps and flow meters.

2.3.3 Torricelli's law

A barrel of wine has a little spout close to the bottom. If the plug in the spout is suddenly removed, the hydrostatic pressure accelerates the wine to stream out with considerable speed. Provided the spout is narrow compared to the size of the barrel, a nearly steady flow will soon establish itself.

Consider a streamline starting near the top of the barrel and running down through the middle of the spout. Near the top at height $z = h$, the fluid is almost at rest, $v \approx 0$. The pressure is atmospheric, $p = p_0$, and the gravitational potential may be taken to be gz. Thus,

$$H_{\text{top}} = gh + \frac{p_0}{\rho_0}. \tag{95}$$

Just outside the spout, the fluid has horizontal velocity v and the pressure is also atmospheric. Hence,

$$H_{\text{bottom}} = \frac{1}{2}v^2 + \frac{p_0}{\rho_0}. \tag{96}$$

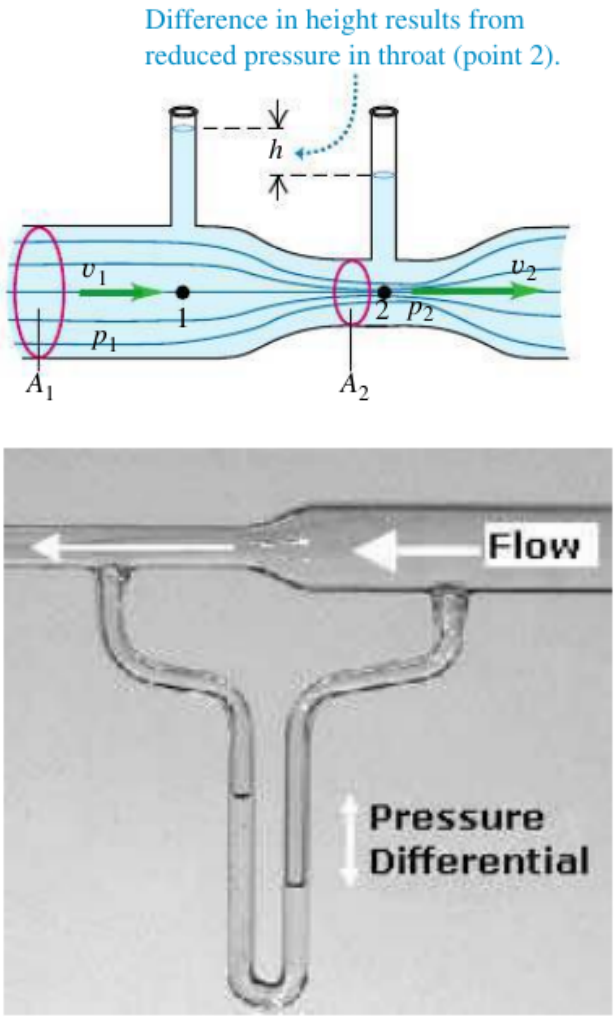


Figure 7: Top: The Venturi meter is used to measure flow speed in a pipe. Bottom: Venturi effect demonstration. Air streams from the right into a narrower tube where it speeds up. The shift in water level in the shunt tube underneath can be used to determine the pressure drop and thus the flow speed.

Equating the two expressions gives

$$\frac{1}{2}v^2 = gh, \tag{97}$$

with the solution

$$v = \sqrt{2gh}. \tag{98}$$

This result is called *Torricelli's law*. Surprisingly, the outflow speed equals the speed acquired by a freely falling particle over the same height.

2.3.4 Quasi-stationary emptying

If the barrel has constant cross-section A_0 (see Fig. 8), the average vertical flow velocity is $v_0 = vA/A_0$, where A is the cross-section of the spout and $v = \sqrt{2gz}$ the outflow velocity when the

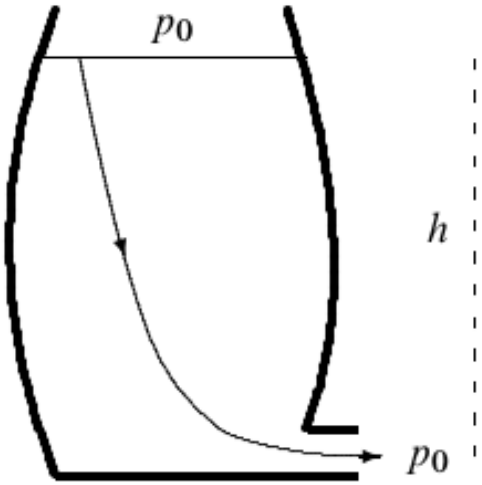


Figure 8: Wine running out of a barrel. The wine emerges with the same speed as it would have obtained by falling freely through the height h of the fluid in the barrel.

water level is z. Since $dz/dt = -v_0$, we obtain

$$\frac{dz}{dt} = -\frac{A}{A_0}\sqrt{2gz}. \tag{99}$$

Integration gives the emptying time

$$t_0 = \int_0^h \frac{dz}{\sqrt{2gz}}\frac{A_0}{A} = \frac{A_0}{A}\sqrt{\frac{2h}{g}}. \tag{100}$$

2.3.5 Case: The Pitot tube

The Pitot tube measures flow speed by converting kinetic pressure into static pressure. In steady state, water rises to height h so that

$$\frac{p_0}{\rho_0} = \frac{1}{2}U^2 + \frac{p}{\rho_0}, \tag{101}$$

which gives

$$\Delta p = \frac{1}{2}\rho_0 U^2 = \rho_0 g h. \tag{102}$$

Hence,

$$U = \sqrt{2gh}. \tag{103}$$

At a speed of $1\,\mathrm{m\ s^{-1}}$ the water rises 5 cm.

2.3.6 Rotating fluid in steady state

This problem has often been solved in a reference frame rotating with the liquid. Here we solve it in the inertial frame, by applying directly the equation of motion in steady state, with constant density ρ and in the presence of a uniform gravitational field.

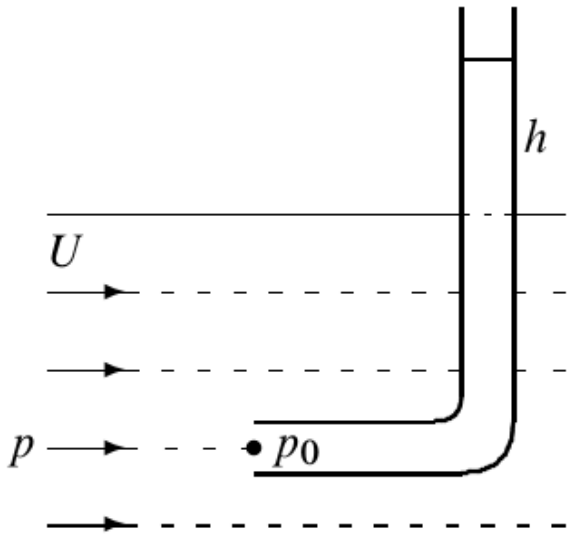


Figure 9: The principle of the Pitot tube. The pressure increase along the stagnating streamline must equal the weight of the raised water column.

In steady state it is true that

$$\vec{v} = \vec{\omega} \times \vec{r},$$

and therefore

$$\vec{\nabla} \times \vec{v} = 2\,\vec{\omega}, \qquad \vec{v} \times (\vec{\nabla} \times \vec{v}) = 2(\vec{\omega} \times \vec{r}) \times \vec{\omega}.$$

A straightforward calculation then shows that the equation of motion reduces to

$$-\omega^2(x\,\hat{e}_x + y\,\hat{e}_y) + \frac{\vec{\nabla} p}{\rho} + g\,\hat{e}_z = 0. \tag{104}$$

From this, the three component equations follow:

$$\frac{\partial p}{\partial x} - \rho\omega^2 x = 0, \tag{105}$$

$$\frac{\partial p}{\partial y} - \rho\omega^2 y = 0, \tag{106}$$

$$\frac{\partial p}{\partial z} + \rho g = 0. \tag{107}$$

Consequently,

$$dp = \frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz = \rho\omega^2(x\,dx + y\,dy) - \rho g\,dz, \tag{108}$$

from which

$$p = p_0 + \frac{\rho\omega^2}{2}(x^2 + y^2) + \rho g(z_0 - z), \tag{109}$$

in agreement with the result obtained in Hydrostatic chapter.

2.3.7 Exercises

1. The inviscid, incompressible flow field exterior to the cylinder shown in Fig.10 is given by

$$v_r = U_\infty \left(1 - \frac{r_c^2}{r^2}\right) \cos\theta, \qquad v_\theta = U_\infty \left(1 + \frac{r_c^2}{r^2}\right) \sin\theta. \tag{110}$$

   If the pressure at $r = \infty$ is zero, i.e., $p_\infty = 0$, find an expression for the pressure neglecting gravity effects:

   (a) Along the negative x-axis
   (b) At the stagnation point
   (c) On the surface of the cylinder
   (d) On the surface of the cylinder at $\theta = 90°$

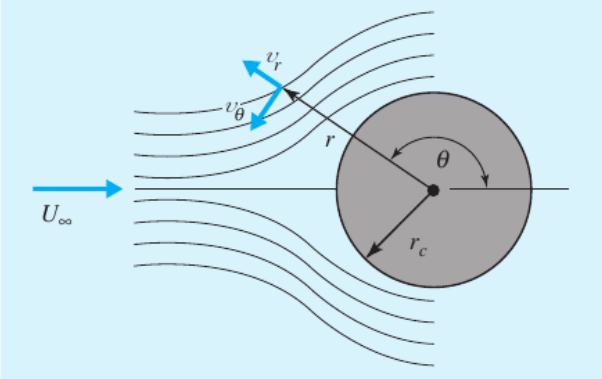


Figure 10: The inviscid, incompressible flow field exterior to the cylinder. Figure taken from Ref. [4].

2. The velocity along the negative x-axis in the inviscid, incompressible flow field exterior to the body shown in Fig. 11 is given by

$$u(x) = U_\infty + \frac{q}{2\pi x}. \tag{111}$$

   If the pressure at $x = -\infty$ is zero, find an expression for the pressure neglecting gravity effects:

   (a) Along the negative x-axis if $U_\infty = 10$ m/s and $q = 20\pi$ m^2/s
   (b) At the stagnation point if $U_\infty = 10$ m/s and $q = 20\pi$ m^2/s
   (c) Along the negative x-axis if $U_\infty = 30$ ft/s and $q = 60\pi$ ft^2/s
   (d) At the stagnation point if $U_\infty = 30$ ft/s and $q = 60\pi$ ft^2/s

3. A short contraction is followed by an expansion, as shown in Fig. 12. The manometer is used to determine the velocity of the fluid providing the viscous effects are negligible. If water is flowing steadily, determine the velocity V_1 if $H = 12$ cm. Losses through the contraction are negligible.

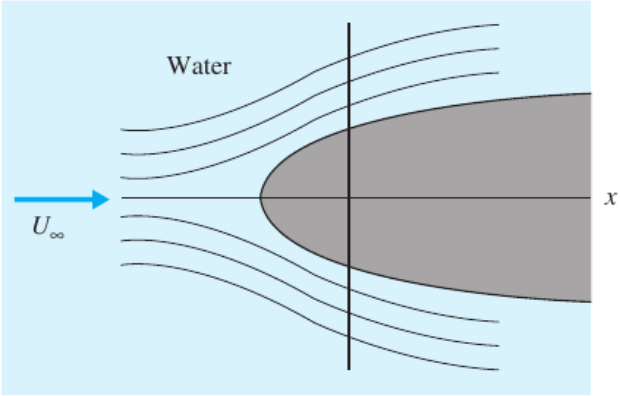


Figure 11: The inviscid, incompressible flow field exterior to the a certain body. Figure taken from Ref. [4].

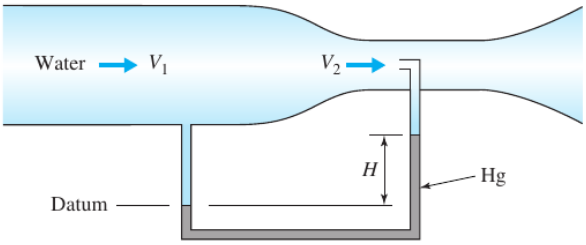


Figure 12: Figure taken from Ref. [4].

4. Air at 120 kPa absolute and 30°C flows vertically upward in a pipe, as shown in Fig. 13. If the water manometer deflection $H = 5$ cm, determine the velocity in the smaller pipe. Assume the air to be incompressible.

5. A manometer, positioned inside a cylinder as shown in Fig. 14, reads 4 cm of water. Estimate U_∞ assuming an inviscid flow. Refer to the velocity field in exercise 1.

6. For the flow shown in Fig. 15, estimate the pressure p_1 and velocity V_1 if $V_2 = 20$ m/s and:

   (a) $H = 1$ cm

   (b) $H = 5$ cm

   (c) $H = 10$ cm

7. In the pipe contraction shown in Fig. 16, water flows steadily with a velocity of $V_1 = 0.5$ m/s and $V_2 = 1.125$ m/s. Two piezometer tubes are attached to the pipe at sections 1 and 2. Determine the height H. Neglect any losses through the contraction.

8. Air at 20°C is drawn into the hose of a vacuum cleaner through a head that is relatively free of obstructions (the flow can be assumed to be inviscid). Estimate the velocity in the hose if the vacuum in the hose measures:

   (a) 2 cm of water

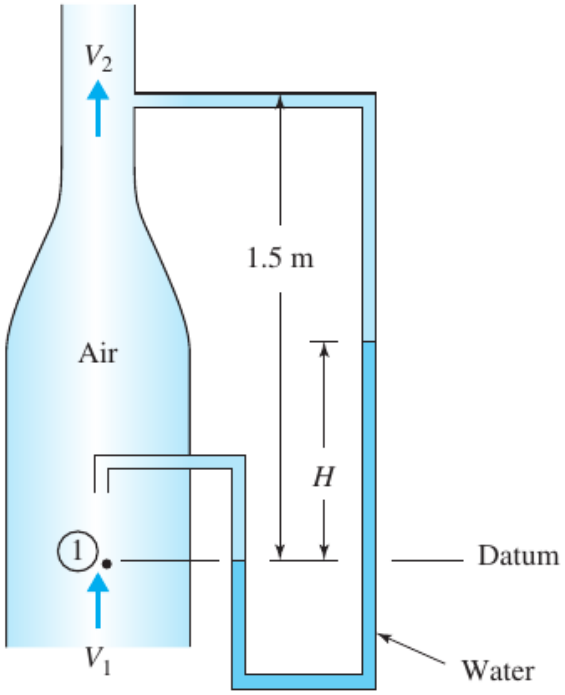


Figure 13: Figure taken from Ref. [4].

(b) 8 cm of water

(c) 1 in. of water

(d) 4 in. of water

9. A wind tunnel is designed to draw in air from the atmosphere and produce a velocity of 100 m/s in the test section. The fan is located downstream of the test section. What pressure is to be expected in the test section if the atmospheric temperature and pressure are:

   (a) $-20°$C, 90 kPa?

   (b) $0°$C, 95 kPa?

   (c) $20°$C, 92 kPa?

   (d) $40°$C, 100 kPa?

10. The pump shown in Fig. 17 creates a flow such that $V = 14$ m/s. Predict the pressure at the gage shown assuming an inviscid flow in the entrance and a uniform flow at the gage. Use a streamline starting at:

    (a) Point A

    (b) Point B

11. The velocity downstream of a sluice gate is assumed to be uniform (Fig. 18). Express V in terms of H and h for this inviscid flow. Use a streamline:

    (a) Along the top downstream

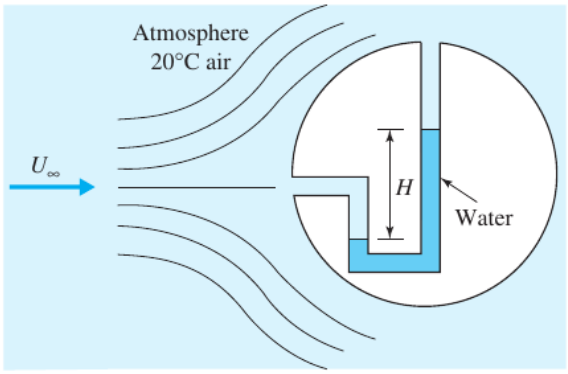


Figure 14: Figure taken from Ref. [4].

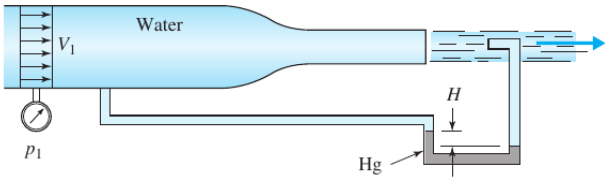


Figure 15: Figure taken from Ref. [4].

(b) Along the bottom downstream

2.4 Vorticity

The value of the Bernoulli field $H(x)$ at a point x depends only on the streamline passing through that point. Different streamlines will in general have different values of H. It is, however, often possible to relate the values of H for bundles of streamlines.

2.4.1 The vorticity field

Using the Euler equation for steady incompressible non-viscous flow, the gradient of the Bernoulli field including gravity becomes (in index notation)

$$\nabla_i H = \frac{1}{2}\nabla_i v^2 + \nabla_i \Phi + \frac{1}{\rho_0}\nabla_i p = v_j \nabla_i v_j - v_j \nabla_j v_i = \left(\vec{v} \times (\vec{\nabla} \times \vec{v})\right)_i. \tag{112}$$

Defining the *vorticity field* as the curl of the velocity field,

$$\vec{\omega} = \vec{\nabla} \times \vec{v}, \tag{113}$$

we obtain for steady flow

$$\vec{\nabla} H = \vec{v} \times \vec{\omega}. \tag{114}$$

The vorticity field is a quantitative measure of the local circulation in the fluid. In any region where the vorticity vanishes we have $\vec{\nabla} H = 0$, so the Bernoulli field must take the same value everywhere in that region, that is $H(x) = H_0$. A flow completely free of vorticity is said to be *irrotational*.

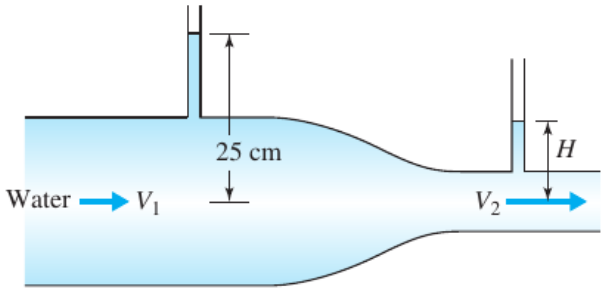


Figure 16: Figure taken from Ref. [4].

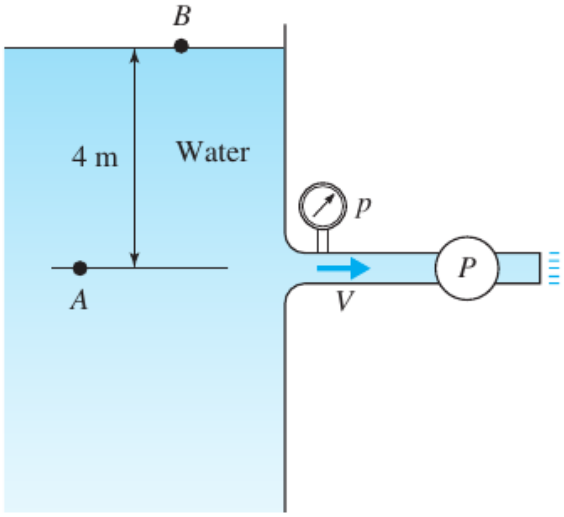


Figure 17: Figure taken from Ref. [4].

2.4.2 Vortex lines

The field lines of the vorticity field are called *vortex lines* (see Fig. 19), and are defined as curves that are everywhere tangent to the vorticity field. Like streamlines, they are solutions of the differential equation at a fixed time t_0,

$$\frac{d\vec{x}}{ds} = \vec{\omega}(\vec{x}, t_0), \tag{115}$$

where s is a running parameter along the curve. Unlike streamlines, the parameter is not time; here s has the dimensions of length$\times$time.

Like streamlines, vortex lines cannot cross each other, and since the vorticity field is a curl, it satisfies

$$\vec{\nabla} \cdot \vec{\omega} = 0.$$

Therefore, vortex lines can neither start nor end anywhere in the fluid; they must continue until they reach the boundary of the flow (possibly at infinity) or close upon themselves.

In steady flow, vortex lines are independent of the chosen time t_0. Bernoulli's theorem, $(\vec{v} \cdot \vec{\nabla})H = 0$, follows immediately by dotting $\vec{\nabla}H = \vec{v} \times \vec{\omega}$ with $\vec{v}$. Similarly, dotting with $\vec{\omega}$ gives

$$(\vec{\omega} \cdot \vec{\nabla})H = 0,$$

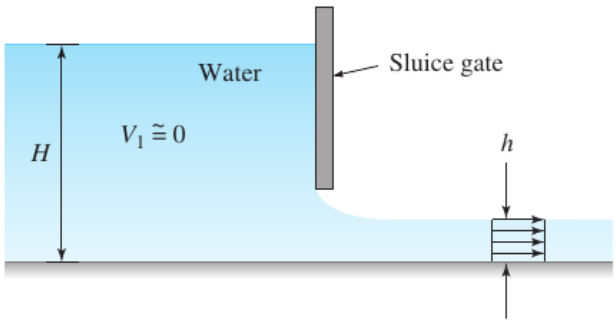


Figure 18: Figure taken from Ref. [4].

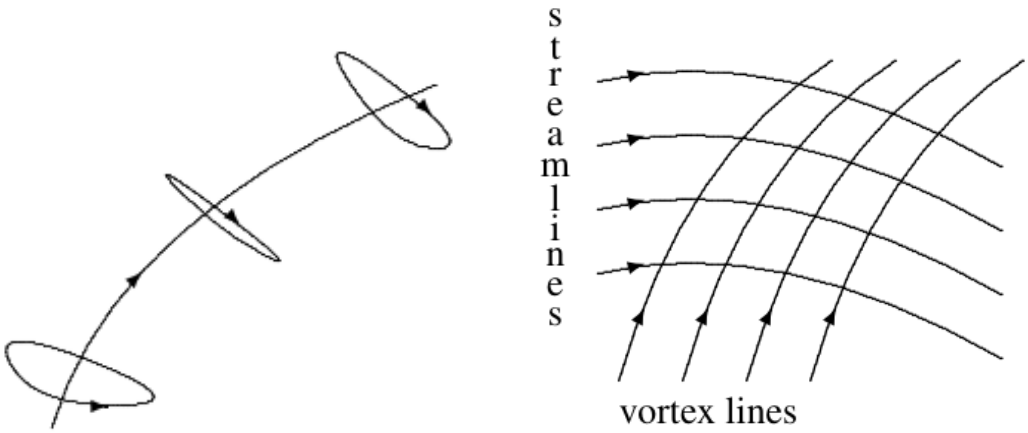


Figure 19: Left: Around a vortex line there is local circulation of fluid. Right: The Bernoulli field is constant on surfaces made from vortex lines and streamlines.

showing that the Bernoulli field is also constant along vortex lines. Together these results show that the Bernoulli field is constant on the two-dimensional surfaces formed by combining vortex lines and streamlines. These are sometimes called *Lamb surfaces* or *Bernoulli surfaces*.

2.4.3 Equation of motion for vorticity

Since the vorticity field is derived from the velocity field, its equation of motion must follow from Euler's equation for the velocity field. Retracing the steps leading to $\vec{\nabla} H = \vec{v} \times \vec{\omega}$ but now allowing for time dependence, we obtain

$$\frac{\partial \vec{v}}{\partial t} = \vec{v} \times \vec{\omega} - \vec{\nabla} H. \tag{116}$$

This reduces to the steady case when $\partial \vec{v}/\partial t = 0$.

Taking the curl of both sides and using the identity that the curl of a gradient vanishes, we arrive at the vorticity equation

$$\frac{\partial \vec{\omega}}{\partial t} = \vec{\nabla} \times (\vec{v} \times \vec{\omega}). \tag{117}$$

A major lesson follows: if the vorticity vanishes initially, $\vec{\omega}(\vec{x}, t) = 0$ throughout a region V at time t, then $\partial \vec{\omega}/\partial t = 0$ there. Consequently, the vorticity field will not change, and if it is zero initially, it will remain zero forever. *Vorticity cannot be generated by the motion of an ideal fluid; it must be present from the outset.*

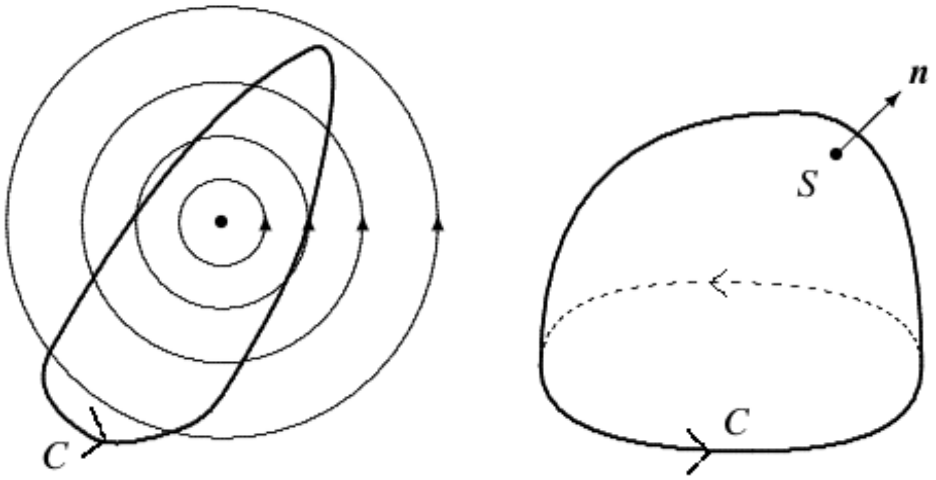


Figure 20: Left: A closed curve C encircling a whirl. Right: A surface S with perimeter C. The normal to the surface is consistent with the orientation of C (here using a right-hand rule).

2.5 Circulation

Vorticity is a local property of a fluid. The corresponding global concept is called *circulation*, defined as the line integral of the velocity field along a closed curve C (see Fig. 20),

$$\Gamma(C,t) = \oint_C \vec{v}(\vec{x},t)\cdot d\vec{\ell}. \tag{118}$$

If C encircles a whirling region, the circulation measures how strongly the fluid rotates around the loop. The sign depends on whether the circulation is taken with or against the whirling motion.

2.5.1 Conservation of circulation

The Lagrangian time derivative of the circulation reads

$$\frac{D\Gamma}{Dt} = \frac{D}{Dt}\oint_C \vec{v}\cdot d\vec{\ell}. \tag{119}$$

When performing the differentiation, one must take into account the change both in $\vec{v}$ and in the shape of the curve, that is,

$$\frac{D\Gamma}{Dt} = \oint_C \frac{D\vec{v}}{Dt}\cdot d\vec{\ell} + \oint_C \vec{v}\cdot\frac{D}{Dt}(d\vec{\ell}). \tag{120}$$

Taking into account equation $d\vec{v} = d\vec{\ell}/dt$, this can be written as

$$\frac{D\Gamma}{Dt} = \oint_C \frac{D\vec{v}}{Dt}\cdot d\vec{\ell} + \oint_C \vec{v}\cdot d\vec{v} = \oint_C \frac{D\vec{v}}{Dt}\cdot d\vec{\ell} + \left.\frac{v^2}{2}\right|_a = \oint_C \frac{D\vec{v}}{Dt}\cdot d\vec{\ell}, \tag{121}$$

that is,

$$\frac{D\Gamma}{Dt} = \oint_C \frac{D\vec{v}}{Dt}\cdot d\vec{\ell}. \tag{122}$$

This expression is known as Kelvin's equation.

Replacing here the equation of motion, whose Lagrangian form is

$$\rho \frac{D\vec{v}}{Dt} + \nabla p + \rho \nabla \Phi = 0, \tag{123}$$

one obtains

$$\frac{D\Gamma}{Dt} = -\oint_C \left(\frac{\nabla p}{\rho} + \nabla \Phi \right) \cdot d\vec{\ell}. \tag{124}$$

In the isentropic case, $\nabla \mathcal{H} = \nabla p/\rho$, so that

$$\frac{D\Gamma}{Dt} = -\oint_C \nabla(\mathcal{H} + \Phi) \cdot d\vec{\ell} = -\oint_C d(\mathcal{H} + \Phi), \tag{125}$$

which implies that the circulation of the isentropic velocity field of an ideal and barotropic fluid is a constant of motion:

$$\Gamma = \text{constant}. \tag{126}$$

The quantity Γ, also known as the vortex intensity, cannot be created or destroyed in an ideal isentropic fluid. In other words, vorticity is a convective property that is conserved in the flow of the fluid.

This is the Hankel–Kelvin circulation theorem (formulated between 1861 and 1869), also known as the law of conservation of velocity circulation or the law of conservation of vortex intensity.

Kelvin's circulation theorem: In an ideal (inviscid), barotropic, and isentropic fluid, the circulation around any closed material curve moving with the fluid is conserved in time.

Physical interpretation In an ideal, isentropic, and barotropic fluid, the forces acting on a fluid element per unit mass reduce to gradients of scalar potentials. In particular, the pressure force can be written as the gradient of the specific enthalpy, and body forces such as gravity derive from a conservative potential.

As a consequence, these forces are incapable of generating or destroying circulation. Since the line integral of a gradient over a closed material curve vanishes, the circulation around any loop that moves with the fluid remains unchanged in time.

From a physical standpoint, this means that vorticity is a convective property: it is transported and deformed by the flow, but it cannot be created in the absence of viscosity, entropy gradients, or non-conservative forces. Any change in circulation must therefore originate from dissipative effects (viscosity), non-barotropic effects ($p \neq p(\rho)$), or external sources beyond the assumptions of ideal and isentropic flows.

It should be emphasized that this result has been obtained using Euler's equation in the form given previously, and therefore relies on the assumption that the flow is isentropic. The theorem does not hold for flows that are not isentropic.

2.5.2 Stokes' theorem and steady flow

The circulation of a vector field around a closed curve (see Fig. 20) equals the flux of its curl through any surface S bounded by the curve:

$$\oint_C \vec{v} \cdot d\vec{\ell} = \int_S (\vec{\nabla} \times \vec{v}) \cdot d\vec{S}. \tag{127}$$

It does not matter which surface S is chosen, provided it has C as boundary and lies entirely within the fluid.

By applying Kelvin's theorem to an infinitesimal closed contour δC and transforming the line integral according to Stokes' theorem, we obtain

$$\oint_C \vec{v} \cdot d\vec{\ell} = \int (\nabla \times \vec{v}) \cdot d\vec{S} \simeq \delta\vec{S} \cdot (\nabla \times \vec{v}) = \text{constant}, \tag{128}$$

where $d\vec{S}$ is a fluid surface element spanning the contour δC. The vector $\nabla \times \vec{v}$ is the vorticity of the fluid flow at a given point. The constancy of the product above can be intuitively interpreted as meaning that the vorticity is transported with the fluid.

2.6 Potential flow

In the absence of viscosity, a flow that is irrotational in a region will stay irrotational. High Reynolds numbers confine vorticity to thin boundary layers, so that the main body of the flow is nearly irrotational.

From $\vec{\nabla} \times \vec{v} = 0$ it follows that the velocity field is the gradient of a scalar potential,

$$\vec{v} = \vec{\nabla}\Psi. \tag{129}$$

The scalar Ψ is called the *velocity potential*.

2.6.1 Incompressible potential flow

In incompressible flow, $\vec{\nabla} \cdot \vec{v} = 0$ implies that Ψ satisfies Laplace's equation:

$$\nabla^2\Psi = 0. \tag{130}$$

Inserting $\vec{\omega} = 0$ and $\vec{v} = \vec{\nabla}\Psi$ into Euler's equation gives

$$\vec{\nabla}\left(H + \frac{\partial\Psi}{\partial t}\right) = 0,$$

so the expression in parentheses depends only on time. Using the Bernoulli function and solving for the pressure, we obtain

$$p = C(t) - \rho_0\left(\frac{1}{2}v^2 + \Phi + \frac{\partial\Psi}{\partial t}\right), \tag{131}$$

where $C(t)$ is an arbitrary function of time.

Potential flow in incompressible fluids is much simpler than flow with vorticity because Laplace's equation is well understood. Any linear superposition of solutions of Laplace's equation is again a solution, and practical problems reduce to finding the solution that satisfies the boundary conditions.

2.7 Application: Cylinder in uniform crosswind

A circular cylinder with axis along the z-axis and radius a is placed in an asymptotically uniform crosswind U along the x-axis. This does not break the translational invariance along the z-axis so that the velocity potential $\Psi(x, y)$ is independent of z. Evidently, the natural choice is cylindrical coordinates (r, ϕ, z).

2.7.1 Velocity potential

Asymptotically, for $r \to \infty$, the velocity potential must approach the field of a constant uniform crosswind, $\Psi \to Ux = Ur\cos\phi$.

The most general solution to the Laplace's equation in cylindrical coordinates is

$$\Psi(r,\varphi) = \sum_{n=0}^{\infty} \left(A_n + \frac{B_n}{r^n} \right) \left(C_n \cos(n\varphi) + D_n \sin(n\varphi) \right), \tag{132}$$

where the integer values of n guarantee the continuity of the potential. Substituting into this equation the far-field potential

$$\Psi = Ux = Ur\cos\varphi, \qquad r \to \infty, \tag{133}$$

one obtains

$$D_n = 0, \qquad A_n C_n = U\delta_{n1}, \tag{134}$$

and writing

$$B_n D_n = E_n, \tag{135}$$

we arrive to

$$\Psi = Ur\cos\varphi + \sum_{n=0}^{\infty} \frac{E_n}{r^n} \cos(n\varphi). \tag{136}$$

It is also necessary to impose the condition that the normal component of the velocity vanishes on the surface of the cylinder,

$$\vec{v} \cdot \hat{\vec{e}}_r \big|_{r=a} = 0, \tag{137}$$

that is,

$$\left. \frac{\partial \Psi}{\partial r} \right|_{r=a} = 0. \tag{138}$$

This condition leads to

$$U\cos\varphi - \sum_{n=1}^{\infty} E_n \frac{n}{a^{n+1}} \cos(n\varphi) = 0, \tag{139}$$

from which

$$E_n = Ua^2\delta_{n1}. \tag{140}$$

Consequently,

$$\Psi(r,\varphi) = U\left(r + \frac{a^2}{r} \right) \cos\varphi. \tag{141}$$

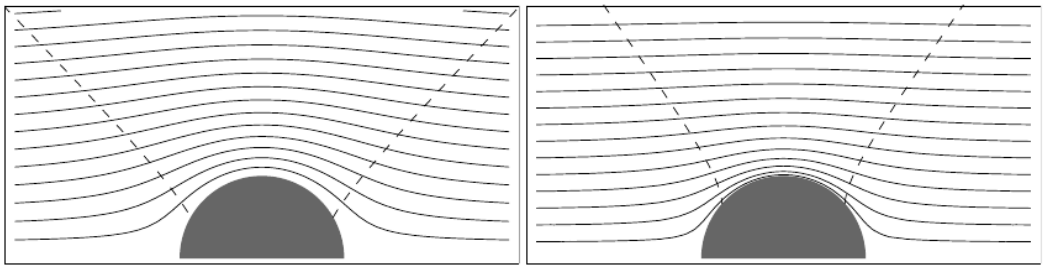


Figure 21: Comparison of potential flows around a cylinder (left) and sphere (right). Only the upper half is shown, and the horizontal bottom may in both cases be replaced by a solid surface. The pressure vanishes at infinity and on the dashed lines (and is negative between). Figure taken from Ref. [3].

2.7.2 Velocity and pressure fields

The velocity field is obtained from derivatives of the potential. The non-vanishing components are

$$v_r = \frac{\partial \Psi}{\partial r} = U \cos\phi \left(1 - \frac{a^2}{r^2}\right), \tag{142}$$

$$v_\phi = \frac{1}{r}\frac{\partial \Psi}{\partial \phi} = -U \sin\phi \left(1 + \frac{a^2}{r^2}\right). \tag{143}$$

The radial velocity vanishes at the surface as expected, while the tangential component at $r = a$,

$$v_\phi\big|_{r=a} = -2U \sin\phi,$$

vanishes only at the stagnation points $\phi = 0, \pi$. See figure 21.

The pressure follows from Bernoulli's theorem. In the absence of gravity and normalized to vanish at infinity, it is

$$p = \frac{1}{2}\rho_0 \left(U^2 - v^2\right) = \frac{1}{2}\rho_0 U^2 \frac{a^2}{r^2}\left(4\cos^2\phi - 2 - \frac{a^2}{r^2}\right). \tag{144}$$

On the surface $r = a$ this reduces to

$$p_a = \frac{1}{2}\rho_0 U^2 (4\cos^2\phi - 3), \tag{145}$$

which is negative for $30° < \phi < 150°$.

The symmetry $p(\phi) = p(-\phi)$ implies that the total force in the y-direction (lift) vanishes, as expected. More surprisingly, the invariance $p(\pi - \phi) = p(\phi)$ implies that the total force in the x-direction (drag) also vanishes. This is known as *d'Alembert's paradox*.

2.7.3 Lift on a half-cylinder

The same solution also describes ideal flow over a cylinder that is half buried in a flat surface, since the normal velocity vanishes in the symmetry plane $y = 0$.

The vertical lift on a stretch of the half-cylinder of length L is

$$\mathcal{L}_y = -\int_0^\pi p_a \, dS_y = -\int_0^\pi p_a \sin\phi \, La \, d\phi = \frac{5}{3}\rho_0 U^2 La. \tag{146}$$

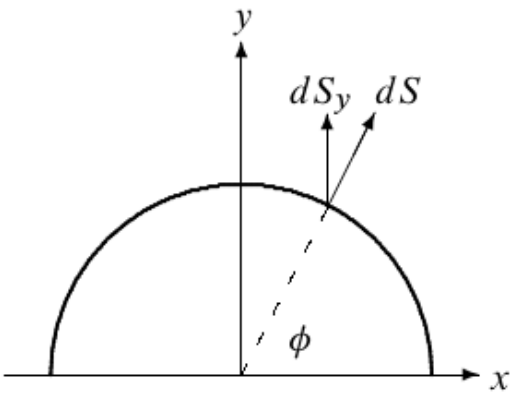


Figure 22: The projection of the surface element $dS = La\,d\phi$ on the y-axis is $dS_y = \sin\phi\,dS$. Figure taken from Ref. [3].

If the average density of the cylinder is $\rho_1 > \rho_0$, the lift-to-weight ratio is

$$\frac{\mathcal{L}_y}{Mg_0} = \frac{5}{3\pi}\,\frac{\rho_0}{\rho_1 - \rho_0}\,\frac{U^2}{ag_0}, \tag{147}$$

including buoyancy. There is therefore a critical wind speed beyond which the lift overcomes the weight and the half-buried cylinder can be lifted out of the ground.

2.7.4 The Magnus effect

An additional irrotational velocity field around a cylinder can be introduced through the potential

$$\Psi = C\phi, \tag{148}$$

which generates a purely azimuthal velocity field,

$$v_r = 0, \qquad v_\phi = \frac{C}{r}. \tag{149}$$

This flow corresponds to circulation around the cylinder and may be produced by a rotating cylinder of radius a with angular velocity Ω, for which

$$C = \Omega a^2. \tag{150}$$

When this circulating flow is superimposed on a uniform flow around the cylinder, the pressure distribution becomes asymmetric. The resulting pressure difference is

$$\Delta p = \rho_0 U \sin\phi \left(1 + \frac{a^2}{r^2}\right)\frac{C}{r}. \tag{151}$$

This asymmetry generates a net lift force known as the *Magnus effect*. The lift per unit length is

$$\mathcal{L}_y = -\rho_0 U L \Gamma, \tag{152}$$

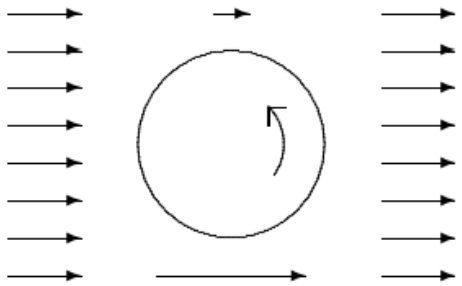


Figure 23: Sketch of the Magnus effect. A rotating cylinder drags air along near its surface (due to even the tiniest viscosity). This increases the wind velocity below the cylinder and decreases it above. By Bernoulli's theorem the pressure will be lower below and higher above, and consequently gives rise to a negative (i.e., downwards) lift. A similar argument explains, for example, why a Ping-Pong ball with top- spin dives toward the table. Figure taken from Ref. [3].

where

$$\Gamma = 2\pi C \tag{153}$$

is the circulation.

The Magnus effect explains the lateral force acting on spinning objects moving through a fluid, such as balls in sports (see Fig. 23).

2.8 Problems

1. A water tank of fixed height H discharges liquid upward through an outlet at the bottom (Fig. 24). What height does the column reach?

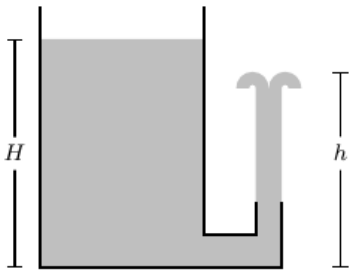


Figure 24: Problem 3.8.1. Figure taken from Ref. [1].

2. Consider the steady flow of an incompressible ideal gas. Using Bernoulli's equation, show that
$$\frac{1}{2}v^2 + \frac{kT}{\mu}\ln\rho + \Phi = \text{constant}. \tag{154}$$

3. An incompressible, non-viscous fluid flows along the interior of a cone with opening angle α, as shown in Fig. 25. Neglecting gravity and considering only pressure differences, study the relation between the pressure and the flow velocity in the steady state.

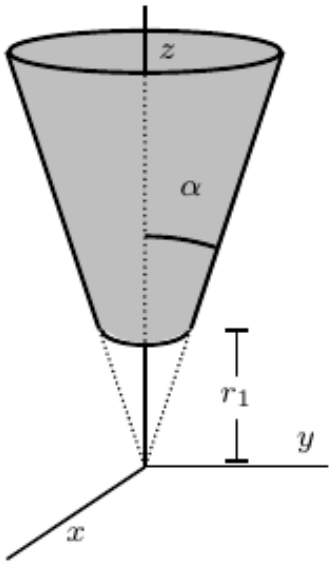


Figure 25: Problem 3.8.3. Figure taken from Ref. [1].

4. The equation of motion of an isentropic fluid, in a rotating reference frame with constant angular velocity $\boldsymbol{\omega}$, can be written in the form

$$\rho \frac{d\vec{v}}{dt} + \nabla \left[H + \mathcal{G} + \frac{\omega^2 r^2}{2} \right] + 2\rho\, \vec{v} \times \boldsymbol{\omega} = 0. \tag{155}$$

Show that the circulation is conserved, that is,

$$\frac{d\Gamma}{dt} = 0. \tag{156}$$

3 Compresible flow

3.1 Sound Waves in an fluids

Sound waves provide the fundamental mechanism by which information propagates in fluids and determine characteristic dynamical timescales, such as the sound-crossing time.

We consider a fluid described by the Euler equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0, \tag{157}$$

$$\frac{d\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} = -\frac{1}{\rho} \nabla p. \tag{158}$$

We assume an equilibrium state with:

$$\rho = \rho_0, \quad p = p_0, \quad \vec{v} = 0,$$

and introduce perturbations:

$$\rho = \rho_0 + \Delta\rho, \tag{159}$$

$$p = p_0 + \Delta p, \tag{160}$$

$$\vec{v} = \Delta \vec{v}. \tag{161}$$

Keeping only first-order terms in perturbations, we obtain:

$$\frac{\partial \Delta\rho}{\partial t} + \rho_0 \nabla \cdot (\Delta \vec{v}) = 0, \tag{162}$$

$$\frac{\partial \Delta \vec{v}}{\partial t} + \frac{1}{\rho_0} \nabla \Delta p = 0. \tag{163}$$

Assuming a barotropic fluid:

$$\Delta p = \left(\frac{dp}{d\rho} \right)_0 \Delta\rho = \frac{K_0}{\rho_0} \Delta\rho,$$

where K_0 is the equilibrium bulk modulus. Combining the linearized equations, we obtain:

$$\frac{\partial^2 \Delta\rho}{\partial t^2} = c_0^2 \nabla^2 \Delta\rho. \tag{164}$$

where we define the sound speed as

$$c_0^2 = \left(\frac{dp}{d\rho} \right)_0 = \frac{K_0}{\rho_0}. \tag{165}$$

Similarly

$$\frac{\partial^2 \Delta p}{\partial t^2} = c_0^2 \nabla^2 \Delta p. \tag{166}$$

Fluid	T (°C)	c_0 ($\mathrm{m\,s^{-1}}$)
Glycerol	25	1920
Sea water	20	1521
Fresh water	20	1482
Lube oil	25	1461
Mercury	25	1449
Ethanol	25	1145
Hydrogen	27	1310
Helium	0	973
Water vapor	100	478
Neon	30	461
Humid air	20	345
Dry air	20	343
Oxygen	30	332
Argon	0	308
Nitrogen	27	363

Table 1: Empirical sound speeds in various liquids (top) and gases (bottom). The temperature of the measurement is also listed.

3.1.1 Adiabatic (isentropic) perturbations

Sound vibrations are normally so rapid that temperature equilibrium is never established. This allows us to assume that the oscillations take place adiabatically, that is, without heat exchange. The bulk modulus of an isentropic ideal gas is

$$K_0 = \gamma p_0, \tag{167}$$

where γ is the adiabatic index. The corresponding isentropic sound velocity is

$$c_0 = \sqrt{\gamma \frac{p_0}{\rho_0}} = \sqrt{\gamma R T_0}. \tag{14.7}$$

In the last step we used the ideal gas law

$$p_0 = R \rho_0 T_0, \tag{168}$$

where

$$R = \frac{R_{\mathrm{mol}}}{M_{\mathrm{mol}}} \tag{169}$$

is the specific gas constant. The isothermal sound velocity is recovered for

$$\gamma = 1. \tag{170}$$

3.1.2 Sound speed in the atmosphere

Adiabatic For dry air at 20°C with

$$\gamma = \frac{7}{5}, \qquad M_{\mathrm{mol}} = 29\ \mathrm{g\,mol^{-1}}, \tag{171}$$

the sound speed becomes

$$c_0 = 343 \text{ m s}^{-1} = 1235 \text{ km h}^{-1}. \tag{172}$$

Since the temperature of the homentropic atmosphere falls linearly with height in the troposphere, the speed of sound varies with height z above the ground as

$$c = c_0\sqrt{1 - \frac{z}{h_2}}, \tag{14.8}$$

where c_0 is the sound speed at sea level and

$$h_2 \approx 30 \text{ km} \tag{173}$$

is the homentropic scale height.

At the flying altitude of modern jet aircraft,

$$z \approx 10 \text{ km}, \tag{174}$$

the sound speed has dropped to

$$c \approx 280 \text{ m s}^{-1} \approx 1000 \text{ km h}^{-1}, \tag{175}$$

disregarding the effect of humidity, which makes it higher.

Beyond the troposphere this expression begins to fail because the homentropic model of the atmosphere fails.

3.1.3 Plane wave solution

An elementary plane density wave propagating along the x-axis with wavelength λ, period τ, and amplitude $\rho_1 > 0$ is described by the density perturbation

$$\Delta\rho = \rho_1 \sin(kx - \omega t), \tag{14.9}$$

where

$$k = \frac{2\pi}{\lambda} \tag{176}$$

is the wavenumber and

$$\omega = \frac{2\pi}{\tau} \tag{177}$$

is the angular frequency.

Substituting this expression into the wave equation gives

$$\omega^2 = c_0^2 k^2, \tag{178}$$

or equivalently

$$c_0 = \frac{\omega}{k} = \frac{\lambda}{\tau}. \tag{179}$$

Surfaces of constant density (or pressure) are planes perpendicular to the direction of propagation, corresponding to

$$kx - \omega t = \text{constant}. \tag{180}$$

Differentiating this condition with respect to time shows that these planes move with velocity

$$\frac{dx}{dt} = \frac{\omega}{k} = c_0, \tag{181}$$

which is the phase velocity of the wave. Therefore, c_0 corresponds to the speed of small-amplitude sound waves in the medium.

From the x-component of the Euler equation, the only non-vanishing component of the velocity field is

$$v_x = v_1 \sin(kx - \omega t), \qquad v_1 = c_0 \frac{\rho_1}{\rho_0}. \tag{14.10}$$

Since

$$v_y = v_z = 0, \tag{182}$$

sound waves in an isotropic fluid are longitudinal, meaning that the fluid motion is parallel to the direction of propagation.

The corresponding displacement field u_x, defined through

$$\frac{\partial u_x}{\partial t} = v_x, \tag{183}$$

becomes

$$u_x = a_1 \cos(kx - \omega t), \qquad a_1 = \frac{v_1}{\omega}, \tag{14.11}$$

where a_1 is the displacement amplitude of the sound wave.

Validity of the approximation. To justify neglecting the advective acceleration, compare the magnitudes of the advective and local acceleration terms:

$$\frac{|(\vec{v} \cdot \vec{\nabla})\vec{v}|}{|\partial \vec{v}/\partial t|} \approx \frac{k v_1^2}{\omega v_1} = \frac{v_1}{c_0}. \tag{14.12}$$

Thus, the approximation is valid when the velocity amplitude satisfies

$$v_1 \ll c_0. \tag{184}$$

Equivalently,

$$\rho_1 \ll \rho_0, \qquad p_1 \ll K_0, \qquad a_1 \ll \frac{\lambda}{2\pi}. \tag{185}$$

3.1.4 The Jeans instability

In the derivation where the propagation of sound waves was examined we neglected gravity, except as a constant external force when we dealt with the propagation of sound waves in an homentropic atmosphere. Now we consider the possibility of self-gravity playing a role, so perform the same analysis as for the sound wave in a uniform medium (in the absence of external gravity), but now involve the extra variable $\Delta\Phi$(the perturbed gravitational potential) and so an extra equation linking $\Delta\Phi$ to the density perturbation $\Delta\rho$ (Poisson's equation). We therefore consider a sound wave passing through a uniform medium (constant p, ρ) with wavelength long enough that we can no longer ignore gravity.

—A perturbation

$$\Delta\rho(\vec{x}, t) \tag{186}$$

on top of an otherwise uniform density ρ_0 generates both a pressure perturbation

$$\Delta p = c_0^2 \Delta\rho \tag{187}$$

and a gravitational field $\vec{g}$. The divergence of the gravitational field satisfies

$$\vec{\nabla} \cdot \vec{g} = -4\pi G \Delta\rho, \tag{14.13}$$

where G is the gravitational constant.

The usual wave equation is then modified to

$$\frac{\partial^2 \Delta\rho}{\partial t^2} = c_0^2 \nabla^2 \Delta\rho + 4\pi G \rho_0 \Delta\rho. \tag{14.14}$$

Assuming a plane-wave solution leads to the dispersion relation

$$\omega^2 = c_0^2 k^2 - 4\pi G \rho_0. \tag{14.15}$$

Ordinary oscillatory waves exist only when

$$\omega^2 > 0, \tag{188}$$

that is,

$$c_0 k > \sqrt{4\pi G \rho_0}. \tag{189}$$

For sufficiently small wavenumbers (large wavelengths), ω becomes imaginary, implying exponentially growing density perturbations. The critical wavelength separating stable and unstable modes is the *Jeans wavelength*,

$$\lambda_J = c_0 \sqrt{\frac{\pi}{G\rho_0}}. \tag{14.16}$$

Perturbations with

$$\lambda > \lambda_J \tag{190}$$

are gravitationally unstable. This phenomenon is known as the *Jeans instability*.

For an ideal gas with molar mass M_{mol} and temperature T_0, the isothermal sound speed is

$$c_0 = \sqrt{RT_0}, \tag{191}$$

where

$$R = \frac{R_{\text{mol}}}{M_{\text{mol}}}. \tag{192}$$

Using typical interstellar values

$$\rho_0 = 3.35 \times 10^{-18}\ \text{kg}\,\text{m}^{-3}, \qquad T_0 = 3\ \text{K}, \tag{193}$$

one finds

$$\lambda_J \approx 1.9 \times 10^{16}\ \text{m}, \tag{194}$$

corresponding to approximately two light years. The mass contained within a spherical region of diameter λ_J is approximately

$$M_J \approx 1.1 \times 10^{31}\ \text{kg}, \tag{195}$$

which is of the order of a few solar masses.

The spatial scales associated with k_J are enormous for reasonable values of density and sound speed. Consider the interstellar medium (ISM) in a galaxy, with

$$\rho \sim m_p\ \text{cm}^{-3}, \tag{196}$$

and a sound speed of approximately

$$c_s \sim 10^6\ \text{cm}\,\text{s}^{-1}, \tag{197}$$

reflecting the temperature of the ISM. The wavelength

$$\lambda_J = \frac{2\pi}{k_J} \tag{198}$$

(the Jeans length) is then of the order of kiloparsecs:

$$\lambda_J = 1.7 \left(\frac{\rho_0}{m_p\ \text{cm}^{-3}} \right)^{-1/2} \left(\frac{c_s}{10^6\ \text{cm}\,\text{s}^{-1}} \right) \text{kpc}. \tag{10.65}$$

This scale is comparable to the characteristic size of galaxies, supporting the idea that galaxies may form through Jeans instability from an approximately homogeneous background medium. Since the initial mass density of the universe was presumably smaller, the characteristic instability scale would have been even larger at earlier times.

The wavelength scale also indicates that a semi-homogeneous object with the size of a galaxy is not expected to become gravitationally unstable under new perturbations, since it is too small to contain perturbations with wavelengths larger than the Jeans scale.

3.2 Steady compressible flow

In steady compressible flow, the velocity, pressure, and density are all independent of time, and the Euler equations take the simpler form,

$$(\vec{v}\cdot\vec{\nabla})\vec{v} = \vec{g} - \frac{1}{\rho}\vec{\nabla}p, \qquad\qquad \vec{\nabla}\cdot(\rho\vec{v}) = 0. \tag{199}$$

Here we shall, for simplicity, assume that the fluid obeys a barotropic equation of state,

$$p = p(\rho), \qquad \text{or} \qquad \rho = \rho(p), \tag{200}$$

leaving us with a closed set of five field equations for the five fields

$$v_x,\ v_y,\ v_z,\ \rho,\ p. \tag{201}$$

Here gravity will mostly be ignored.

3.2.1 Effective incompressibility

We now demonstrate that, in steady flow, a fluid behaves effectively as incompressible when the flow speed is much smaller than the local speed of sound. The ratio between the local flow speed v and the local sound speed c is called the Mach number,

$$\mathrm{Ma} = \frac{|\vec{v}|}{c}. \tag{14.18}$$

A steady flow is therefore approximately incompressible when

$$\mathrm{Ma} \ll 1, \tag{202}$$

whereas compressibility effects become important when

$$\mathrm{Ma} \gtrsim 1. \tag{203}$$

For a barotropic fluid,

$$\vec{\nabla}p = \frac{dp}{d\rho}\vec{\nabla}\rho = c^2\vec{\nabla}\rho, \tag{204}$$

where

$$c = \sqrt{\frac{dp}{d\rho}} \tag{205}$$

is the local speed of sound.

Using the continuity equation

$$\vec{\nabla}\cdot(\rho\vec{v}) = \rho\vec{\nabla}\cdot\vec{v} + (\vec{v}\cdot\vec{\nabla})\rho = 0, \tag{206}$$

together with the Euler equation (neglecting gravity), one obtains

$$\vec{\nabla}\cdot\vec{v} = -\frac{1}{\rho}(\vec{v}\cdot\vec{\nabla})\rho = -\frac{1}{\rho c^2}(\vec{v}\cdot\vec{\nabla})p = \frac{\vec{v}\cdot(\vec{v}\cdot\vec{\nabla})\vec{v}}{c^2}. \tag{14.19}$$

Applying the Schwarz inequality gives

$$|\vec{\nabla}\cdot\vec{v}| \leq \frac{|\vec{v}|^2}{c^2}|\vec{\nabla}\vec{v}| = \mathrm{Ma}^2|\vec{\nabla}\vec{v}|, \tag{14.20}$$

where

$$|\vec{\nabla}\vec{v}| = \sqrt{\sum_{ij}(\nabla_i v_j)^2} \tag{207}$$

is the norm of the velocity-gradient tensor.

Thus, when

$$\mathrm{Ma}^2 \ll 1, \tag{208}$$

the divergence of the velocity field is much smaller than the velocity gradients, making the incompressibility condition

$$\vec{\nabla}\cdot\vec{v} = 0 \tag{209}$$

a good approximation.

Typically, flows may be regarded as incompressible when

$$\mathrm{Ma} \lesssim 0.3. \tag{210}$$

Example [Mach numbers]: Typical everyday motions generate very small Mach numbers. Moving your hands through air produces velocities of only a few meters per second, corresponding to

$$\mathrm{Ma} \approx 0.01. \tag{211}$$

Driving a car at

$$120\ \mathrm{km\,h^{-1}} \approx 33\ \mathrm{m\,s^{-1}} \tag{212}$$

corresponds to

$$\mathrm{Ma} \approx 0.12. \tag{213}$$

A passenger aircraft flying at an altitude of approximately

$$10\ \mathrm{km} \tag{214}$$

with speed

$$900\ \mathrm{km\,h^{-1}} \approx 250\ \mathrm{m\,s^{-1}} \tag{215}$$

has

$$\mathrm{Ma} \approx 0.9, \tag{216}$$

since the speed of sound decreases with altitude.

Even when the flow remains subsonic, compressibility effects become important near aircraft wings and fuselages. Supersonic aircraft typically operate around

$$\mathrm{Ma} \sim 2 - 3, \tag{217}$$

whereas the Space Shuttle reenters the atmosphere at hypersonic speeds near

$$\mathrm{Ma} \sim 25. \tag{218}$$

The strong compression of air in front of such vehicles produces shock waves and sonic booms.



Figure 26: A static airfoil in an air stream coming in horizontally from the left. The pictured streamline (dashed) ends at the forward stagnation point.

3.2.2 Bernoulli's theorem for barotropic fluids

For compressible fluids, Bernoulli's theorem remains valid in modified form. For a barotropic fluid with

$$\rho = \rho(p), \tag{219}$$

the Bernoulli function becomes

$$H = \frac{1}{2}v^2 + \Phi + w(p), \tag{14.21}$$

where

$$w(p) = \int \frac{dp}{\rho(p)} \tag{14.22}$$

is the pressure potential.

For an isentropic ideal gas with adiabatic index γ, the pressure potential is proportional to the temperature,

$$w = c_p T, \qquad c_p = \frac{\gamma}{\gamma - 1} R, \tag{14.23}$$

where c_p is the specific heat at constant pressure and

$$R = \frac{R_{\text{mol}}}{M_{\text{mol}}} \tag{220}$$

is the specific gas constant.

Consequently, in the absence of gravity, a decrease in velocity along a streamline in isentropic flow is accompanied by an increase in temperature, pressure, and density.

3.2.3 Stagnation temperature rise

An object moving through an ideal fluid has at least one stagnation point at the front where the fluid comes to rest relative to the object. There is also at least one stagnation point at the rear of the body, although vortex formation and turbulence generally disturb the flow strongly in that region, producing unsteady eddies and tangled streamlines. As a result, Bernoulli's theorem often cannot be reliably applied at the rear of the body.

At the forward stagnation point the gas is compressed and its temperature becomes higher than that of the surrounding fluid (see Fig. 26). In the reference frame where the object is at

rest and the fluid moves asymptotically with constant speed and temperature, the flow is steady. Using the modified Bernoulli equation for a barotropic fluid in the absence of gravity,

$$\frac{1}{2}v^2 + c_p T = c_p T_0, \tag{221}$$

where T_0 is the stagnation temperature and T is the temperature at a point along the streamline where the fluid velocity is v.

The increase in temperature due to adiabatic compression is therefore

$$\Delta T = T_0 - T = \frac{v^2}{2c_p}. \tag{222}$$

The stagnation temperature rise depends only on the relative velocity between the body and the fluid, and not directly on the temperature, pressure, or density of the gas. A lower molar mass implies a larger specific heat and therefore a smaller stagnation temperature increase.

3.3 Problems

1. (a) Show that an ideal small-amplitude pressure wave in a barotropic medium satisfies

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho_0 v^2 + \frac{c_0^2 \Delta\rho^2}{2\rho_0}\right) = -\nabla \cdot (\Delta p\, \vec{v}). \tag{223}$$

   (b) Use this result to justify that the energy density in a wave is

$$\varepsilon = \frac{1}{2}\rho_0 v^2 + \frac{c_0^2 \Delta\rho^2}{2\rho_0}. \tag{224}$$

   Show that it contains both kinetic and compressional energy.

   (c) Show that the average energy density over one period of a harmonic wave is

$$\langle \varepsilon \rangle = \frac{1}{2}\rho_0 v_1^2, \tag{225}$$

   where v_1 is the velocity amplitude.

2. A Pitot tube is mounted at the front of an aircraft moving with speed

$$U = 850 \text{ km h}^{-1} \tag{226}$$

   relative to the atmosphere, which may be assumed to be an ideal gas with

$$\gamma = 1.4, \qquad T = -50°\text{C}, \qquad p = 400 \text{ hPa}. \tag{227}$$

   Calculate the relative temperature and pressure increases in the Pitot tube.

3. Show that for unsteady, compressible potential flow in a barotropic fluid with

$$\rho = \rho(p), \tag{228}$$

the equations of motion may be chosen to be

$$\frac{\partial \Psi}{\partial t} + \frac{1}{2}v^2 + \Phi + w(p) = 0, \tag{229}$$

and

$$\frac{\partial \rho}{\partial t} + (\vec{v} \cdot \nabla)\rho = -\rho \nabla^2 \Psi, \tag{230}$$

where

$$\vec{v} = \nabla \Psi, \tag{231}$$

and

$$w(p) = \int \frac{dp}{\rho(p)}. \tag{232}$$

4. Consider a non-viscous barotropic fluid in an external time-independent gravitational field $\vec{g}(\vec{x})$ satisfying

$$\nabla \cdot \vec{g} = 0. \tag{233}$$

Let $\rho_0(\vec{x})$ and $p_0(\vec{x})$ be the density and pressure in hydrostatic equilibrium.

(a) Show that the wave equation for small-amplitude pressure oscillations around hydrostatic equilibrium becomes

$$\frac{1}{c_0^2}\frac{\partial^2 \Delta p}{\partial t^2} = \nabla^2 \Delta p - (\vec{g} \cdot \nabla)\frac{\Delta p}{c_0^2}, \tag{234}$$

where $c_0(\vec{x})$ is the local sound speed in hydrostatic equilibrium.

(b) Estimate under which conditions the extra term can be neglected in standard gravity for an atmospheric sound wave of wavelength λ.

4 Viscosity

Disturbing a fluid at rest slightly, setting it into motion with spatially varying velocity field, will to first order of approximation generate stresses that depend linearly on the spatial derivatives of the velocity field. Fluids with a linear relationship between stress and velocity gradients are said to be Newtonian, and the coefficients in this relationship are material constants that characterize the strength of viscosity.

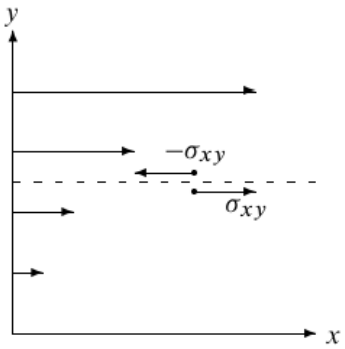


Figure 27: Shear viscosity in laminar (layered) flow. The fluid above the dashed line moves slightly faster than the fluid below and exerts a positive shear stress σ_{xy} on the fluid beneath. By Newton's third law, the fluid below exerts an opposite shear stress $-\sigma_{xy}$ on the fluid above. Figure taken from Ref. [3].

4.1 Shear viscosity

Consider a fluid flowing steadily along the x-direction with a velocity field $v_x(y)$ that is independent of x but may vary with y. Such a field could, for example, be created by enclosing a fluid between moving plates, and is an elementary example of *laminar* or layered flow. If the velocity field has no y-dependence, so that the fluid is in uniform motion along the x-axis, there should not be any internal stresses. If, on the other hand, the velocity grows with y, so that $dv_x(y)/dy > 0$, we expect the fluid immediately above a plane $y = \text{const}$ to drag along the fluid immediately below because of friction and thus exert a positive shear stress, $\sigma_{xy}(y) > 0$, on this plane (see Fig. 27).

It seems reasonable to expect that a larger velocity gradient will evoke stronger stress. In Newton's law of viscosity the shear stress is made proportional to the gradient,

$$\sigma_{xy}(y) = \eta \frac{dv_x(y)}{dy}. \tag{235}$$

The proportionality constant η is called the coefficient of *shear viscosity*, the *dynamic viscosity*, or simply the *viscosity*. It is a measure of how strongly the moving layers of fluid are coupled by friction. In compressible fluids there is also a bulk viscosity coefficient, but in practice it is usually much smaller and often negligible.

The viscosities of real fluids range over many orders of magnitude. Since dv_x/dy has dimension of inverse time, the unit for viscosity is $\text{Pa} \cdot \text{s}$ (pascal–seconds).

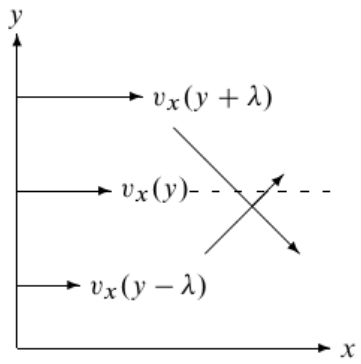


Figure 28: Layers of gas moving with different velocities give rise to shear forces because they exchange molecules with different average velocities. Figure taken from Ref. [3].

4.1.1 Molecular origin of viscosity in gases

In gases where molecules are far apart, internal stress arises from molecular bombardment of surfaces, transferring momentum across them. In liquids the stress is caused partly by molecular motion and partly by intermolecular forces. We therefore restrict the discussion to gases.

Molecules in a gas move nearly randomly at speeds much larger than the macroscopic flow velocity $v(x,t)$. In laminar planar flow with velocity $v_x(y)$ and gradient dv_x/dy, a molecule of mass m crossing a surface element dS_y from above carries more momentum in the x-direction than one crossing from below. The result is a net transfer of momentum downward.

Let λ be the mean free path and τ the typical time between collisions. Disregarding numerical factors of order unity, a layer of thickness λ above dS_y carries an excess momentum

$$dP_x \approx \rho\lambda^2 \frac{dv_x}{dy}\, dS_y. \tag{236}$$

The shear stress follows as momentum transferred per unit time and area:

$$\sigma_{xy} \approx \frac{dP_x}{\tau dS_y},$$

which recovers Newton's law with estimate

$$\eta \approx \rho\frac{\lambda^2}{\tau} \approx \rho\lambda v_{\mathrm{mol}}. \tag{237}$$

Using $\lambda/\tau \approx v_{\mathrm{mol}}$ and the rms molecular speed $v_{\mathrm{mol}} = \sqrt{3RT}$, this explains the magnitude of gas viscosity.

4.1.2 Temperature dependence of viscosity

The viscosity of any material depends on temperature. Most liquids become thinner when heated (viscosity decreases), while gases become more viscous at higher temperatures.

For an ideal gas one finds

$$\eta = \eta_0\sqrt{\frac{T}{T_0}}. \tag{238}$$

The viscosity of gases is independent of pressure and increases slightly faster than $\sqrt{T}$ because of molecular attraction.

4.1.3 Kinematic viscosity

The estimate above suggests defining the *kinematic viscosity*

$$\nu = \frac{\eta}{\rho}. \tag{239}$$

It has units of $\mathrm{m}^2\,\mathrm{s}^{-1}$. In ideal gases we have $\rho \sim p/T$, so that

$$\nu \sim \frac{T^{3/2}}{p},$$

and the kinematic viscosity decreases with temperature for isentropic flow.

Unlike the dynamic viscosity, it is the kinematic viscosity ν that appears in the Navier–Stokes equation. Under velocity-driven flows, air behaves as if it were more viscous than water by an order of magnitude, which explains why air seems "thicker" in certain aerodynamic contexts.

4.1.4 Velocity-driven planar flow

Assume laminar planar flow with velocity $v_x(y,t)$. Since there is no advection term $(\vec{v}\cdot\vec{\nabla})v_x$, Cauchy's equation reduces to

$$\rho\frac{\partial v_x}{\partial t} = \frac{\partial \sigma_{xy}}{\partial y} = \eta\frac{\partial^2 v_x}{\partial y^2}. \tag{240}$$

Dividing by ρ gives

$$\frac{\partial v_x}{\partial t} = \nu\frac{\partial^2 v_x}{\partial y^2}. \tag{241}$$

This is a diffusion equation for velocity.

4.1.5 Case: Steady planar flow

In steady state,

$$\frac{\partial^2 v_x}{\partial y^2} = 0,$$

so

$$v_x = A + By.$$

For plates at $y = 0$ and $y = d$ with velocities 0 and U, respectively, boundary conditions yield

$$v_x(y) = \frac{U}{d}y. \tag{242}$$

The shear stress is

$$\sigma_{xy} = \eta\frac{U}{d}. \tag{243}$$

4.2 Newtonian and Non-Newtonian Fluids

The mechanical response of a fluid under deformation is characterized by the relation between the shear stress and the rate of deformation. This relation is the central subject of *rheology*. Fluids are commonly classified as either Newtonian or non-Newtonian according to how their viscosity behaves under shear.

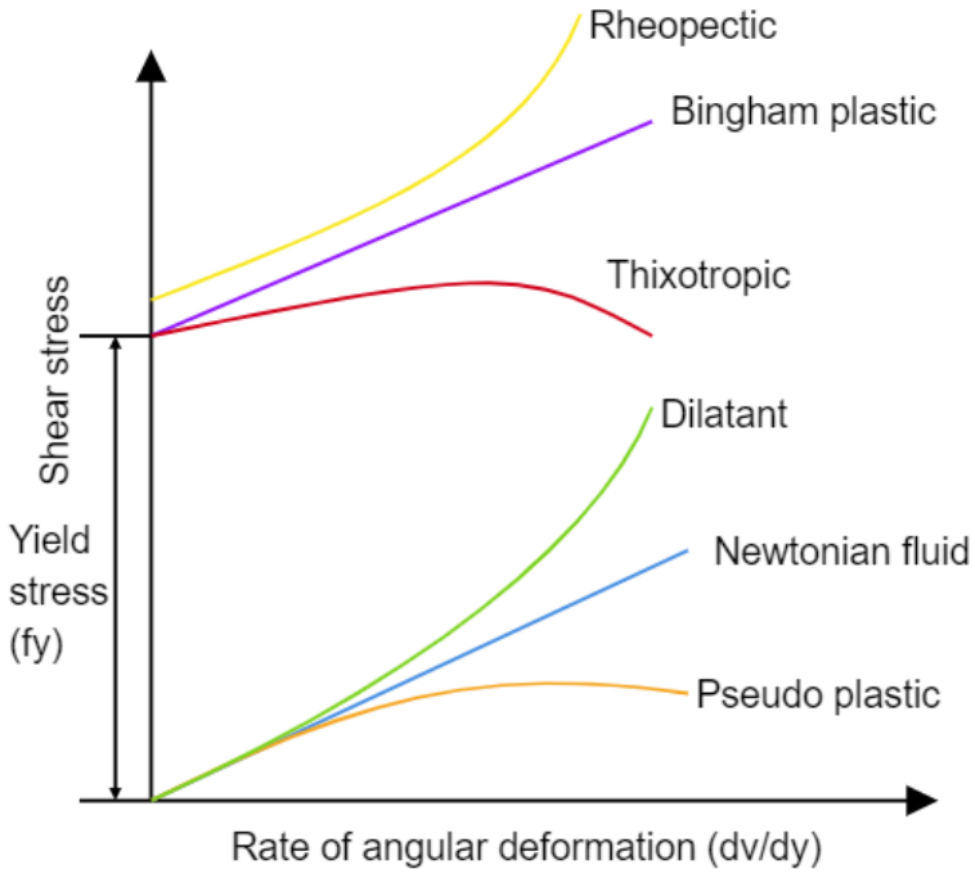


Figure 29: Newtonian and Non-Newtonian Fluids. See video.

4.2.1 Newtonian fluids

A Newtonian fluid obeys Newton's law of viscosity, according to which the shear stress is proportional to the velocity gradient perpendicular to the flow direction:

$$\sigma = \eta \frac{dv}{dy}, \tag{244}$$

The defining characteristic of a Newtonian fluid is that the viscosity η is independent of the shear rate. Consequently, the relation between shear stress and shear rate is linear. Examples of Newtonian fluids include: water, air, alcohol, glycerin, light oils, dilute gas mixtures.

For these fluids, viscosity depends mainly on temperature and pressure, but not on how strongly the fluid is stirred or deformed. Newtonian models provide an excellent approximation for many engineering applications involving ordinary liquids and gases.

4.2.2 Non-Newtonian fluids

Many real fluids do not satisfy Newton's law of viscosity. In these cases the viscosity depends on the shear rate, on the duration of the applied stress, or even on the deformation history of the material. Such fluids are called *non-Newtonian fluids*.

For a non-Newtonian fluid, the relation between shear stress and shear rate is generally non-linear:

$$\sigma \neq \eta \frac{dv}{dy}, \tag{245}$$

with no single constant viscosity capable of describing the flow.

Non-Newtonian behavior is common in: polymer solutions, paints, blood, toothpaste, mud, ketchup, shampoos, starch suspensions.

These materials often possess complex microscopic structures that reorganize under stress, producing shear-dependent rheological properties.

Non-Newtonian fluids are usually divided into several categories.

Shear-thinning fluids

In shear-thinning (or pseudoplastic) fluids, the apparent viscosity decreases as the shear rate increases:

$$\frac{d\eta_{\text{app}}}{d\dot{\gamma}} < 0, \tag{246}$$

where $\dot{\gamma}$ denotes the shear rate.

Typical examples include: ketchup, paint, blood, polymer solutions.

This behavior is advantageous in applications such as paints, which become less viscous during brushing or spraying but recover a larger viscosity afterward, reducing dripping.

Shear-thickening fluids

In shear-thickening (or dilatant) fluids, the apparent viscosity increases with shear rate:

$$\frac{d\eta_{\text{app}}}{d\dot{\gamma}} > 0. \tag{247}$$

A well-known example is a suspension of cornstarch in water. Under sudden impact the fluid behaves almost like a solid because the particle network resists rapid deformation.

Bingham plastics

Some materials behave as rigid solids until a threshold stress, called the *yield stress*, is exceeded. Once this occurs, the material flows approximately as a viscous fluid:

$$\sigma = \sigma_y + \eta_p \dot{\gamma}, \tag{248}$$

where:

- σ_y is the yield stress,
- μ_p is the plastic viscosity.

Examples include: toothpaste, mayonnaise, drilling muds.

Time-dependent fluids

Some fluids exhibit viscosity changes that depend on the duration of the applied shear.

- **Thixotropic fluids:** viscosity decreases with time under constant shear.
- **Rheopectic fluids:** viscosity increases with time under constant shear.

Many industrial suspensions and drilling fluids are thixotropic.

Viscoelastic fluids

Certain fluids combine viscous and elastic properties. When deformed, they may partially recover their original shape after the stress is removed. Polymer melts and biological fluids often exhibit viscoelasticity. Their behavior cannot be described solely by viscosity and requires more sophisticated constitutive equations.

4.2.3 Power-law model

A common empirical description of non-Newtonian fluids is the power-law model:

$$\sigma = K\dot{\gamma}^{\,n}, \tag{249}$$

where:

- K is the consistency index,
- n is the flow behavior index.

Different values of n correspond to different rheological behaviors:

$$n = 1 \quad \text{Newtonian fluid,} \tag{250}$$
$$n < 1 \quad \text{shear-thinning fluid,} \tag{251}$$
$$n > 1 \quad \text{shear-thickening fluid.} \tag{252}$$

Non-Newtonian behavior usually originates from internal microstructure:

- long polymer chains,
- suspended particles,
- droplets,
- colloidal aggregates,
- biological cells.

Under shear these structures may align, deform, stretch, or reorganize, modifying the effective resistance to flow. In polymeric fluids, for example, molecular alignment under shear often produces shear thinning.

Newtonian fluids possess a constant viscosity and exhibit a linear relation between shear stress and shear rate. Non-Newtonian fluids exhibit more complex rheological behavior in which the apparent viscosity depends on shear rate, time, or deformation history. Although Newtonian models are sufficient for many simple liquids and gases, most complex industrial and biological fluids are fundamentally non-Newtonian and require more elaborate constitutive models.

4.3 Dynamics of incompressible Newtonian fluids

In Newtonian fluids the shear stress σ_{xy} is directly proportional to the velocity gradient $\nabla_y v_x$—also called the *shear strain rate*— with proportionality constant equal to the shear viscosity η. Most non-Newtonian fluids become "thinner" as the shear strain rate increases, meaning the shear stress grows slower than linearly. Even the most Newtonian of fluids, water, is thinner at shear strain rates above $10^{12}\,\mathrm{s}^{-1}$. Only a few fluids (for example some starches stirred in water) appear to thicken with increasing strain rate. The science of the general flow properties of materials is called *rheology*.

We shall in this section establish the general dynamics for incompressible, isotropic, and homogeneous Newtonian fluids.

4.3.1 Isotropic viscous stress

Newton's law of viscosity (15.1) is a linear relation between the shear stress and the velocity gradient, only valid in a particular flow geometry. As for Hooke's law for elasticity, we want a definition of viscous stress that takes the same form in any flow geometry and in any Cartesian coordinate system.

Most fluids are not only Newtonian but also *isotropic*. In an isotropic fluid at rest there are no internal directions, and the stress tensor is determined by the pressure,

$$\sigma_{ij} = -p\,\delta_{ij}.$$

When such a fluid is set in motion, the velocity field $\vec{v}(\vec{x},t)$ defines a direction at every point. Stress is produced by velocity gradients, hence viscous stresses are determined by the tensor $\nabla_i v_j$. In an incompressible fluid,

$$\sum_i \nabla_i v_i = \vec{\nabla}\cdot\vec{v} = 0,$$

so the most general symmetric tensor that can be built from velocity gradients is

$$\sigma_{ij} = -p\,\delta_{ij} + \eta\left(\nabla_i v_j + \nabla_j v_i\right). \tag{253}$$

This is the natural generalization of Newton's law of viscosity for incompressible flow in any Cartesian coordinate system. For steady planar flow $\vec{v} = (v_x(y), 0, 0)$ the only nonzero components are

$$\sigma_{xy} = \sigma_{yx} = \eta\nabla_y v_x,$$

confirming that η is indeed the shear viscosity.

4.3.2 The Navier–Stokes equations

The right-hand side of Cauchy's equation of motion equals the effective density of force

$$f_i^* = f_i + \sum_j \nabla_j \sigma_{ij}.$$

Using the stress tensor above and $\vec{\nabla}\cdot\vec{v} = 0$,

$$\sum_j \nabla_j\sigma_{ij} = -\nabla_i p + \eta\left(\sum_j \nabla_i\nabla_j v_j + \sum_j \nabla_j^2 v_i\right) = -\nabla_i p + \eta\nabla^2 v_i.$$

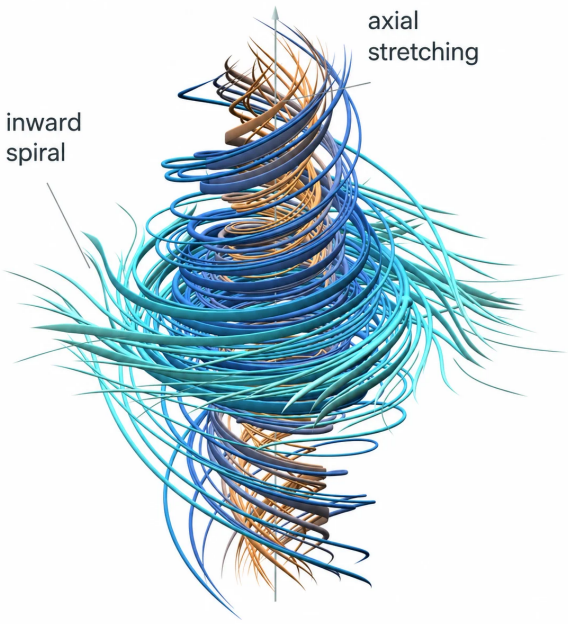


Figure 30: A snapshot of local incompressible motion. Orange marks faster angular rotation; teal marks slower rotation. Circulating speed also depends on radius. The trajectories show inward spiraling and axial stretching. Figure taken from OpenAI webpage.

Assuming the fluid is homogeneous, i.e. η and ρ are constants, we obtain the Navier–Stokes equations for incompressible Newtonian fluids:

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{v} = \vec{g} - \frac{1}{\rho_0}\vec{\nabla} p + \nu \nabla^2 \vec{v}, \qquad \vec{\nabla} \cdot \vec{v} = 0. \tag{254}$$

Here ρ_0 is the constant density, $\nu = \eta/\rho_0$ the kinematic viscosity, and $\vec{g} = \vec{f}/\rho_0$ the acceleration field of volume forces (e.g. gravity).

Although compact, these equations contain the full complexity of real fluid dynamics and generally cannot be solved in closed form except for highly symmetric geometries and idealized conditions.

4.3.3 Boundary conditions

The Navier–Stokes equation is first order in time and requires initial values and boundary conditions.

Density. The density may be discontinuous across a material interface and therefore provides no boundary condition.

Velocity. The normal component

$$v_n = \vec{v} \cdot \vec{n}$$

must be continuous across an interface between incompressible fluids. The tangential component

$$\vec{v}_t = \vec{n} \times (\vec{v} \times \vec{n})$$

must also be continuous, leading to the *no-slip condition*. The full velocity field is therefore continuous across any interface,

$$[\vec{v}] = 0. \tag{255}$$

An exception is an interface between a liquid and vacuum, where velocity may be discontinuous.

Stress. Newton's Third Law requires continuity of the stress vector:

$$[\boldsymbol{\sigma} \cdot \hat{n}] = 0. \tag{256}$$

At a vacuum boundary, $\boldsymbol{\sigma} \cdot \hat{n} = 0$, since vacuum cannot exert force.

Pressure. Pressure need not be continuous across all interfaces, but for incompressible viscous flow near a solid boundary it must coincide with the wall pressure.

4.4 Classification of flows

The most interesting phenomena in fluid dynamics arise from the competition between inertia and viscosity, represented in the Navier–Stokes equation by the advective acceleration $(\vec{v} \cdot \vec{\nabla})\vec{v}$ and the viscous diffusion term $\nu\nabla^2\vec{v}$. Inertia attempts to continue motion, whereas viscosity acts as a brake.

If inertia is dominant, the viscous term may be neglected and Euler's equation for ideal flow is recovered, describing lively, non-viscous or ideal flow. If viscosity dominates, the advective term may be dropped and the equations reduce to those describing slow *creeping flow*.

4.4.1 The Reynolds number

A dimensionless estimate of whether inertia or viscosity dominates is given by the Reynolds number, defined as the ratio of the advective to viscous terms. For a characteristic velocity scale U and length scale L,

$$|\vec{\nabla}\vec{v}| \sim \frac{U}{L}, \qquad |\nabla^2\vec{v}| \sim \frac{U}{L^2}.$$

Thus,

$$\mathrm{Re} \approx \frac{|(\vec{v} \cdot \vec{\nabla})\vec{v}|}{|\nu\nabla^2\vec{v}|} = \frac{U^2/L}{\nu U/L^2} = \frac{UL}{\nu}. \tag{257}$$

Alternatively, the Reynolds number may be viewed as the ratio

$$\mathrm{Re} \approx \frac{t_{\mathrm{diff}}}{t_{\mathrm{flow}}},$$

where $t_{\mathrm{diff}} \sim L^2/\nu$ is the diffusion time and $t_{\mathrm{flow}} \sim L/U$ is the flow time.

For $\mathrm{Re} \ll 1$ the flow is slow and viscous (creeping). For $\mathrm{Re} \gg 1$ inertia dominates and the flow is lively. At intermediate values, turbulence may develop.

4.4.2 Examples

Bathtub turbulence Typical values $U \sim 1$ m/s and $L \sim 1$ m give

$$\mathrm{Re} \sim 10^6,$$

so turbulence is inevitable.

Water pipe For a pipe of diameter $d = 1.25$ cm and flow rate $Q = 100$ cm^3/s,

$$U = \frac{Q}{\pi a^2} \approx 0.8 \text{ m/s}, \qquad \text{Re} \approx 10^4,$$

which is deep in the turbulent regime. For olive oil under otherwise identical conditions we get $\text{Re} \approx 0.15$, and the flow would be creeping.

System	Fluid	Size (m)	Velocity (m/s)	Reynolds number
Ship (Queen Mary 2)	Water	345	15	5.2×10^9
Submarine (Ohio class)	Water	170	12	2.0×10^9
Jet airplane (Boeing 747)	Air	71	250	1.2×10^9
Blue whale	Water	33	10	3.3×10^8
Car	Air	5	30	9.7×10^6
Swimming human	Water	2	1	2.0×10^6
Jogging human	Air	1	3	1.9×10^5
Herring	Water	0.3	1	3.0×10^5
Golf ball	Air	0.043	40	1.1×10^5
Ping-pong ball	Air	0.040	10	2.6×10^4
Fly	Air	0.01	1	6.5×10^2
Flea	Air	0.001	3	1.9×10^2
Gnat	Air	0.001	0.1	6.5
Bacterium	Water	10^{-6}	10^{-5}	10^{-5}

4.4.3 Hydrodynamic similarity

Two flows are said to be *hydrodynamically similar* if they have the same Reynolds number and congruent geometry.

For steady incompressible flow with $\vec{g} = 0$,

$$(\vec{v} \cdot \vec{\nabla})\vec{v} = -\frac{1}{\rho_0}\vec{\nabla} p + \nu \nabla^2 \vec{v}. \tag{258}$$

Introduce dimensionless variables

$$\vec{v} = U\tilde{\vec{v}}, \qquad \vec{x} = L\tilde{\vec{x}}, \qquad p = \rho_0 U^2 \tilde{p}, \qquad \vec{\nabla} = \frac{1}{L}\tilde{\vec{\nabla}}. \tag{259}$$

Then the Navier–Stokes equation becomes

$$(\tilde{\vec{v}} \cdot \tilde{\vec{\nabla}})\tilde{\vec{v}} = -\tilde{\vec{\nabla}}\tilde{p} + \frac{1}{\text{Re}}\tilde{\nabla}^2 \tilde{\vec{v}}. \tag{260}$$

The only remaining parameter is the Reynolds number. Thus, any two flows with equal Reynolds numbers and identical geometry have the same dimensionless solutions.

The viscous force on an object of size L scales as

$$D \sim \sigma L^2 \sim \eta \frac{U}{L} L^2 = \eta U L = \rho \nu U L = \rho U^2 L^2 \frac{1}{\text{Re}}.$$

For stones in air and water with equal Reynolds numbers,

$$\frac{D_{\text{air}}}{D_{\text{water}}} = \left(\frac{\eta\nu}{\eta\nu}\right)_{\text{air/water}} \approx 0.27.$$

4.4.4 Example: Robofly

Using hydrodynamic similarity, one can study insect flight using scaled models. A wing of size $L \approx 4\,\mathrm{mm}$ flapping at $50\,\mathrm{Hz}$ has

$$U \approx \pi f L \approx 1.3\ \mathrm{m/s}, \qquad \mathrm{Re} \approx 160.$$

The same flow can be reproduced using mineral oil of higher viscosity.

4.4.5 Example: High-pressure wind tunnels

In early flight research, Reynolds numbers were increased by raising pressure. Since η is pressure-independent and $\rho \propto p$,

$$\mathrm{Re} \propto \rho.$$

The Variable Density Tunnel operated at 20 atm and enabled realistic testing on small-scale models.

Flows with the same Reynolds number may still differ in geometry. Hence Re predicts general behavior (laminar vs turbulent), not detailed flow structures.

4.5 Dynamics of compressible Newtonian fluids

The velocity divergence, $\nabla \cdot \vec{v}$, is the only scalar that can be formed from linear combinations of the velocity gradients. Therefore, the only additional term allowed in the stress tensor must be proportional to $\delta_{ij}\,\nabla \cdot \vec{v}$. The proportionality constant is conventionally written as $\zeta - \frac{2}{3}\eta$, where η is the shear viscosity and ζ is the bulk (or expansion) viscosity. The stress tensor for a compressible isotropic Newtonian fluid is then

$$\sigma_{ij} = -p\,\delta_{ij} + \eta\left(\nabla_i v_j + \nabla_j v_i - \frac{2}{3}\,\nabla \cdot \vec{v}\,\delta_{ij}\right) + \zeta\,\nabla \cdot \vec{v}\,\delta_{ij}. \tag{261}$$

Interpreting this expression as a first-order expansion in the velocity field, the pressure p is identified with the thermodynamic pressure of the fluid at rest. The tracelessness of the shear term implies that the mechanical pressure is

$$p_{\mathrm{mech}} = p - \zeta\,\nabla \cdot \vec{v}, \tag{262}$$

showing that bulk viscosity generates an additional dynamic pressure proportional to the local expansion or contraction of the fluid.

4.5.1 The Navier–Stokes equations

Inserting the stress tensor into Cauchy's equation of motion, we obtain the most general form of the Navier–Stokes equation (for constant η and ζ),

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right) = \vec{f} - \nabla p + \eta\nabla^2\vec{v} + \left(\zeta + \frac{1}{3}\eta\right)\nabla(\nabla\cdot\vec{v}). \tag{263}$$

If the viscosities have a spatial variation, for example due to temperature, the expression becomes more complicated. Together with the equation of continuity,

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0, \tag{264}$$

we have obtained four dynamic equations for the four fields v_x, v_y, v_z, and ρ, while the pressure is determined by the thermodynamic equation of state,

$$p = p(\rho, T).$$

For isothermal or isentropic flow the temperature is given algebraically, whereas in the general case we also need a differential heat equation to specify the dynamics of the temperature field.

4.5.2 Boundary conditions

The principal results of the discussion of boundary conditions for incompressible fluids remain valid for compressible fluids, namely the continuity of the velocity field and the stress vector across a material interface,

$$[\vec{v}] = 0, \qquad [\boldsymbol{\sigma} \cdot \hat{n}] = 0. \tag{265}$$

The proof of the continuity of pressure at a solid wall can, however, not be carried through in this case.

4.6 Problems

1. (a) Calculate the temperature dependence of the kinematic viscosity for isentropic processes in ideal gases.
   (b) What is the exponent of the temperature for monatomic, diatomic, and multiatomic gases?

2. Estimate the Reynolds number for:
   (a) an ocean current,
   (b) a waterfall,
   (c) a weather cyclone,
   (d) a hurricane,
   (e) a tornado,
   (f) lava running down a mountainside,
   (g) plate tectonic motion.

3. Consider an incompressible fluid of constant density ρ occupying a volume V. The total kinetic energy is

$$E_{\text{kin}} = \frac{1}{2}\rho \int_V v^2 \, dV. \tag{266}$$

   The Navier–Stokes equation may be written as

$$\rho \frac{D\vec{v}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\sigma}^T, \tag{267}$$

   where $\boldsymbol{\sigma}$ is the viscous stress tensor.

(a) Show that the local kinetic-energy balance equation can be written as

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho v^2\right) = -\nabla\cdot\vec{J} - \sigma'_{ik}\frac{\partial v_i}{\partial x_k}, \tag{268}$$

where

$$\vec{J} = \rho\vec{v}\left(\frac{v^2}{2} + \frac{p}{\rho}\right) - \vec{v}\cdot\boldsymbol{\sigma} \tag{269}$$

is the mechanical-energy flux.

(b) Integrate the previous equation over a volume V and use the divergence theorem to show that

$$\frac{dE_{\text{kin}}}{dt} = -\oint_{\partial V}\vec{J}\cdot d\vec{S} - \int_V \sigma'_{ik}\frac{\partial v_i}{\partial x_k}\,dV. \tag{270}$$

Interpret physically each term.

(c) Assuming that the fluid occupies all space and that the velocity vanishes sufficiently rapidly at infinity, show that

$$\frac{dE_{\text{kin}}}{dt} = -\int_V \sigma'_{ik}\frac{\partial v_i}{\partial x_k}\,dV. \tag{271}$$

(d) Using the constitutive relation for an incompressible Newtonian fluid,

$$\sigma'_{ik} = \eta\left(\frac{\partial v_i}{\partial x_k} + \frac{\partial v_k}{\partial x_i}\right), \tag{272}$$

prove that

$$\frac{dE_{\text{kin}}}{dt} = -\frac{\eta}{2}\int_V \left(\frac{\partial v_i}{\partial x_k} + \frac{\partial v_k}{\partial x_i}\right)^2 dV. \tag{273}$$

Hence show that viscosity always decreases the mechanical energy of the flow.

4. Consider a static fluid with uniform density ρ_0 and pressure p_0. Assume that the fluid obeys the equations

$$\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{v}) = 0, \tag{274}$$

and

$$\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\vec{\nabla})\vec{v} = -\frac{1}{\rho}\vec{\nabla}p - \frac{\eta}{\rho}\left[\nabla^2\vec{v} + \frac{1}{3}\vec{\nabla}(\vec{\nabla}\cdot\vec{v})\right] - \frac{\zeta}{\rho}\vec{\nabla}(\vec{\nabla}\cdot\vec{v}), \tag{275}$$

where η and ζ are the coefficients of shear and bulk viscosity, which may be assumed constant. Show that the fluid supports small-amplitude plane-wave perturbations of the form

$$\delta v \approx Ae^{-\gamma x}e^{i(kx-\omega t)}, \tag{276}$$

$$\delta\rho = \rho - \rho_0 \approx Be^{-\gamma x}e^{i(kx-\omega t)}, \tag{277}$$

where $\gamma \ll k$ and

$$\gamma \approx \frac{\omega^2}{2\rho_0 c_s^3}\left(\frac{4}{3}\eta + \zeta\right), \qquad c_s \approx \frac{\omega}{k}, \tag{278}$$

and

$$c_s^2 = \frac{dp}{d\rho} \tag{279}$$

is the sound speed.

Give a physical explanation of this result.

5 Steady, incompressible, viscous flow

Most of the fluids encountered in daily life, air, water, gasoline, and oil, are effectively incompressible as long as flow velocities are well below the speed of sound, and often they flow steadily through the channels and pipes that we use to guide them. In looking for exact solutions for viscous flow we shall make the simplifying assumptions that the flow is incompressible and steady, satisfying the Navier–Stokes equations,

$$(\vec{v}\cdot\nabla)\vec{v} = \vec{g} - \frac{1}{\rho_0}\nabla p + \nu\nabla^2\vec{v}, \qquad \nabla\cdot\vec{v} = 0, \tag{280}$$

where ρ_0 is the constant density of the fluid and ν the constant kinematic viscosity.

5.1 Pressure-driven channel flow

Here we analyze velocity-driven planar flow between infinitely extended parallel plates where both plates are fixed and fluid is driven between them by a pressure gradient. For simplicity we assume that there is no gravity; gravity-driven flow will be analyzed in the following section.

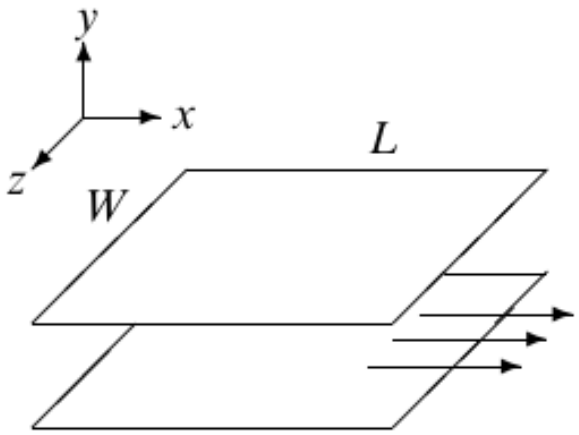


Figure 31: Channel flow between plates of width W parallel and length L. Figure taken from Ref. [3].

The coordinate system is chosen with the x-axis pointing along the direction of flow and the y-axis orthogonal to the plates (see Fig. 31). A velocity field respecting the planar symmetry is of the form

$$\vec{v} = \big(v_x(y), 0, 0\big) = v_x(y)\,\hat{e}_x. \tag{281}$$

An infinitely extended flow like this is of course unphysical, but should nevertheless offer an approximation to the real flow between plates of finite extent, provided the dimensions of the plates are sufficiently large compared to their mutual distance.

With the assumed form of the flow, the incompressibility condition is automatically fulfilled,

$$\nabla\cdot\vec{v} = \nabla_x v_x(y) = 0.$$

The advective acceleration vanishes likewise,

$$(\vec{v}\cdot\nabla)\vec{v} = v_x(y)\,\nabla_x v_x(y)\,\hat{e}_x = 0.$$

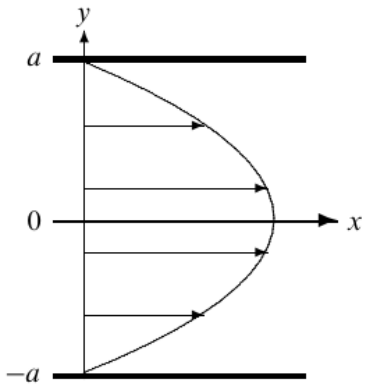


Figure 32: Characteristic parabolic steady velocity profile between the infinitely extended plates with distance $d = 2a$. Figure taken from Ref. [3].

The Navier–Stokes equation (16.1) then takes the form

$$\nabla p = \eta\, \hat{e}_x\, \nabla^2 v_x(y). \tag{282}$$

From the y- and z-components of this equation we get $\nabla_y p = \nabla_z p = 0$, implying that the pressure cannot depend on y and z, or in other words, $p = p(x)$, and the x-component of the above equation becomes

$$\frac{dp(x)}{dx} = \eta\, \frac{d^2 v_x(y)}{dy^2}. \tag{283}$$

The left-hand side depends only on x and the right-hand side only on y, and this is only possible if both sides take the same constant value independent of both x and y. Denoting the common value by $-G$, we may immediately solve each of the equations

$$\frac{dp}{dx} = -G \qquad \text{and} \qquad \eta\, \frac{d^2 v_x}{dy^2} = -G$$

with the result

$$p = p_0 - Gx, \qquad v_x = -\frac{G}{2\eta} y^2 + Ay + B. \tag{284}$$

Here p_0, A, and B are integration constants. The only freedom left in the planar flow problem lies in these constants, which are fixed by the boundary conditions of the specific flow configuration.

boundary conditions. Let the plates be positioned a distance $d = 2a$ apart, for example at $y = -a$ and $y = a$. Applying the no-slip boundary conditions $v_x(-a) = v_x(a) = 0$ to the general solution, we obtain $A = 0$ and $B = Ga^2/2\eta$, and the velocity field becomes (see Fig. 32)

$$v_x = \frac{G}{2\eta} \left(a^2 - y^2\right). \tag{285}$$

It has a characteristic parabolic shape with the maximal velocity

$$v_x^{\max} = \frac{Ga^2}{2\eta}$$

in the middle of the gap, $y = 0$.

Driving pressure and discharge rate. For a real system with finite plates of length L in the direction of flow and width W with a tiny mutual distance, $d \ll L, W$, the flow should be reasonably well described by this solution, except near the edges of the plates (see Figure 32).

Two global quantities are immediately measurable for a real system with finite plates. One is the pressure drop,

$$\Delta p = p(0) - p(L),$$

along the length of the plate L. It is called the driving pressure, and may be determined by means of a suitable manometer. The other is the total volume ΔV of (incompressible) fluid discharged during a time T. From these we may calculate the pressure gradient G and the total volumetric discharge rate Q,

$$G = \frac{\Delta p}{L}, \qquad Q = \frac{\Delta V}{T}. \tag{286}$$

Apart from edge effects, both G and Q ought to be constants, independent of the chosen length L or the collection time T.

The solution relates these quantities. Integrating the velocity field over the area $A = Wd$, orthogonal to the flow, we find

$$Q = \int_A \vec{v} \cdot d\vec{S} = \int_{-a}^{a} v_x(y)\, W\, dy = \frac{2GWa^3}{3\eta}. \tag{287}$$

From the discharge rate we may calculate the average velocity of the flow,

$$U = \frac{Q}{A} = \frac{Ga^2}{3\eta}. \tag{288}$$

It is not surprising that for fixed pressure gradient the average velocity grows with the plate distance, $d = 2a$, because the friction from the walls becomes less and less important with increasing plate distance. The maximal flow velocity is 50% higher than the average,

$$v_x^{\max} = \frac{Ga^2}{2\eta} = \frac{3}{2}U.$$

Since the velocity changes from zero at the walls to 50% more than the average in the middle between them, the Reynolds number for pressure-driven flow between parallel plates a distance d apart is conventionally defined as

$$\mathrm{Re} = \frac{Ud}{\nu}. \tag{289}$$

This definition makes sense independent of whether the incompressible flow is laminar or turbulent, because the discharge rate Q, and thus the average velocity $U = Q/A$, is the same everywhere along the channel.

Example. Oil with $\eta = 2 \times 10^{-2}\,\mathrm{Pa\,s}$ and $\rho_0 = 800\,\mathrm{kg\,m^{-3}}$ between plates $d = 1\,\mathrm{cm}$ apart is driven by a pressure drop $\Delta p = 10^3\,\mathrm{Pa}$ over a distance $L = 1\,\mathrm{m}$. The average velocity becomes in this case $U \approx 0.4\,\mathrm{m\,s^{-1}}$, corresponding to a Reynolds number of $\mathrm{Re} \approx 167$.

Drag and power The no-slip condition requires the fluid to be at rest at the bounding walls. It nevertheless exerts a shear stress on the walls, which may be calculated from the planar solution at $y = -a$:

$$\sigma_{\text{wall}} = \sigma_{xy}\big|_{y=-a} = \eta\,\nabla_y v_x\big|_{y=-a} = Ga. \tag{290}$$

This wall shear stress may be viewed as analogous to static friction between solid bodies.

The total shear force on the walls, called the drag, is obtained by multiplying the wall shear stress by the total wall area of both plates,

$$D = \sigma_{\text{wall}}\, 2LW = GLWd = \Delta p\, A, \tag{291}$$

where $A = Wd$ is the area of the opening through which the fluid flows. That the wall drag should equal the total pressure force on the fluid could have been foreseen because the momentum of the fluid between the plates is constant in steady flow, implying that the total force on the fluid,

$$F_x = -D + \Delta p\, A,$$

must vanish.

The rate of work, or power, of the external pressure forces is similarly obtained:

$$P = \int_A \vec{v}\cdot\Delta p\, d\vec{S} = \Delta p \int_A \vec{v}\cdot d\vec{S} = \Delta p\, Q = \frac{2G^2a^3}{3\eta}\, WL. \tag{292}$$

Since the fluid is at rest at the bounding walls, the wall shear stresses perform no work. The work performed on the fluid by the driving pressure is not lost but dissipated into heat by the viscous friction. If the walls are insulating, this heat is washed along with the flow and leaves the channel through the exit.

5.2 Gravity-driven planar flow

Gravity may also drive the flow between parallel plates if they are inclined an angle θ to the horizon. We choose again a coordinate system with the x-axis in the direction of flow and the y-axis orthogonal to the plates. Assuming constant gravity, the gravitational field is also inclined an angle θ to the negative y-axis. See left panel of Fig. 33.

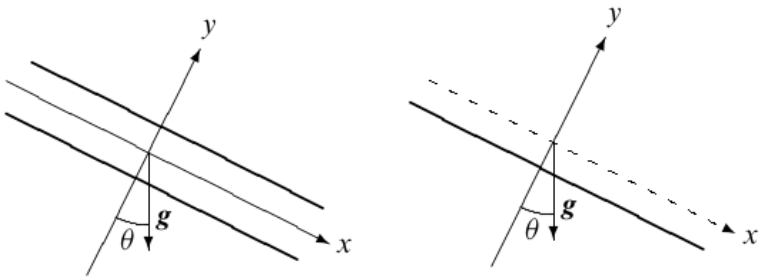


Figure 33: Left: Parallel plates inclined an angle θ to the horizon. Right: Flow with an open surface on a plate inclined an angle θ to the horizon. Figures taken from Ref. [3].

The y- and z-components of the Navier–Stokes equation take the form

$$\nabla_y p = -\rho_0 g_0 \cos\theta, \qquad \nabla_z p = 0. \tag{293}$$

The solution is

$$p = p_0(x) - \rho_0 g_0 y \cos\theta,$$

where $p_0(x)$ is an arbitrary function of x. Inserting this into the x-component of (16.1) we get

$$\frac{1}{\rho_0}\frac{dp_0(x)}{dx} = g_0 \sin\theta + \nu \frac{d^2 v_x(y)}{dy^2}. \tag{294}$$

As in the preceding section it follows that $\nabla_x p_0(x)$ is a constant, and assuming that there is no pressure difference between the ends of the plates at $x = 0$ and $x = L$, it follows that $p_0(x)$ must also be constant, $p_0(x) = p_0$. Applying the no-slip boundary conditions,

$$v_x(-a) = v_x(a) = 0,$$

the complete solution becomes

$$v_x = \frac{g_0 \sin\theta}{2\nu}\left(a^2 - y^2\right), \qquad p = p_0 - \rho_0 g_0 y \cos\theta. \tag{295}$$

Here p_0 is the constant pressure in the central plane $y = 0$. The pressure is for all θ simply the hydrostatic pressure in a constant field of gravity of strength $g_0 \cos\theta$.

5.3 Inclined flow with an open surface

A liquid layer of constant thickness a flowing down an inclined plate with an open surface is another example of purely gravity-driven flow (see right panel of Fig. 33). On the inclined plate, $y = -a$, the no-slip condition again demands $v_x(-a) = 0$, whereas on the open surface, $y = 0$, the pressure must be constant and the stress must vanish,

$$\sigma_{xy} = \eta \frac{dv_x}{dy} = 0.$$

Since these boundary conditions are fulfilled, the velocity and pressure are given by the same expressions as above.

The velocity profile now has its maximum at the open surface. The average velocity is

$$U = \frac{1}{a}\int_{-a}^{0} v_x(y)\, dy = \frac{g_0 a^2 \sin\theta}{3\nu}. \tag{296}$$

The volumetric discharge rate in a swath of width W orthogonal to the flow is

$$Q = UaW.$$

The Reynolds number is naturally defined in terms of the layer thickness,

$$\mathrm{Re} = \frac{Ua}{\nu}. \tag{297}$$

Introducing the convenient velocity parameter

$$U_0 = \left(\tfrac{1}{3}\nu g_0 \sin\theta\right)^{1/3}, \tag{298}$$

we may express all the variables in the convenient form

$$U = \mathrm{Re}^{2/3} U_0, \qquad a = \mathrm{Re}^{1/3}\frac{\nu}{U_0}, \qquad W = \frac{Q}{\nu\,\mathrm{Re}}. \tag{299}$$

Given the inclination angle θ, the viscosity ν, and the discharge rate Q, the Reynolds number Re is the only variable. The length L of the plate is irrelevant as long as $L \gg a$.

Example (Liquid film). A water film of thickness $a = 0.1\,\mathrm{mm}$ flowing down an inclined plate at $\theta = 30°$ has average velocity $U = 16\,\mathrm{mm\,s^{-1}}$ and Reynolds number $\mathrm{Re} = 1.6$. A 10 mm layer of glycerol flowing down the same slope has much larger average velocity $15\,\mathrm{cm\,s^{-1}}$ but a similar Reynolds number of 1.3.

5.4 Laminar pipe flow

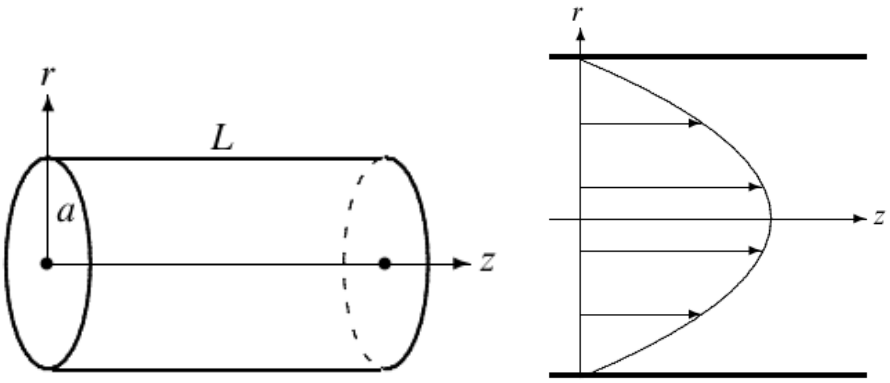


Figure 34: Left: Section of a circular pipe of inner radius a and length L. The pressure is higher at $z = 0$ than at $z = L$, and the pressure drop $\Delta p = p(0) - p(L)$ drives the fluid through the pipe. Right: Velocity profile for laminar flow through a circular pipe. Figures taken from Ref. [3].

An infinitely long circular cylindrical tube is invariant both under translations along its axis and rotations around it (see left panel of Fig. 34). In a coordinate system with the z-axis coincident with the cylinder axis, one velocity field that respects this symmetry is

$$\vec{v} = (0, 0, v_z(r)) = v_z(r)\,\hat{e}_z, \tag{300}$$

where

$$r = \sqrt{x^2 + y^2}$$

is the radial distance from the cylinder axis.

The gradient operator in cylindrical coordinates takes the form

$$\nabla = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\phi \frac{1}{r}\frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z}. \tag{301}$$

Applying this operator to the assumed velocity field, we find

$$\nabla \cdot \vec{v} = \hat{e}_r \cdot \partial_r v_z(r) \hat{e}_z = 0, \qquad (\vec{v} \cdot \nabla)\vec{v} = v_z\, \partial_z v_z(r)\, \hat{e}_z = 0.$$

In the absence of gravity, $\vec{g} = 0$, the Navier–Stokes equation now simplifies to

$$\nabla p = \eta\, \hat{e}_z \nabla^2 v_z(r) = \eta\, \hat{e}_z \left(\frac{d^2 v_z}{dr^2} + \frac{1}{r}\frac{dv_z}{dr} \right). \tag{302}$$

From the x- and y-components of this equation we get $\nabla_x p = \nabla_y p = 0$, and consequently the pressure can only depend on z, that is,

$$p = p(z).$$

Since the left-hand side depends only on z whereas the right-hand side depends only on r, neither side can depend on r or z. Denoting the common constant value by $-G$, we obtain the ordinary differential equations

$$\frac{dp}{dz} = -G, \qquad \eta \frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = -G. \tag{303}$$

Integrating, we find

$$p = p_0 - Gz, \qquad v_z = -\frac{G}{4\eta}r^2 + A\log r + B, \tag{304}$$

where p_0, A, and B are integration constants.

Consider now a pipe with inner radius a and diameter $d = 2a$. The fluid velocity cannot be infinite at $r = 0$, implying $A = 0$ in the general solution. The no-slip boundary condition requires that $v_z(a) = 0$, and this fixes the last integration constant to $B = Ga^2/4\eta$, so that the velocity profile becomes

$$v_z = \frac{G}{4\eta}\left(a^2 - r^2\right). \tag{305}$$

See right panel of Fig. 34. As for pressure-driven channel flow, it is parabolic and reaches, as one would expect, its maximal value

$$v_z^{\max} = \frac{Ga^2}{4\eta}$$

at the center of the pipe.

Driving pressure and discharge rate Two global quantities are—as for channel flow—immediately measurable for a pipe of finite length L. One is the pressure drop between entry and exit,

$$\Delta p = p(0) - p(L),$$

and the other is the total volume ΔV of (incompressible) fluid discharged during a time T. From these we may calculate the pressure gradient G and the total volumetric discharge rate Q,

$$G = \frac{\Delta p}{L}, \qquad Q = \frac{\Delta V}{T}. \tag{306}$$

Apart from edge effects, both G and Q should be constant when the pipe is sufficiently long and the discharge is collected over a sufficiently long time. The volumetric discharge rate may immediately be calculated by integrating the velocity field over the cross-section,

$$A = \pi a^2,$$

of the pipe,

$$Q = \int_A \vec{v}\cdot d\vec{S} = \int_0^a v_z(r)\, 2\pi r\, dr = \frac{G}{4\eta}\pi\int_0^a (a^2 - r^2)\, 2r\, dr. \tag{307}$$

Carrying out the integral, this becomes the famous *Hagen–Poiseuille law*, relating the driving pressure gradient and the discharge rate,

$$Q = \frac{\pi a^4}{8\eta}\, G. \tag{308}$$

As could have been expected, the discharge rate grows linearly with the pressure gradient, and inversely with viscosity. The dramatic fourth-power growth with radius could of course have been deduced from dimensional arguments since it is the only missing factor. It expresses the fairly common observation that much more water flows through a wider pipe for a given driving pressure because the shear wall stress becomes less able to hold back the fluid.

The velocity of the flow averaged over the cross-section of the pipe may be calculated from the rate of discharge,

$$U = \frac{Q}{\pi a^2} = \frac{Ga^2}{8\eta}. \tag{309}$$

The maximal velocity at the center of the pipe is twice the average velocity,

$$v_z^{\max} = 2U.$$

The central flow must be faster than for channel flow, because in a pipe there is less room in the central region to compensate for the slow-down of fluid at the pipe wall.

Reynolds number The Reynolds number should be understood as a dimensionless characterization of the ratio of advective to viscous forces in terms of the speed and geometry of the general flow and becomes

$$\mathrm{Re} = \frac{Ud}{\nu}. \tag{310}$$

Empirically pipe flow remains laminar until turbulence sets in at a Reynolds number between 2,000 and 4,000, with 2,300 as a "nominal" value for smooth pipes. At that point the otherwise linear relationship between volume discharge Q and pressure gradient G becomes nonlinear. For a smooth pipe under very carefully controlled conditions, the transition to turbulence can in fact be delayed until a Reynolds number on the order of 100,000. Above that value the flow is so sensitive to disturbances that it becomes practically impossible to avoid turbulence, even if the Poiseuille solution is believed to be formally stable toward infinitesimal perturbations (in the rz-plane) for all Reynolds numbers.

Example (Aortic flow). Human blood is not a particularly Newtonian fluid, but its viscosity may approximately be taken to be $\eta = 2.7 \times 10^{-3}\,\mathrm{Pa\,s}$ and its density near that of water. The stroke volume of the resting heart is about $70\,\mathrm{cm}^3$, and with a resting heart rate of 60 beats per minute, the aortic blood flow rate (averaged over a heartbeat) is $Q \approx 70\,\mathrm{cm}^3\,\mathrm{s}^{-1}$. Since the aortic root diameter is $2a \approx 25\,\mathrm{mm}$, the average blood velocity becomes $U \approx 14\,\mathrm{cm\,s}^{-1}$ and the Reynolds number $\mathrm{Re} \approx 1{,}300$, well below the turbulent region. The pressure gradient becomes $G \approx 20\,\mathrm{Pa\,m}^{-1}$, showing that the pressure drop in the large arteries is small compared to systolic blood pressure, $p \approx 120\,\mathrm{mmHg} \approx 16{,}000\,\mathrm{Pa}$.

Example (Hypodermic syringe). A hypodermic syringe has a cylindrical chamber with a diameter of about $2b = 1\,\mathrm{cm}$ and a hollow needle with an internal diameter of about $2a = 0.5\,\mathrm{mm}$. During an injection, about $5\,\mathrm{cm}^3$ of the liquid (here assumed to be water) is gently pressed through the needle in a time $\Delta t = 10\,\mathrm{s}$, such that the volume rate is about $Q = 0.5\,\mathrm{cm}^3\,\mathrm{s}^{-1}$. The average fluid velocity in the needle becomes $U \approx 2.5\,\mathrm{m\,s}^{-1}$, corresponding to a Reynolds number $\mathrm{Re} \approx 1{,}300$, which is in the laminar region below the onset of turbulence. One is thus justified in using the Poiseuille solution for the flow through the needle. The pressure gradient necessary to drive this flow is found from (16.32) and becomes rather large, $G \approx 3.3\,\mathrm{bar\,m}^{-1}$. For a needle of length

$L \approx 5\,\mathrm{cm}$, the pressure drop becomes $\Delta p \approx 0.16\,\mathrm{bar} \approx 16{,}000\,\mathrm{Pa}$, about the same as the systolic blood pressure. The pressure drop in the syringe chamber can be completely ignored, because the chamber's diameter is 20 times that of the needle.

Example (Water pipe). Household water supply has to reach the highest floor in apartment buildings with pressure "to spare." Pressures must therefore be of the order of bars when water is not tapped. The typical discharge rate from a kitchen faucet is around $Q \approx 100\,\mathrm{cm}^3\,\mathrm{s}^{-1}$, leading to an average velocity in a half-inch pipe of about $U \approx 0.8\,\mathrm{m\,s}^{-1}$. The Reynolds number becomes about $\mathrm{Re} \approx 10{,}000$, which is well inside the turbulent regime. The pressure gradient calculated from the Hagen–Poiseuille law, $G \approx 160\,\mathrm{Pa\,m}^{-1}$, is for this reason untrustworthy.

5.5 Laminar cylindric flow

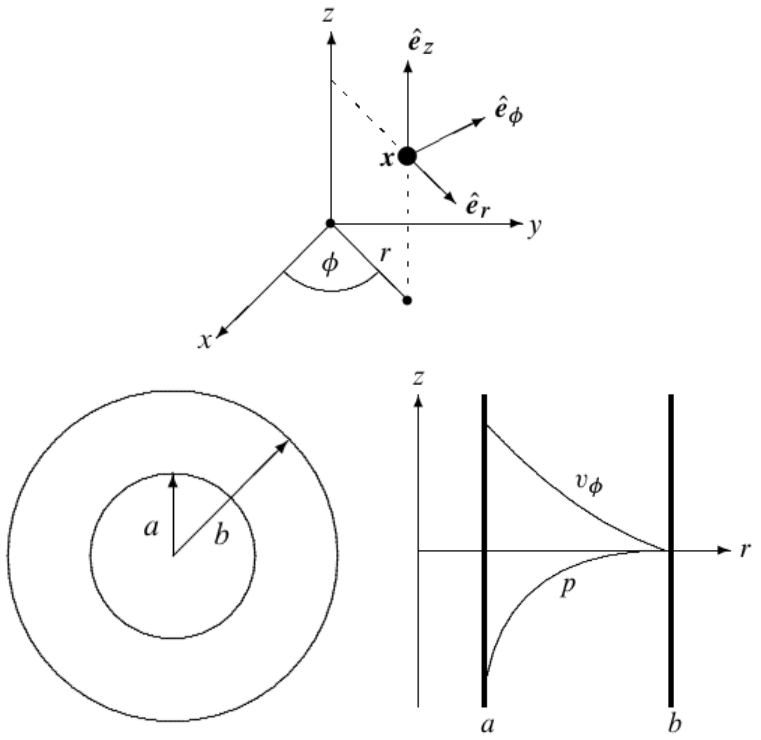


Figure 35: Caption

Infinitely extended coaxial cylinders are both translationally symmetric along the rotation axis and rotationally symmetric around it. In cylindrical coordinates a maximally symmetric circulating flow takes the form

$$\vec{v} = v_\phi(r)\,\hat{e}_\phi, \tag{311}$$

where $\hat{e}_\phi$ is the tangential unit vector and r the distance from the axis. The field lines are concentric circles.

Again we turn to the nabla operator in cylindrical coordinates to calculate the tensor gradient of the velocity field. Since there is no z-dependence we find by means of the relations

$$\partial_\phi \hat{e}_\phi = -\hat{e}_r, \qquad \partial_\phi \hat{e}_r = \hat{e}_\phi,$$

$$\nabla\vec{v} = \left(\hat{e}_r\frac{\partial}{\partial r} + \hat{e}_\phi\frac{1}{r}\frac{\partial}{\partial\phi}\right)\left(v_\phi(r)\hat{e}_\phi\right) = \hat{e}_r\hat{e}_\phi\frac{dv_\phi}{dr} - \hat{e}_\phi\hat{e}_r\frac{v_\phi}{r}. \tag{312}$$

The trace of this tensor yields the divergence that vanishes,

$$\mathrm{Tr}[\nabla\vec{v}] = \nabla\cdot\vec{v} = 0,$$

because $\nabla\vec{v}$ only has off-diagonal components in the cylindrical basis. This is in agreement with the elementary observation that streamlines neither diverge nor converge in this flow. Dotting the tensor gradient from the left with $\vec{v}$, the advective acceleration becomes

$$(\vec{v}\cdot\nabla)\vec{v} = \vec{v}\cdot(\nabla\vec{v}) = -\hat{e}_r\,\frac{v_\phi^2}{r}. \tag{313}$$

One should not be surprised: the centripetal acceleration in a circular motion with velocity v_ϕ is indeed directed radially inwards and of size v_ϕ^2/r. Finally, dotting the tensor gradient from the left with ∇, we obtain the Laplacian:

$$\nabla^2\vec{v} = \left(\hat{e}_r\frac{\partial}{\partial r} + \hat{e}_\phi\frac{1}{r}\frac{\partial}{\partial\phi}\right)\left(\hat{e}_r\hat{e}_\phi\frac{dv_\phi}{dr} - \hat{e}_\phi\hat{e}_r\frac{v_\phi}{r}\right) = \hat{e}_\phi\frac{d^2v_\phi}{dr^2} + \hat{e}_\phi\frac{1}{r}\frac{dv_\phi}{dr} - \hat{e}_\phi\frac{v_\phi}{r^2}. \tag{314}$$

In the absence of gravity, the Navier–Stokes equation (16.1) then becomes

$$-\rho_0\hat{e}_r\frac{v_\phi^2}{r} = -\nabla p + \eta\,\hat{e}_\phi\frac{d}{dr}\left(\frac{1}{r}\frac{d(rv_\phi)}{dr}\right), \tag{315}$$

where we in the last term have rewritten the Laplacian in a convenient way.

Projecting it on the three cylindrical basis vectors, $\hat{e}_r$, $\hat{e}_\phi$, and $\hat{e}_z$, we obtain

$$-\rho_0\frac{v_\phi^2}{r} = -\frac{\partial p}{\partial r}, \tag{316}$$

$$0 = -\frac{1}{r}\frac{\partial p}{\partial\phi} + \eta\,\frac{d}{dr}\left(\frac{1}{r}\frac{d(rv_\phi)}{dr}\right), \tag{317}$$

$$0 = -\frac{\partial p}{\partial z}. \tag{318}$$

The first equation expresses that the radial pressure gradient must deliver the centripetal force required by the circular motion of the fluid. The last equation shows that the pressure is independent of z, and from the second equation we see by differentiation with respect to ϕ that $\partial^2 p/\partial\phi^2 = 0$. This means that p can at most be linear in ϕ, that is, of the form $p = p_0(r) + p_1(r)\phi$. But here we must require $p_1 = 0$, for otherwise the pressure would have different values for $\phi = 0$ and $\phi = 2\pi$, and that is impossible. The pressure does not depend on ϕ but only on r, and thus it disappears completely from (164).

With the pressure out of the way, the integration of (164) has become almost trivial, with the general result

$$v_\phi = Ar + \frac{B}{r}, \tag{319}$$

where A and B are integration constants. Inserting this into (163) and integrating over r, we find the pressure

$$\frac{p}{\rho_0} = C + \frac{1}{2}A^2r^2 - \frac{1}{2}\frac{B^2}{r^2} + 2AB\log r, \tag{320}$$

where C is a third integration constant.

The Couette solution Suppose the fluid is contained between two long coaxial material cylinders with radii a and $b > a$. We shall for simplicity only study the case where the outer cylinder is held fixed and the inner cylinder rotates like a spindle with constant angular velocity Ω. The boundary conditions,

$$v_\phi(a) = a\Omega, \qquad v_\phi(b) = 0,$$

then determine A and B, and the velocity profile becomes

$$v_\phi = \frac{\Omega a^2}{r}\,\frac{b^2 - r^2}{b^2 - a^2}. \tag{321}$$

The velocity field decreases monotonically from its value $a\Omega$ at the inner cylinder to zero at the outer.

The pressure is found to be

$$p^* = p_0 + \frac{1}{2}\rho_0 \left(\frac{\Omega a^2}{b^2 - a^2}\right)^2 \left(r^2 - \frac{b^4}{r^2} + 4b^2 \log\frac{b}{r}\right), \tag{322}$$

where p_0 is its value at the outer cylinder. The pressure increases monotonically toward p_0.

Stress, torque, and power The velocity gradient shows that the only non-vanishing shear stress component is

$$\sigma_{\phi r} = \eta\,(\hat{e}_\phi \cdot (\nabla\vec{v}) \cdot \hat{e}_r + \hat{e}_r \cdot (\nabla\vec{v}) \cdot \hat{e}_\phi) = \eta\left(\frac{dv_\phi}{dr} - \frac{v_\phi}{r}\right), \tag{323}$$

which upon insertion of the Couette solution becomes

$$\sigma_{\phi r} = -2\eta\Omega\,\frac{a^2 b^2}{b^2 - a^2}\,\frac{1}{r^2}. \tag{324}$$

It represents the viscous friction between the layers of circulating fluid, and the sign is negative because the fluid *outside* the radius r acts as a brake on the motion of the fluid *inside*.

In order to maintain the steady rotation of the inner cylinder (of length L) it is necessary to act on it with a moment of force or torque (and with an opposite moment on the outer cylinder). More generally, multiplying the shear stress $\sigma_{\phi r}$ with the moment arm r and the area $2\pi rL$ of the cylinder at r, we obtain the torque with which the fluid inside r acts on the fluid outside,

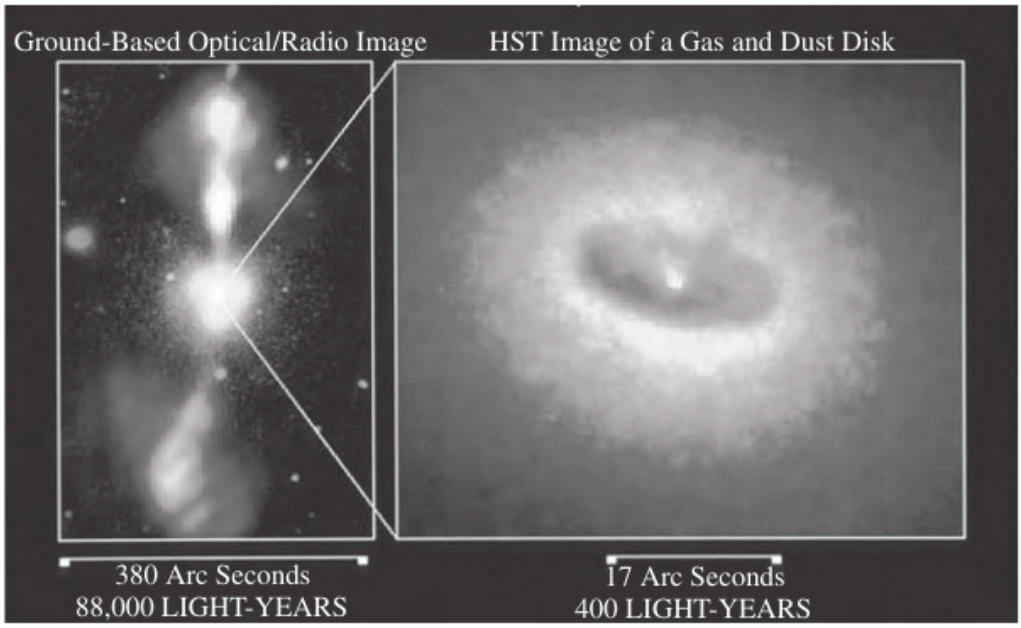
$$\mathcal{M}_z = r(-\sigma_{\phi r})\,2\pi rL = 4\pi\eta\Omega L\,\frac{a^2 b^2}{b^2 - a^2}. \tag{325}$$

We could have foreseen that $\mathcal{M}_z$ would be independent of r from angular momentum conservation. In steady flow, the total angular momentum of a layer of fluid contained between any two cylindrical surfaces is constant and there is no transport of angular momentum through the cylindrical surfaces because they are parallel with the velocity. Consequently the total moment of force has to vanish, implying that the moment acting on the inside of the layer must be equal and opposite to the moment that is acting on the outside of the layer.

The rate of work that must be done to keep the inner cylinder rotating is obtained by multiplying the stress on the fluid, $-\sigma_{\phi r}$ at $r = a$, by the velocity $a\Omega$ and the area of the cylinder,

$$P = (-\sigma_{\phi r})\,a\Omega\,2\pi aL = \mathcal{M}_z\Omega = 4\pi\eta\Omega^2 L\,\frac{a^2 b^2}{b^2 - a^2}. \tag{326}$$

Since the kinetic energy of the fluid is constant in steady flow and since no fluid enters or leaves the system, this must equal the rate of energy dissipation in the fluid.



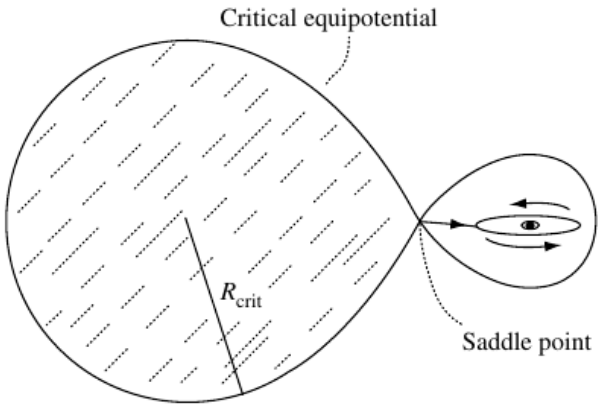


Figure 36: A giant disc of cold gas and dust fuels a possible black hole at the core of the galaxy NGC 4261 (right panel). Hot gas is ejected along the disc axis from the vicinity of the black hole creating the radio jets which are shown in the left panel. (NASA/STScI). Schematic diagram of a close binary system containing a neutron star or white dwarf (on the right) and a giant star (on the left) which fills the critical equipotential and donates mass to its companion through the saddle point. The donated mass possesses angular momentum as a result of the binary's orbital motion and hence forms an accretion disc around the compact star.

5.6 Application: Accretion Disks

One of the most important applications of fluid dynamics in astrophysics is the study of *accretion disks*. These systems arise whenever gas possessing nonzero angular momentum is gravitationally bound to a massive central object. Accretion disks are found in a wide variety of astronomical environments, including protoplanetary systems, close binary stars, neutron stars, stellar-mass black holes, and the supermassive black holes located at the centers of galaxies.

The origin of the orbiting gas depends on the particular astrophysical setting. In protoplanetary systems, the disk forms from the collapse of a rotating molecular cloud. In close binary stars, gas may be transferred from one star to its companion. Around black holes, the accreted material may originate from stellar winds, tidally disrupted stars, or interstellar gas. Despite these different formation mechanisms, all accretion disks share a common property: the gas possesses angular momentum and therefore cannot fall directly toward the central object.

As collisions and shocks dissipate random motions, the gas settles into a thin disk perpendicular to the mean angular momentum vector. The fluid elements then move on nearly circular orbits around the central mass. For a particle orbiting a mass M at a distance R, the balance between gravity and centrifugal force implies

$$\Omega^2 R = \frac{GM}{R^2}, \tag{327}$$

where Ω is the angular velocity. Hence

$$\Omega(R) = \left(\frac{GM}{R^3}\right)^{1/2}. \tag{328}$$

This relation is known as *Keplerian rotation*. It immediately implies that

$$\frac{d\Omega}{dR} < 0, \tag{329}$$

so that neighboring rings of gas rotate at different angular velocities. Consequently, an accretion disk is a natural example of a *circular shear flow*.

The existence of shear allows viscous stresses to develop between adjacent rings of gas. The faster-moving inner rings exert a torque on the slower-moving outer rings, transferring angular momentum outward. As a result, material in the inner regions loses angular momentum and gradually spirals inward, while material farther out gains angular momentum and drifts outward.

This process has profound consequences. As gas moves inward, its gravitational potential energy decreases. The lost potential energy is converted into thermal energy through viscous dissipation and is eventually radiated away. Accretion disks are therefore among the most efficient energy sources in the Universe. Indeed, the enormous luminosities of quasars and active galactic nuclei are powered by the release of gravitational energy through accretion onto supermassive black holes.

A key point is that accretion cannot occur in the absence of viscosity. Without viscous stresses, each fluid element would remain on its Keplerian orbit indefinitely, conserving its angular momentum. Viscosity provides the mechanism by which angular momentum is redistributed throughout the disk, thereby enabling matter to move toward the central object.

Why molecular viscosity is insufficient

Although viscosity plays a crucial role in accretion disks, the ordinary molecular viscosity of astrophysical gases is far too small to explain the observed accretion rates. A characteristic viscous timescale is

$$t_\nu \sim \frac{R^2}{\nu}, \tag{330}$$

where ν is the kinematic viscosity. Using microscopic estimates for ν, one obtains timescales that are often much longer than the ages of observed disks.

This discrepancy suggests that angular momentum must be transported by an *effective viscosity* much larger than the molecular viscosity.

The α-disk model

A simple and highly successful phenomenological description was proposed by Shakura and Sunyaev. In this model, the effective viscosity is written as

$$\nu = \alpha c_s H, \tag{331}$$

where:

- c_s is the local sound speed,
- H is the vertical scale height of the disk,
- α is a dimensionless parameter satisfying $0 < \alpha < 1$.

The parameter α encapsulates our ignorance of the detailed mechanism responsible for angular momentum transport. Observations and numerical simulations typically suggest values in the range

$$\alpha \sim 10^{-3} - 10^{-1}. \tag{332}$$

Magnetorotational instability

The modern explanation for the origin of this enhanced viscosity is the *magnetorotational instability* (MRI), discovered by Balbus and Hawley in 1991.

The MRI operates in weakly magnetized rotating fluids whenever

$$\frac{d\Omega}{dR} < 0, \tag{333}$$

which is precisely the condition satisfied by Keplerian disks. Small magnetic perturbations connect neighboring fluid elements and cause angular momentum to be transferred outward. The instability rapidly develops into turbulence, which acts as an effective viscosity and drives accretion.

The MRI is now believed to be the dominant mechanism responsible for angular momentum transport in many astrophysical disks.

Astrophysical examples

Accretion disks appear in a remarkable variety of systems:

- Protoplanetary disks around young stars, where planets form.
- Cataclysmic variables, in which a white dwarf accretes matter from a companion star.
- X-ray binaries containing neutron stars or stellar-mass black holes.
- Active galactic nuclei powered by accretion onto supermassive black holes.
- Quasars, among the most luminous objects in the Universe.

Despite spanning more than ten orders of magnitude in mass and size, these systems are governed by the same physical principles: gravity, angular momentum conservation, and viscous transport.

Connection with Kepler's third law

The Keplerian angular velocity may be written as

$$\Omega = \sqrt{\frac{GM}{R^3}}. \tag{334}$$

For a solar-mass star,

$$M = M_\odot, \tag{335}$$

and an orbital radius of

$$R = 1 \text{ AU}, \tag{336}$$

the orbital period is

$$T = \frac{2\pi}{\Omega} \approx 1 \text{ yr}, \tag{337}$$

recovering Kepler's third law. Accretion disks therefore provide a direct connection between the fluid dynamics of rotating gases and the orbital mechanics of planetary systems.

In summary, accretion disks constitute one of the most important astrophysical manifestations of viscous shear flow. They illustrate how viscosity, although often weak, can profoundly alter the dynamics of a fluid by transporting angular momentum and enabling the conversion of gravitational potential energy into radiation.

5.7 Problems

1. Consider pressure-driven laminar flow between infinitely extended plates moving with constant velocity U with respect to each other in the same direction as the pressure gradient. Determine under which conditions the maximal flow velocity will be found between the plates.

2. Consider pressure-driven laminar flow between infinitely extended plates moving with constant velocity U with respect to each other in a direction orthogonal to the pressure gradient. Determine the planar flow field.

3. Why is it impossible to make a pressure-driven planar fluid sheet with an open surface?

4. A thin fluid sheet (water) of density ρ_1, viscosity η_1, and thickness d_1 runs steadily down a plate inclined at an angle θ to the horizon. Another plate with the same inclination is placed at a distance $d_1 + d_2$ from the first, and there is another fluid (air) in the gap with density ρ_2, viscosity η_2, and thickness d_2. Assuming laminar flow, calculate:

   (a) the pressure field,

   (b) the velocity fields in the water and in the air between the plates.

5. Calculate:

   (a) the drag,

   (b) the dissipation,

   for pressure-driven planar flow expressed in terms of the average velocity U.

6. Calculate:

   (a) the drag,

   (b) the dissipation,

   for gravity-driven planar flow with an open surface on an inclined plate in terms of the average velocity.

7. Consider laminar gravity-driven flow through a vertical pipe.

   (a) Determine the radius of the pipe as a function of the Reynolds number when the fluid is water.

   (b) Determine the largest radius and

   (c) the entry length just before the onset of turbulence.

8. (a) Show that in pipe flow the only dimensionless combination of the parameters Q, a, ρ_0, and η is the Reynolds number (up to trivial transformations).

   (b) Can you construct another dimensionless parameter using

   $$G = \frac{\Delta p}{L}$$

   instead of Q?

9. Pipe resistance is defined as the ratio of driving pressure to mass discharge,

   $$R = \frac{\Delta p}{\rho_0 Q}.$$

(a) Show that pipe resistance is additive for pipes connected in series.

(b) Show that the reciprocals of the resistances are additive for pipes connected in parallel.

10. Show that for the Poiseuille profile,

$$\int_0^a v_z(r)^n \, 2\pi r \, dr = \frac{(2U)^n \pi a^2}{n+1}, \tag{338}$$

where U is the average velocity.

11. (a) Show that for a cylindrical tube with hollow annular cross-section defined by

$$a < r < b,$$

the solution is given by the general Poiseuille solution with

$$A = \frac{G}{4\eta} \frac{b^2 - a^2}{\ln b - \ln a}, \tag{339}$$

$$B = \frac{G}{4\eta} \frac{a^2 \ln b - b^2 \ln a}{\ln b - \ln a}. \tag{340}$$

(b) Determine the drag on the tube per unit length when a fluid is driven through the tube under a pressure gradient G.

12. A pipe has elliptical cross-section with major and minor semi-axes a and b.

(a) Show that laminar pressure-driven flow of an incompressible fluid with pressure gradient G is given by

$$v_z = \frac{G}{2\eta} \frac{a^2 b^2}{a^2 + b^2} \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right), \qquad p = p_0 - Gz. \tag{341}$$

(b) Show that the fields of the circular pipe and parallel plates are contained in this expression.

(c) Calculate the volume flow rate. (Hint: Use elliptic coordinates

$$x = ar \cos\theta, \qquad y = br \sin\theta,$$

with area element

$$dA = abr \, d\theta \, dr.$$

)

(d) Calculate the average flow velocity.

(e) Calculate the drag over a length L.

13. An infinitely long pipe has an annular cross-section in the interval

$$a < r < b.$$

The inner wall $r = a$ moves with velocity U, whereas the outer wall $r = b$ is at rest. The space between the walls contains a fluid with viscosity η and there is no pressure gradient along the pipe.

(a) Determine the flow v_z in the pipe.

(b) Calculate the volume discharge rate Q. Show that $Q \to 0$ as $a \to b$.

(c) Calculate the drag on a stretch L of the outer wall.

14. A permeable circular cylindrical pipe carrying fluid under pressure leaks fluid through its surface into the surrounding ocean at a constant rate. In the steady state, determine the maximally symmetric form of the radial flow pattern outside the pipe.

15. Show that the coefficients of the general circulating flow may be written as

$$A = \frac{\Omega_b b^2 - \Omega_a a^2}{b^2 - a^2}, \tag{342}$$

$$B = \frac{a^2 b^2 (\Omega_a - \Omega_b)}{b^2 - a^2}, \tag{343}$$

where Ω_a and Ω_b are the angular velocities of the inner and outer cylinders, respectively.

16. A cylinder with radius R_1 moves parallel to its axis with velocity u inside a coaxial cylinder with radius R_2. Determine the motion of a fluid occupying the space between the cylinders.

17. For the plane Couette flow, if in addition to the movement of the upper plate there is also an applied negative pressure gradient $\partial p/\partial x_1$, obtain the velocity distribution. Also obtain the volume flow rate per unit width.

18. Two layers of liquids with viscosities μ_1 and μ_2, densities ρ_1 and ρ_2, respectively, and with equal depths b, flow steadily between two fixed horizontal parallel plates. Find the velocity distribution for this steady unidirectional flow.

19. Consider the flow of an incompressible viscous fluid through the annular space between two concentric horizontal cylinders. The radii are a and b.

(a) Find the flow field if there is no variation of pressure in the axial direction and if the inner and the outer cylinders have axial velocities v_a and v_b, respectively.

(b) Find the flow field if there is a pressure gradient in the axial direction and both cylinders are fixed.

Take body forces to be zero.

20. Show that for the velocity field

$$v_x = v(y, z), \qquad v_y = v_z = 0,$$

the Navier–Stokes equations, with $\vec{f}_{ext} = 0$, reduce to

$$\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = \frac{1}{\mu}\frac{dp}{dx} = \beta = \text{constant}. \tag{344}$$

21. Given the velocity field in the form

$$v_x = v = A\left(\frac{y^2}{a^2} + \frac{z^2}{b^2}\right) + B, \qquad v_y = v_z = 0,$$

find A and B for the steady laminar flow of a Newtonian fluid in a pipe having an elliptical cross-section given by

$$\frac{y^2}{a^2} + \frac{z^2}{b^2} = 1.$$

Assume no body forces, and use the governing equation obtained in the previous problem.

22. The space between two concentric spherical shells is filled with an incompressible Newtonian fluid. The inner shell, of radius r_i, is fixed; the outer shell, of radius r_o, rotates with angular velocity Ω about a diameter. Find the velocity distribution. Assume the flow to be laminar without secondary flow.

23. For the plane Couette flow, if in addition to the movement of the upper plate there is also an applied negative pressure gradient $\partial p/\partial x_1$, obtain the velocity distribution. Also obtain the volume flow rate per unit width.

24. Two layers of liquids with viscosities μ_1 and μ_2, densities ρ_1 and ρ_2, respectively, and with equal depths b, flow steadily between two fixed horizontal parallel plates. Find the velocity distribution for this steady unidirectional flow.

25. Consider the flow of an incompressible viscous fluid through the annular space between two concentric horizontal cylinders. The radii are a and b.

    (a) Find the flow field if there is no variation of pressure in the axial direction and if the inner and the outer cylinders have axial velocities v_a and v_b, respectively.

    (b) Find the flow field if there is a pressure gradient in the axial direction and both cylinders are fixed.

    Take body forces to be zero.

26. Show that for the velocity field

$$v_x = v(y, z), \qquad v_y = v_z = 0,$$

the Navier–Stokes equations, with $\vec{f}_{ext} = 0$, reduce to

$$\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = \frac{1}{\mu}\frac{dp}{dx} = \beta = \text{constant}. \tag{345}$$

27. Given the velocity field in the form

$$v_x = v = A\left(\frac{y^2}{a^2} + \frac{z^2}{b^2}\right) + B, \qquad v_y = v_z = 0,$$

find A and B for the steady laminar flow of a Newtonian fluid in a pipe having an elliptical cross-section given by

$$\frac{y^2}{a^2} + \frac{z^2}{b^2} = 1.$$

Assume no body forces, and use the governing equation obtained in the previous problem.

28. The space between two concentric spherical shells is filled with an incompressible Newtonian fluid. The inner shell, of radius r_i, is fixed; the outer shell, of radius r_o, rotates with angular velocity Ω about a diameter. Find the velocity distribution. Assume the flow to be laminar without secondary flow.

29. Consider the following velocity field in cylindrical coordinates for an incompressible fluid:

$$v_r = v(r), \qquad v_\theta = 0, \qquad v_z = 0. \tag{346}$$

(a) Show that

$$v_r = \frac{A}{r}, \tag{347}$$

where A is a constant, so that the equation of conservation of mass is satisfied.

(b) If the rate of mass flow through the circular cylindrical surface of radius r and unit length in the z-direction is Q_m, determine the constant A in terms of Q_m.

30. Given the following velocity field in cylindrical coordinates for an incompressible fluid:

$$v_r = v(r, \theta), \qquad v_\theta = 0, \qquad v_z = 0. \tag{348}$$

Show that:

(a)

$$v_r = \frac{f(\theta)}{r}, \tag{349}$$

where $f(\theta)$ is any function of θ.

(b) In the absence of body forces,

$$\frac{d^2 f}{d\theta^2} + 4f + \frac{\rho f^2}{\mu} + k = 0, \tag{350}$$

and

$$p = 2\mu \frac{f}{r^2} + \frac{k\mu}{2r^2} + C, \tag{351}$$

where k and C are constants.

31. Given the velocity field of a linearly viscous fluid:

$$v_1 = kx_1, \qquad v_2 = -kx_2, \qquad v_3 = 0. \tag{352}$$

(a) Show that the velocity field is irrotational.

(b) Find the stress tensor.

(c) Find the acceleration field.

(d) Show that the velocity field satisfies the Navier–Stokes equations by finding the pressure distribution directly from the equations. Neglect body forces and take $p = p_0$ at the origin.

(e) Use Bernoulli's equation to find the pressure distribution.

(f) Find the rate of dissipation of mechanical energy into heat.

(g) If $x_2 = 0$ is a fixed boundary, what condition is not satisfied by the velocity field?

32. Obtain the vorticity vector for the plane Poiseuille flow.

33. Obtain the vorticity vector for the Hagen–Poiseuille flow.

34. For a two-dimensional flow of an incompressible fluid, the velocity components can be expressed in terms of a scalar function ψ, known as the Lagrange stream function, by

$$v_x = \frac{\partial \psi}{\partial y}, \qquad v_y = -\frac{\partial \psi}{\partial x}. \tag{353}$$

(a) Show that the equation of conservation of mass is automatically satisfied for any $\psi(x, y)$ that has continuous second partial derivatives.

(b) Show that for two-dimensional flow of an incompressible fluid,

$$\psi = \text{constant} \tag{354}$$

are streamlines.

(c) If the velocity field is irrotational, then

$$v_i = -\frac{\partial \varphi}{\partial x_i}. \tag{355}$$

Show that the curves of constant velocity potential

$$\varphi = \text{constant} \tag{356}$$

and the streamlines

$$\psi = \text{constant} \tag{357}$$

are orthogonal to each other.

(d) Obtain the only nonzero vorticity component in terms of ψ.

35. Obtenga el perfil de velocidad para el flujo axial entre dos cascarones concéntricos de sección circular y radios interior a_1 y exterior a_2. Demuestre que la velocidad y el caudal son

$$v_z = \frac{\Delta P}{4\eta(z - z_0)} \left[a_2^2 - r^2 + \frac{a_2^2 - a_1^2}{\ln(a_2/a_1)} \ln\left(\frac{r}{a_2}\right) \right], \tag{358}$$

$$Q = \frac{\pi \Delta P}{8\eta(z - z_0)} \left[a_2^4 - a_1^4 - \frac{(a_2^2 - a_1^2)^2}{\ln(a_2/a_1)} \right] = A\langle v_z \rangle = \pi(a_2^2 - a_1^2)\langle v_z \rangle. \tag{359}$$

36. Un cilindro de radio a_1 se coloca en el eje de un cascarón cilíndrico de radio a_2. El cascarón exterior gira con velocidad angular ω constante, y en el espacio entre los dos hay un fluido viscoso. Este instrumento se conoce como viscosímetro de Couette–Hatschek.

Asumiendo simetría en z, considerando ρ y η constantes, y despreciando la gravedad, demuestre que

$$v_\varphi = \frac{\omega r}{1-\left(\frac{a_1}{a_2}\right)^2}\left[1-\left(\frac{a_1}{r}\right)^2\right], \tag{360}$$

$$T_{r\varphi} = -\eta r\frac{\partial}{\partial r}\left(\frac{v_\varphi}{r}\right)_{r=a_1} = \frac{2\eta\omega a_2^2}{a_2^2-a_1^2}. \tag{361}$$

Calcule el torque por unidad de longitud sobre el cilindro interior. Pruebe que

$$T_{rr} = T_{\varphi\varphi} = T_{zz} = 0 \tag{362}$$

y que

$$T_{r\varphi} = -\eta r\frac{\partial}{\partial r}\left(\frac{v_\varphi}{r}\right) = -\frac{\eta\omega}{1-(a_1/a_2)^2}\frac{\partial}{\partial r}\left[1-\frac{a_1^2}{r^2}\right] = -\frac{2\eta\omega a_1^2}{1-(a_1/a_2)^2}\frac{1}{r^2}. \tag{363}$$

El torque necesario para hacer rotar el cilindro a velocidad ωa_2 es

$$\tau = a_2 F = a_2\int T_{r\varphi}\,dA_r = \frac{4\eta\omega a_1^2\pi L}{1-(a_1/a_2)^2}. \tag{364}$$

La medida del torque y la velocidad angular permiten determinar la viscosidad del fluido.

Demuestre, además, que de la ecuación de movimiento

$$\frac{\partial P}{\partial r} = \frac{\rho v_\varphi^2}{r} \tag{365}$$

se sigue que

$$\frac{\partial P}{\partial r} = \frac{\rho\omega^2}{\left[1-(a_1/a_2)^2\right]^2}\frac{1}{r}\left[1-\frac{a_1^2}{r^2}\right]^2. \tag{366}$$

37. Resuelva el problema anterior si ambos cilindros giran con velocidades angulares ω_1 y ω_2. Demuestre que

$$v_\varphi = \frac{1}{1-\left(\frac{a_1}{a_2}\right)^2}\left[\left(\omega_2-\omega_1\left(\frac{a_1}{a_2}\right)^2\right)r + \frac{(\omega_1-\omega_2)a_1^2}{r}\right]. \tag{367}$$

Considere los siguientes casos particulares:

(a) $\omega_1 = 0$,

(b) $\omega_1 = \omega_2$,

(c) $a_2 \to \infty$ con $\omega_2 \to 0$,

(d) $a_1 = 0$.

Demuestre que

$$a_1 T_{r\varphi}|_{r=a_1} = \eta \left[\frac{\partial v_z}{\partial r} - \frac{v_z}{r} \right] = -2\eta a_2^2 \frac{\omega_1 - \omega_2}{a_2^2 - a_1^2}, \tag{368}$$

$$\frac{F}{L} = T_{r\varphi}|_{r=a_1} \times 2\pi a_1, \tag{369}$$

$$\left(\frac{F}{L} \right) a_1 = 4\pi\eta \frac{\omega_1 - \omega_2}{a_1^2 - a_2^2} a_1^2 a_2^2. \tag{370}$$

6 Creeping flow: the Stokes flow

At low Reynolds number, $\mathrm{Re} \ll 1$, the advective acceleration can be left out of the Navier–Stokes equations for incompressible flow, so that they for steady flow become

$$\nabla p = \eta \nabla^2 \vec{v}, \qquad \nabla \cdot \vec{v} = 0. \tag{371}$$

This approximation is usually called *Stokes flow*. Gravity, $\vec{g} = -\nabla\Phi$, has for simplicity been left out but is easy to include by replacing p by $p + \rho_0 \Phi$. From the divergence of the first equation it follows that the pressure must satisfy the Laplace equation,

$$\nabla^2 p = 0.$$

Note also that

$$\nabla \times \nabla^2 \vec{v} = 0. \tag{372}$$

Creeping flow is mathematically (and numerically) much easier to handle than general flow because of the absence of non-linear terms that tend spontaneously to break the natural symmetry of the solutions in time as well as space, with turbulence as the extreme result. The linearity of the creeping flow equations may sometimes be used to express solutions to complicated flow problems as linear superpositions of simpler solutions.

6.1 Creeping flow around a solid ball

A solid spherical ball of radius a moving at constant speed U through a viscous fluid is the centerpiece of creeping flow, going back to Stokes in 1851. Provided the Reynolds number

$$\mathrm{Re} = \frac{2aU}{\nu} \tag{373}$$

is sufficiently small, the pressure and velocity fields must satisfy the Stokes equations in the rest frame of the ball.

In general, since the pressure only appears in Eq.(371) in the combination p/η, and since the no-slip boundary condition on the surface of the body not involve the pressure, the velocity field cannot depend on the viscosity η, but only on the asymptotic velocity U and on the geometry and orientation of the body. The linearity of creeping flow guarantees that the velocity field must everywhere be proportional to the magnitude of the asymptotic velocity, $|\vec{v}| \sim U$.

The first step in obtaining the solution is to recognize that the perfect spherical symmetry of the sphere is only broken by the asymptotic velocity vector $\vec{U}$, and that the velocity field and the pressure must be linear in $\vec{U}$, as discussed above. Since the only other vector at play in the formulation of the problem is the radius vector $\vec{x}$ from the center of the sphere, the scalar pressure must be proportional to $\vec{U} \cdot \vec{x}$. In spherical coordinates with $\vec{U} = U\hat{e}_z$, we have

$$\vec{U} \cdot \vec{x} = rU\cos\theta,$$

leading to a pressure field of the form

$$p = \eta U \cos\theta \, q(r), \tag{374}$$

where $q(r)$ is a function of only the radial distance $r = |\vec{x}|$. The explicit factor η is included because, as argued above, the pressure must be proportional to the viscosity.

The velocity field must similarly be a linear combination,

$$\vec{v} = A(r)\vec{U} + B(r)(\vec{U}\cdot\vec{x})\vec{x},$$

of the vectors $\vec{U}$ and $(\vec{U}\cdot\vec{x})\vec{x}$ with coefficients that can only depend on r. Using that $\vec{x}\cdot\hat{e}_r = r$ and $\vec{x}\cdot\hat{e}_\theta = 0$, together with $\vec{U}\cdot\hat{e}_r = U\cos\theta$ and $\vec{U}\cdot\hat{e}_\theta = -U\sin\theta$, it follows that the spherical field components must be of the form

$$v_r = U\cos\theta\, f(r), \qquad v_\theta = -U\sin\theta\, g(r), \qquad v_\phi = 0, \tag{375}$$

where $f(r)$ and $g(r)$ are functions of r only.

Using that the Laplacian of the pressure has to vanish, we obtain from the spherical expression for the scalar Laplacian,

$$\nabla^2 p = \eta U\cos\theta\left(\frac{d^2q}{dr^2} + \frac{2}{r}\frac{dq}{dr} - \frac{2q}{r^2}\right) = 0. \tag{376}$$

Requiring the expression in parentheses to vanish,

$$\frac{d^2q}{dr^2} + \frac{2}{r}\frac{dq}{dr} - \frac{2q}{r^2} = 0, \tag{377}$$

we have obtained an ordinary second-order differential equation that is homogeneous in r. One may verify that the two linearly independent solutions are $q \sim r$ and $q \sim 1/r^2$, so that

$$q = \frac{C}{r^2} + Dr, \tag{378}$$

where C and D are integration constants.

The divergence of the velocity field must vanish. Then we get

$$\nabla\cdot\vec{v} = \cos\theta\left(\frac{df}{dr} + \frac{2f}{r} - \frac{2g}{r}\right)U = 0, \tag{379}$$

leading to

$$g = f + \frac{1}{2}r\frac{df}{dr}. \tag{380}$$

This leaves only the function $f(r)$ to be determined. From the Stokes equation and the radial part of the vector Laplacian, The Laplacian of a vector field becomes

$$\begin{aligned}\nabla^2\vec{U} = {}& \hat{e}_r\left(\frac{\partial^2 U_r}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 U_r}{\partial\theta^2} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 U_r}{\partial\phi^2} + \frac{2}{r}\frac{\partial U_r}{\partial r} + \frac{\cot\theta}{r^2}\frac{\partial U_r}{\partial\theta} - \frac{2}{r^2}\frac{\partial U_\theta}{\partial\theta} - \frac{2}{r^2\sin\theta}\frac{\partial U_\phi}{\partial\phi} - \frac{2U_r}{r^2} - \frac{U_\theta}{r^2\tan\theta}\right)\\ &+ \hat{e}_\theta\left(\frac{\partial^2 U_\theta}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 U_\theta}{\partial\theta^2} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 U_\theta}{\partial\phi^2} + \frac{2}{r}\frac{\partial U_\theta}{\partial r} + \frac{\cot\theta}{r^2}\frac{\partial U_\theta}{\partial\theta} + \frac{1}{r^2}\frac{\partial U_r}{\partial\theta} - \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial U_\phi}{\partial\phi} - \frac{U_\theta}{r^2\sin^2\theta}\right)\\ &+ \hat{e}_\phi\left(\frac{\partial^2 U_\phi}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 U_\phi}{\partial\theta^2} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 U_\phi}{\partial\phi^2} + \frac{2}{r}\frac{\partial U_\phi}{\partial r} + \frac{\cot\theta}{r^2}\frac{\partial U_\phi}{\partial\theta} + \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial U_\theta}{\partial\phi} + \frac{2}{r^2\sin\theta}\frac{\partial U_r}{\partial\phi} - \frac{U_\phi}{r^2\sin^2\theta}\right).\end{aligned} \tag{381}$$

we obtain

$$\frac{d^2f}{dr^2} + \frac{2}{r}\frac{df}{dr} + \frac{4g}{r^2} - \frac{4f}{r^2} = \frac{dq}{dr}, \tag{382}$$

and inserting q and g from above, we arrive at

$$\frac{d^2f}{dr^2} + \frac{4}{r}\frac{df}{dr} = -\frac{2C}{r^3} + D. \tag{383}$$

Let $y(r) = df/dr$. The equation becomes

$$\frac{dy}{dr} + \frac{4}{r}y = -\frac{2C}{r^3} + D. \tag{384}$$

This is a first-order linear equation. The integrating factor is

$$\mu(r) = \exp\left(\int \frac{4}{r}\,dr\right) = r^4.$$

Multiplying by $\mu(r)$,

$$\frac{d}{dr}\left(r^4 y\right) = -2Cr + Dr^4. \tag{385}$$

Integrating,

$$r^4 y = -Cr^2 + \frac{D}{5}r^5 + K, \tag{386}$$

and therefore

$$\frac{df}{dr} = -\frac{C}{r^2} + \frac{D}{5}r + \frac{K}{r^4}. \tag{387}$$

Integrating once more, we obtain

$$f(r) = A + \frac{B}{r^3} + \frac{C}{r} + \frac{D}{10}r^2, \tag{388}$$

where A and B are integration constants.

Inserting this into the expression for g, we find

$$g = A - \frac{B}{2r^3} + \frac{C}{2r} + \frac{1}{5}Dr^2. \tag{389}$$

The framed formulas constitute the complete generic solution for this problem.

The field

Finally we must apply boundary conditions to determine the four unknown integration constants A, B, C, and D. For $r \to \infty$, the field must approach the homogeneous field $\vec{U}$, implying $f \to 1$ and $g \to 1$, or $A = 1$ and $D = 0$. At the surface of the sphere, $r = a$, the field must vanish because of the no-slip condition, leading to

$$B = \tfrac{1}{2}a^3, \qquad C = -\tfrac{3}{2}a.$$

Putting it all together, the spherical velocity components and the pressure become

$$v_r = \left(1 - \frac{3a}{2r} + \frac{a^3}{2r^3}\right) U \cos\theta, \tag{390}$$

$$v_\theta = -\left(1 - \frac{3a}{4r} - \frac{a^3}{4r^3}\right) U \sin\theta, \tag{391}$$

$$v_\phi = 0, \tag{392}$$

$$p = -\frac{3}{2}\eta \frac{a}{r^2} U \cos\theta. \tag{393}$$

The pressure is forward–backward asymmetric, highest on the part of the sphere that turns toward the incoming flow ($\theta = \pi$). This asymmetry is in marked contrast to the potential flow solution, for which the pressure is symmetric. The flow pattern is independent of the viscosity of the fluid, whereas the pressure is proportional to η, as expected for creeping flow problems where the boundary conditions do not involve the pressure.

Stokes' law

The asymmetric pressure distribution and the viscous shear stresses create a drag on the sphere in the direction of the asymptotic flow. To find it we first calculate the stress components σ_{rr} and $\sigma_{\theta r}$ from the standard Newtonian stress tensor. From the velocity gradient matrix in spherical coordinates, we obtain

$$\sigma_{rr} = -p + 2\eta \frac{\partial v_r}{\partial r} = \frac{3a}{2r^2}\left(3 - \frac{2a^2}{r^2}\right) \eta U \cos\theta, \tag{394}$$

$$\sigma_{\theta r} = \eta\left(\frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} + \frac{\partial v_\theta}{\partial r}\right) = -\frac{3a^3}{2r^4}\eta U \sin\theta. \tag{395}$$

The only other non-vanishing stress components are

$$\sigma_{\theta\theta} = \sigma_{\phi\phi} = \frac{2a^3}{2r^4}\eta U \cos\theta.$$

At the surface of the sphere, $r = a$, the stress components become

$$\sigma_{rr}\big|_{r=a} = \frac{3\eta U}{2a}\cos\theta, \qquad \sigma_{\theta r}\big|_{r=a} = -\frac{3\eta U}{2a}\sin\theta. \tag{396}$$

From these we get the surface stress vector

$$\boldsymbol{\sigma} \cdot \hat{e}_r\big|_{r=a} = \hat{e}_r \sigma_{rr} + \hat{e}_\theta \sigma_{\theta r} = \frac{3\eta U}{2a}\hat{e}_z. \tag{397}$$

Surprisingly it is of constant magnitude and points everywhere in the direction of the asymptotic flow across the entire surface.

The total reaction force is obtained by multiplying the constant stress vector by the area $4\pi a^2$ of the sphere. Since the force is proportional to U, it is a pure drag,

$$D = 6\pi\eta a U. \tag{398}$$

This is the famous *Stokes' law* from 1851.

Terminal velocity

Although Stokes' law has been derived in the rest frame of the sphere, it is also valid in the rest frame of the asymptotic fluid. The terminal velocity of a falling solid sphere may be obtained by equating the force of gravity (minus buoyancy) with the Stokes drag,

$$(\rho_1 - \rho_0)\frac{4}{3}\pi a^3 g_0 = 6\pi\eta a U, \tag{399}$$

where g_0 is the gravitational acceleration and ρ_1 is the average density of the sphere. Solving for U, we find

$$U = \frac{2}{9}\left(\frac{\rho_1}{\rho_0} - 1\right)\frac{a^2 g_0}{\nu}, \tag{400}$$

where $\nu = \eta/\rho_0$ is the kinematic viscosity.

6.2 Life at Low Reynolds Number

Most of the examples discussed in this chapter involve macroscopic flows, such as water flowing through pipes, air moving around aircraft, or large-scale atmospheric and oceanic motions. In these situations, inertial effects are often important and the Reynolds number may be large. However, the physical world changes dramatically when one considers sufficiently small organisms or devices moving in a viscous fluid.

A classic discussion of this subject was presented by E. M. Purcell in his celebrated article *Life at Low Reynolds Number* (1977). In this paper, Purcell explores the consequences of fluid motion in the limit

$$Re = \frac{UL}{\nu} \ll 1, \tag{401}$$

where viscous forces dominate over inertial forces. In this regime, the Navier–Stokes equations reduce to the Stokes equations,

$$-\nabla p + \eta\nabla^2\vec{v} = 0, \tag{402}$$

which are linear and do not contain acceleration terms.

As a consequence, the motion of the fluid exhibits properties that are entirely different from our everyday experience. In particular, inertia becomes negligible: if an external force ceases to act, the fluid and any immersed object stop almost instantaneously. Furthermore, the equations governing the flow are invariant under time reversal. This remarkable property leads to Purcell's famous *scallop theorem*, which states that a swimmer performing a reciprocal motion (one that looks identical when played backward in time) cannot achieve net propulsion in a fluid at very low Reynolds number.

Purcell illustrates these ideas through the motion of microorganisms such as bacteria, whose typical Reynolds numbers can be as small as

$$Re \sim 10^{-5}. \tag{403}$$

Such organisms cannot swim by the same mechanisms employed by fish or humans. Instead, they rely on non-reciprocal motions, such as rotating helical flagella, to move through the surrounding fluid.

Another important lesson from Purcell's article concerns the competition between advection and diffusion. At microscopic scales, diffusion is often more effective than active swimming for transporting nutrients and chemical signals. Consequently, locomotion at low Reynolds number is not primarily a means of overcoming inertia but rather a strategy for exploring the local environment and locating regions with favorable conditions.

Purcell's article remains one of the most influential and pedagogically valuable papers in fluid mechanics. It provides a striking illustration of how the behavior of a physical system can change qualitatively when the governing dimensionless parameters are varied, and it serves as an excellent example of the power of dimensional analysis and physical reasoning.

6.3 Problems

1. A spherical particle begins to fall from rest in constant gravity. Determine how the particle reaches the terminal speed given by Eq. (17.27) under the assumption that the Reynolds number is always small.

2. Tiny unicellular animals in the sea die and their carcasses settle slowly toward the bottom. Assume that an animal's carcass is spherical with diameter 10 μm and average density 1.2 times that of water. Calculate:

   (a) the Reynolds number,

   (b) the time it takes for the carcass to settle to the bottom of the deep sea (depth 4 km).

3. A microscopic spherical grain of volcanic (or cosmic) dust of radius a enters the homentropic atmosphere at a height z.

   (a) Determine the time t_0 it takes for the grain to settle to the ground under the assumptions that:

   - the density of the grain is much larger than the density of air,
   - Stokes' law is valid at any height,
   - the actual density and viscosity of air at that height are used.

   (b) Evaluate the settling time numerically for reasonable values of the parameters.

4. Calculate the pressure contribution to the drag from Stokes' solution for a sphere.

5. An object with constant drag coefficient C_D falls from rest in constant gravity. Determine how the particle reaches terminal speed under the assumption that buoyancy can be disregarded.

6. Considere un fluido viscoso en movimiento entre dos placas paralelas circulares separadas una distancia L. La placa inferior ($z = 0$) está en reposo y la superior ($z = L$) gira con velocidad angular constante Ω. No considere gravedad ni efectos de borde.

   (a) Escriba las ecuaciones de continuidad y movimiento. ¿De qué factores dependen la velocidad y la presión?

(b) Demuestre que la ecuación de movimiento, después de separar variables como

$$v_\theta = A(r)B(z),$$

toma la forma

$$\frac{1}{A}\frac{d}{dr}\left[\frac{1}{r}\frac{d}{dr}(rA)\right] + \frac{1}{B}\frac{d^2B}{dz^2} = 0. \tag{404}$$

Considere constantes de separación α^2 y $-\alpha^2$, respectivamente.

(c) Analice el caso $\alpha = 0$ y demuestre que

$$v_\theta = Cr(az + b).$$

(d) Para $-\alpha^2$, demuestre que la ecuación radial es

$$\frac{d^2A}{dr^2} + \frac{1}{r}\frac{dA}{dr} - \left(\alpha^2 + \frac{1}{r^2}\right)A = 0. \tag{405}$$

La solución a esta ecuación es una función de Bessel modificada $I_1(\alpha r)$. ¿Por qué no se utiliza la función de Bessel modificada $K_1(\alpha r)$?

Resuelva la ecuación para $B(z)$. Escriba la solución general, incluyendo la solución para $\alpha = 0$:

$$v_\theta = I_1(\alpha r)\left[D\sin(\alpha z) + E\cos(\alpha z)\right] + Cr(az + b). \tag{406}$$

(e) Aplicando las condiciones de frontera en $z = 0, L$, demuestre que el perfil de velocidad es

$$v_\theta = \sum_{n=1}^{\infty} D_n I_1\left(\frac{n\pi r}{L}\right)\sin\left(\frac{n\pi z}{L}\right) + \Omega r\left(\frac{z}{L}\right). \tag{407}$$

References

[1] A. Sepúlveda, *Hidrodinámica*. Fondo Editorial EIA, Colombia, 2019.

[2] L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, vol. 6 of *Course of Theoretical Physics*. Pergamon Press, Oxford, UK, 2nd ed., 1987.

[3] B. Lautrup, *Physics of Continuous Matter: Exotic and Everyday Phenomena in the Macroscopic World*. CRC Press, Boca Raton, 2 ed., 2011.

[4] M. C. Potter, D. C. Wiggert, and B. H. Ramadan, *Mechanics of Fluids*. Cengage Learning, Stamford, CT, 4th ed., 2011.

[5] F. M. White, *Fluid Mechanics*. McGraw-Hill, New York, NY, 7th ed., 2011. University of Rhode Island.

[6] H. D. Young, R. A. Freedman, A. L. Ford, and F. W. Sears, *Sears and Zemansky's University Physics: With Modern Physics*. Pearson Addison-Wesley, San Francisco, 13th ed., 2012.